

ENGINEERING DESIGN HANDBOOK

VCH Sizing Framework

A Sizing Framework for Submerged Hydronic Systems
Thermal and Hydraulic Design — Second Edition, Revised & Expanded

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— **Sambridda Ranabhat**

Kathmandu, Nepal

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Abstract

This document presents the Velocity-Constrained Hydronic (VCH) Sizing Framework—a structured, iterative engineering methodology for the thermal and hydraulic design of submerged hydronic heating and cooling systems used in bulk liquid thermal processing applications. The framework addresses the fundamental inadequacy of linear design approaches by integrating thermal demand analysis, thermofluid selection, vessel geometry, hydraulic resistance characterisation, heat transfer evaluation, coil optimisation, environmental loss quantification, thermal driver sizing, and thermal battery design into a unified eleven-step engineering workflow culminating in a formal design verification gate.

The methodology introduces two distinct operating mode demands—the *batch processing target*, which accounts for the full sensible and latent energy required to process incoming cold raw material within a defined operational window, and the *balance target*, which quantifies the steady-state energy required to maintain a vessel already at operating temperature against environmental heat loss. The governing demand is established as an independent fixed design target before any hardware is selected, eliminating the circular dependency inherent in frameworks that derive their design target from the hardware they are sizing.

Practical engineering constraints are embedded throughout, including velocity-based erosion ceilings enforced against commercially available Nominal Pipe Sizes, a compound coil packing efficiency correction accounting for fabrication tolerances, structural support requirements, weld access, thermal expansion, and dead-zone prevention, and a worst-case ambient temperature reference applied consistently across all thermal boundary calculations. The framework incorporates a rigorous series thermal resistance network for environmental heat loss quantification, Effectiveness-NTU heat transfer evaluation with separate steady-state and cold-start coefficient assessments, and a thermal battery sizing procedure that produces a complete custom tank geometry bounded by hydraulic priming and driver stability constraints.

The resulting methodology provides system designers, mechanical engineers, and mechatronics integrators with a standardised, auditable, and physically grounded design protocol that is inherently safeguarded against long-term fouling degradation, hydraulic instability, and thermal underperformance.

1 Introduction

The thermal management of bulk liquid storage and processing vessels is a foundational requirement across the chemical, food processing, agricultural commodity, and heavy manufacturing industries. The engineering challenge appears deceptively straightforward: heat a volume of fluid from one temperature to another within a defined time. In practice, however, the design of the internal hydronic system is governed by a tightly coupled set of thermal, hydraulic, geometric, and control constraints that interact in ways a simple linear calculation cannot capture. This complexity is compounded when the process fluid undergoes a phase transition—such as the melting of solid or slurry commodities—where thermophysical properties shift by orders of magnitude.

The traditional approach to hydronic coil sizing treats the problem as a sequential thermodynamic exercise. Each step is performed in isolation. The result is a design that is thermodynamically plausible on paper but fails in practice for reasons that are entirely predictable: pipe diameters derived from unconstrained calculations do not correspond to commercially available sizes; coils sized to theoretical thermal minimums degrade rapidly under organic fouling; and conventional calculations assume static fluid properties, completely ignoring the massive thermal resistance of solid boundary layers.

The Velocity-Constrained Hydronic (VCH) Sizing Framework was developed to resolve this gap. The methodology emerged from the design of a real industrial thermal circulation system involving submerged hydronic coils for viscous, phase-changing fluid heating, where the inadequacy of conventional sizing tools became apparent during translation from theoretical design to physical fabrication.

In this second edition, the framework has been significantly expanded and refined. It departs from traditional linear design approaches in four fundamental ways:

- **Independent Thermal Targets and Energy-Weighted Phase Allocation.** The required process heat load is fixed at the outset. When phase-changing media are processed, the framework partitions the thermal energy budget into distinct latent and sensible zones, using independent overall heat transfer coefficients combined based on energy-weighted duration fractions.
- **Commercial Geometric and Hydraulic Constraints.** All pipe geometry is constrained to commercially available Nominal Pipe Sizes from the outset. A velocity ceiling is assigned before the coil is designed, governing every hydraulic, shear-stress, and automation decision that follows. Fouling resistances are evaluated dynamically as functions of fluid velocity, wall roughness, fluid purity, surface temperature, and operational cycling.
- **Hydrodynamic State Integration.** The methodology accounts for the physical boundary layer variations between stagnant vessels and mechanically agitated tanks, integrating impeller dynamics and shear-driven forced convection correlations where applicable.
- **Multi-Condition Evaluation and Simplified Linear Sizing.** The framework evaluates performance under both worst-case cold-start and stabilised steady-state operating conditions. In the final sizing optimisation stage, a first-order linear shortcut ($\varepsilon \approx \text{NTU}$ for $\text{NTU} < 0.1$) bypasses complex transcendental iterations, allowing direct, non-iterative calculation of the required surface area.

2 Scope and Operational Boundaries

To preserve the mathematical validity, physical safety, and operational predictability of the VCH Sizing Framework, its engineering boundaries must be precisely defined. Applying the framework outside its validated boundary conditions introduces risks of mechanical failure, control instability, and capital loss.

2.1 Intended Applications

The VCH Sizing Framework is explicitly compiled, mathematically modelled, and physically validated for systems operating within the following technical domains:

- **Submerged Coil Heat Exchangers.** Designed specifically for internal helical or serpentine coil configurations completely submerged within a bulk process fluid contained in a cylindrical vessel. Both stagnant and mechanically agitated hydrodynamic regimes are accommodated.
- **Single-Phase Closed-Loop Hydronic Systems.** Tailored for circulation systems utilising single-phase liquid thermal transport media—including treated water, ethylene glycol blends, synthetic thermal oils, and food-grade heat transfer fluids—operating below their boiling points and above their freezing thresholds at system operating pressure.
- **Multiphase and Sensible Process Thermal Processing.** Validated for applications involving controlled temperature elevation, temperature maintenance, or latent phase-change transitions. The framework explicitly models phase-change heat transfer through segmented energy-budget partitioning and weighted overall heat transfer coefficient calculations.
- **High-Viscosity Fluid Thermal Processing.** Optimised for the non-linear temperature-viscosity behaviour characteristic of heavy organic products such as edible plant oils, heavy hydrocarbons, waxes, and liquid-phase polymers.
- **Multi-Vessel Industrial Plant Layouts.** Capable of sizing a centralised thermal driver and thermal battery to serve multiple parallel process vessels simultaneously, with total capacity governed by simultaneous peak loading across all active nodes.

2.2 Unintended Applications

The foundational assumptions of the VCH Sizing Framework render it invalid for the following equipment classifications:

- **Phase-Change Utility Systems.** Steam boilers, flash evaporators, or any thermal asset where the internal utility thermofluid undergoes a liquid-to-vapour phase transformation.
- **Cryogenic and Deep-Refrigeration Systems.** The material selection matrices, baseline fouling factors, and thermal expansion parameters are structurally unsuited to cryogenic regimes.

- **Multiphase Transport Regimes.** Slug, plug, or annular multiphase flows inside the pipe network, where erosion mechanics are driven by particulate or droplet impact.
- **Direct-Fired and Combustion Systems.** Direct-fired thermal oxidizers, radiant furnace tubes, or combustion-driven heat recovery configurations operating under radiation-dominated heat flux profiles.
- **Open-Loop Configurations.** Systems where the utility fluid is continuously consumed or discharged rather than recirculated.

2.3 Note on Cooling Applications

The thermodynamic and hydraulic principles underlying this framework are applicable to cooling loops via symmetric logic, provided the inverse directional constraints are properly accounted for:

- *Inverted Ambient References:* The worst-case design envelope shifts to peak summer ambient conditions.
- *Altered Fouling Mechanisms:* Cooling coils risk accelerating the crystallisation of waxes and fats directly onto the cold external surface.
- *Chiller Volumetric Limits:* Transient control loops must account for compressor anti-recycle constraints.

3 Governing Assumptions

To establish analytical tractability and ensure a mathematically rigorous foundation, the VCH Sizing Framework relies on the following standardised engineering assumptions.

1. **Piecewise Steady-State Operational Baseline.** Phase-change transitions are modelled using a piecewise steady-state approach across segregated thermal energy zones. Dynamic macro-transients are consolidated through a design factor of safety, worst-case cold-start heat transfer coefficients, and thermal battery volumetric sizing.
2. **Incompressible Fluid Flow.** The thermofluid is assumed to be an incompressible liquid operating entirely in a single liquid phase, validating the direct application of the continuity equation and the Darcy-Weisbach friction loss model.
3. **Newtonian Fluid Approximation.** Both the internal thermofluid and the external bulk process fluid are treated as Newtonian fluids within their active operational temperature bands, allowing the use of standard dimensionless groups (Reynolds, Prandtl, Nusselt) and the Colebrook-White friction factor correlation.
4. **Uniform Bulk Process Temperature.** The bulk process fluid is assumed to maintain a uniform macro-scale bulk temperature at any distinct step within the processing cycle, allowing the external thermal resistance to be modelled using a localised convective boundary layer coefficient.
5. **Quasistatic Piecewise Fluid Properties.** Thermophysical properties are treated as constant within a given calculation pass, extracted as piecewise values evaluated at the mean bulk operating temperature of the respective thermal zone.
6. **Single-Phase Hydronic Transport.** Zero localised boiling, flashing, or vapour-fraction generation along the internal pipe walls is assumed.
7. **Negligible Radiative Heat Losses.** External heat dissipation is assumed to be dominated by conduction through the insulation layer and subsequent convection to the ambient air.
8. **Known Worst-Case Ambient Temperature.** The historically worst-case ambient temperature for the installation site is predefined as a static boundary condition, applied consistently across all steps where ambient temperature impacts vessel heat loss.
9. **Available Process Fluid Thermophysical Properties.** The thermophysical properties of the process fluid are assumed to be known or reliably estimable from empirical literature.

4 Glossary of Terms

Batch Processing Mode The peak thermal demand operating state of a process vessel during which raw material is actively being heated from its incoming condition to the target operating temperature.

Balance Mode The steady-state minimum demand operating state during which the vessel is already at operating temperature and the system maintains thermal equilibrium against environmental heat loss only.

Coil Packing Efficiency (η_{pack}) The compound dimensionless multiplier applied to the theoretical maximum coil length to account for real-world fabrication, structural, and operational constraints. It is the product of five sub-factors: fabrication quality, support structure, weld clearance, thermal expansion, and dead-zone prevention.

Dean Number (De) A dimensionless parameter quantifying the intensity of the secondary counter-rotating flow structures (Dean vortices) generated by fluid flowing through a curved or helical pipe. Governs the selection of the appropriate helical friction factor and Nusselt number correlations.

Dean Vortices Secondary counter-rotating flow cells generated by the centrifugal force field acting on fluid traversing a helical pipe bend. These vortices increase the effective friction factor and enhance the internal convective heat transfer coefficient relative to straight pipe flow.

Design Factor of Safety (FOS) A dimensionless multiplier applied to the governing process heat load to establish the design thermal target. Accounts for long-term thermal performance degradation due to fouling, uncertainty in heat transfer coefficient estimation, manufacturing tolerance effects, and cold-start surge demand. Minimum value 1.5, recommended value 2.0, maximum recommended value 2.5.

Design Thermal Target (Q_{design}) The fixed independent thermal power benchmark against which all hardware decisions in the framework are evaluated. Defined as $Q_{\text{required}} \times \text{FOS}$. Established in Step 1 and carried forward unchanged through all subsequent steps.

Effectiveness (ε) The dimensionless ratio of the actual heat transfer rate to the thermodynamic maximum possible heat transfer rate, bounded strictly between 0 and 1.

Fouling Resistance (R_f) The thermal resistance introduced by deposit layers accumulating on internal or external pipe surfaces over the operational service life.

Hydronic Priming Volume (V_{hydronic}) The minimum thermofluid inventory required to completely fill all internal wetted passages of the closed circulation loop. The unconditional minimum volume floor for the thermal battery.

Logarithmic Mean Temperature Difference (LMTD / ΔT_{lm}) The effective thermal driving force between two fluid streams in a heat exchanger, accounting for the exponential change in temperature differential along the length of the exchange surface.

Nominal Pipe Size (NPS) A standardised pipe designation system defined by ASME B36.10M and B36.19M specifying commercially available external diameters, wall thick-

nesses, and internal flow areas.

Number of Transfer Units (NTU) A dimensionless parameter quantifying the thermal sizing of a heat exchanger relative to the fluid heat capacity rate. For $C_r = 0$, $\text{NTU} = -\ln(1 - \varepsilon)$.

Overall Heat Transfer Coefficient (U) The composite thermal conductance of the complete heat exchange interface between the thermofluid and the process fluid per unit of external pipe surface area.

Pump Affinity Laws Governing relationships between pump rotational speed and hydraulic performance: flow rate is proportional to speed, head is proportional to speed squared, and power is proportional to speed cubed.

Thermal Battery The custom-fabricated utility fluid reservoir that stores conditioned thermofluid and supplies it continuously to the circulation loop serving the process vessels.

Thermal Driver The primary commercial heating or cooling source—electric immersion heater assembly, fired thermal fluid heater, industrial heat pump, or vapour compression chiller—that conditions the thermofluid to the required supply temperature.

Thermal Time Constant (S_t) A stability metric quantifying the thermal inertia of the thermal battery relative to the peak process demand: $S_t = \rho V_{\text{battery}} C_p / Q_{\text{peak}}$ (seconds).

Variable Frequency Drive (VFD) An electronic motor speed controller that regulates pump rotational speed by varying the frequency of the electrical supply to the pump motor. Used to clamp fluid velocity to or below the allowable ceiling.

Velocity Ceiling (v_{max}) The maximum allowable steady-state fluid velocity within the internal heat exchange coil. Determined by the combination of thermofluid class and pipe material. Enforced as a hard constraint on all pump selection, VFD configuration, and automation control logic.

ΔT_{coil} The temperature difference between the thermofluid at the coil inlet and outlet. Capped at 10% of $(T_{\text{operating}} - T_{\text{worst ambient}})$.

ΔT_{drive} The temperature difference between the thermofluid inside the coil and the process fluid in the tank—the thermal driving force across the coil wall.

ΔT_{loss} The temperature difference between the process fluid at operating temperature and the worst-case ambient temperature, driving the steady-state environmental heat loss.

5 How to Use This Document

This document is structured as a sequential engineering design guide. The eleven steps of the core methodology are designed to be executed in order. Each step produces a defined set of outputs that serve as inputs to the steps that follow.

5.1 Document Structure

Front Matter

Comprises the Abstract, Introduction, Scope and Operational Boundaries, Governing Assumptions, Glossary, and How to Use section.

Framework Overview

Provides a concise description of the structured engineering methodology, detailing the core temperature differential definitions and integrating the non-iterative linear optimisation shortcut for low-NTU regimes.

Core Methodology

Details the sequential engineering procedure. Each step follows a standardised modular format: technical overview, numbered subsections, verification gates, and a step output summary.

Appendix A

Governing Mathematical Derivations.

Appendix B

Thermophysical Fluid Property Matrices.

Appendix C

Commercial Piping Standards and Material Profiles.

Appendix D

Distributed Process Automation and Control Architecture — First Edition Baseline (retained as legacy reference).

Appendix E

PLC Implementation and Discrete-Time Control Formulations — First Edition Baseline (retained as legacy reference).

Appendix F

Advanced Process Automation and Control Architecture — Second Edition.

Appendix G

Advanced PLC Implementation and Discrete-Time Control Formulations — Second Edition.

Appendix H

Sizing Case Study and Worked Example — First Edition Baseline (retained as historical reference).

Appendix I

Industrial Multiphase Sizing Case Study & Field Validation (Palm Oil Melting & Retention System, MEPL, Kathmandu).

5.2 Feedback Loops

The framework contains three formal iteration loops:

The Hydraulic Velocity Loop

Triggered in Step 5 when the velocity ceiling cannot be satisfied by VFD adjustment alone. Returns to Step 4 to increment the NPS.

The Thermal Deficiency Loop

Triggered in Steps 6 and 7 when the coil cannot deliver Q_{design} within the vessel envelope. Returns to Step 4 to increment the NPS.

The Q_{loss} Feedback Loop

Triggered in Step 9 when the rigorous heat loss calculation deviates from the placeholder by more than 15%. Returns to Step 1 to recalculate Q_{design} with the updated loss value.

5.3 First-Time Users

Engineers using the VCH framework for the first time are advised to read the Framework Overview and the Glossary before beginning the core methodology. The three temperature differential definitions— ΔT_{coil} , ΔT_{drive} , and ΔT_{loss} —are foundational and should be understood clearly before Step 1 is started.

5.4 Experienced Users

For engineers familiar with the framework executing a subsequent design, the Step Output Summaries at the end of each step provide a compact checklist of all values that must be carried forward.

6 Overview

Revised & Extended Methodology — Second Edition

6.1 Governing Design Philosophy

The framework operates on four foundational principles:

1. The thermal target is defined *before* the hardware. $Q_{\text{design}} = Q_{\text{required}} \times \text{FOS}$ is fixed before any hardware is selected.
2. All pipe geometry is locked to commercially available Nominal Pipe Sizes in accordance with ASME B36.10M and B36.19M. Arbitrary theoretical diameters are not permitted.
3. Fluid velocity is treated as a hard ceiling, not a guideline. All pump selection, VFD configuration, and control automation logic is bounded by v_{max} .
4. A unified worst-case ambient reference temperature is applied throughout—the historically coldest recorded ambient temperature for heating applications; the historically hottest for cooling.

6.2 Temperature Differential Definitions

Three distinct temperature differentials govern the framework and must not be conflated:

ΔT_{drive} — the temperature difference between the thermofluid inside the coil and the process fluid in the tank. This is the thermal driving force across the coil wall and governs the rate of heat transfer. Used in LMTD and Effectiveness-NTU calculations.

ΔT_{coil} — the temperature difference between the thermofluid at the coil inlet and the thermofluid at the coil outlet. Capped at 10% of $(T_{\text{operating}} - T_{\text{worst ambient}})$ to ensure that fluid thermophysical properties remain sufficiently uniform for single-pass lumped-parameter calculations to remain valid.

ΔT_{loss} — the temperature difference between the process fluid and the worst-case ambient temperature. Drives the steady-state environmental heat loss through the vessel walls and insulation.

6.3 Framework Execution Sequence

Step 1 — Establish the Design Thermal Target

The total sensible and latent energy required to bring the process fluid from its initial condition to its target operating temperature, compressed into the active operating window, defines the required continuous thermal power. Multiplied by the design factor of safety to establish Q_{design} .

Step 2 — Select and Characterise the Thermofluid

The secondary thermal transport medium is selected based on the operating temperature range, viscosity profile, material compatibility, thermal stability, and regulatory

constraints. Thermophysical properties are extracted at the bulk operating temperature.

Step 3 — Develop a Conceptual Vessel Geometry

A baseline vessel geometry is established from the client's target process volume using standard compound vessel geometries with conical or hemi-ellipsoidal caps.

Step 4 — Select NPS and Design the Maximum Coil Configuration

The velocity ceiling $v_{\max}$ is formally assigned. The maximum helical coil configuration is modelled and de-rated by a compound packing efficiency factor η_{pack} . The result is the maximum practical coil length $L_{\text{practical}}$.

Step 5 — Select a Pump for the Maximum Coil Configuration

Hydraulic resistance is characterised using Darcy-Weisbach with helical curvature corrections (Ito's correlations). A pump is selected by overlaying its performance curve onto the system resistance curve. A VFD is applied where necessary to clamp velocity below $v_{\max}$.

Step 6 — Determine Total Transmissible Thermal Power

The maximum deliverable thermal power is evaluated at the confirmed operating flow rate using the Effectiveness-NTU method. Phase-change systems are handled via an energy-weighted overall heat transfer coefficient (U_{weighted}).

Step 7 — Optimise the Required Coil Length

The ε -NTU relationship is inverted analytically to yield a closed-form expression for the minimum coil length L_{required} as a direct function of Q_{design} , U , $C_{\min}$, and the pipe outer diameter. The optimised helical pitch and turn count are back-calculated from the confirmed length, and the system hydraulic curve is reviewed at the reduced coil geometry.

Step 8 — Correct Vessel Volume for Coil Displacement

The volume displaced by the submerged coil assembly is quantified. Vessel dimensions are adjusted via vertical shell extension (preferred) or radial expansion to recover the lost capacity.

Step 9 — Size the Thermal Driver

The Q_{loss} placeholder is retired and replaced with a rigorous series thermal resistance network calculation. The total thermal driver capacity Q_{total} is established for simultaneous peak loading across all vessels.

Step 10 — Size the Thermal Battery

The utility fluid reservoir is sized between a minimum (governed by hydronic priming and driver stability constraints) and a maximum (governed by peak batch autonomy). Full custom tank geometry is derived.

Step 11 — Design Verification and Release

All governing constraints from Steps 1–10 are consolidated into a structured audit across four domains: thermal, hydraulic, geometric, and control stability. The design is released only when every check is documented as passed.

6.4 Closed-Loop Iteration Logic

- If the fluid velocity at the natural operating point exceeds $v_{\max}$ and cannot be resolved by VFD reduction, the designer returns to Step 4, increments the NPS upward by one standard dimension, and repeats the hydraulic characterisation.
- If the optimised transmissible thermal power remains insufficient to meet Q_{design} despite coil maximisation, the designer returns to Step 4 and selects an expanded pipe diameter.
- If the corrected vessel volume required to offset coil displacement exceeds the available plant footprint, the designer returns to Step 8 and prioritises vertical shell extension over radial expansion.

This overview presents the conceptual and logical structure of the VCH framework. Governing equations, thermophysical property reference data, commercial pipe dimensional standards, and material selection matrices are compiled in the associated appendices.

7 Core Methodology

Step 1: Establishment of the Design Thermal Target

The design process begins not with hardware selection, but with a precise, independently defined thermal target. This distinction is fundamental to the VCH framework's integrity. All subsequent pipe sizing, coil geometry, pump selection, and thermal driver procurement decisions are evaluated against this target as a fixed external benchmark. It is *never* derived from the system's own hardware capacity.

Step 1 establishes two distinct thermal power demands that characterise the two primary operating modes of any batch thermal processing installation.

7.1 The Two Operating Modes

Batch Processing Mode represents the peak thermal demand condition. The energy requirement must overcome both the sensible thermal deficit of the incoming material and, where applicable, the latent energy barrier of a phase transition.

Balance Mode represents the steady-state minimum demand condition. The vessel is already at operating temperature and the system is simply maintaining thermal equilibrium against continuous environmental heat loss. In most industrial installations, batch processing mode will govern the design.

7.2 Batch Target — Q_{batch}

The batch thermal target quantifies the continuous thermal power required to complete the processing of a defined incoming material quantity within the specified operational window.

7.2.1 Incoming Material Condition

The framework assumes the raw material arrives at the worst-case ambient temperature. For heating applications, this is the historically coldest recorded ambient temperature.

7.2.2 Total Batch Energy Components

For materials that undergo a solid-to-liquid phase transition, the total batch energy comprises three sequential contributions:

Component 1 — Sensible Heating of the Solid Phase:

$$Q_{\text{solid}} = m \cdot C_{p,\text{solid}} \cdot (T_{\text{melt}} - T_{\text{worst ambient}}) \quad (1)$$

Component 2 — Latent Heat of Fusion:

$$Q_{\text{melt}} = m \cdot L_f \quad (2)$$

Component 3 — Sensible Heating of the Liquid Phase:

$$Q_{\text{liquid}} = m \cdot C_{p,\text{liquid}} \cdot (T_{\text{operating}} - T_{\text{melt}}) \quad (3)$$

Total Batch Energy:

$$Q_{\text{total}} = Q_{\text{solid}} + Q_{\text{melt}} + Q_{\text{liquid}} \quad (4)$$

Example: A facility processing palm oil—which melts at approximately 35 °C and arrives solid during winter months—would evaluate all three components. A facility processing a liquid petroleum derivative that arrives above its pour point year-round would evaluate Components 1 and 3 only, with T_{melt} replaced by $T_{\text{worst ambient}}$.

7.2.3 Conversion to Continuous Thermal Power

$$Q_{\text{batch}} = \frac{Q_{\text{total}}}{t_{\text{run}}} \quad (5)$$

where t_{run} is the available processing duration (seconds); the default is 7 hours = 25 200 s.

Specifying an aggressively short processing window has a compounding effect: it increases Q_{batch} , which increases Q_{design} , which drives larger coil geometry, higher pump capacity, and a more powerful thermal driver. The t_{run} specification is a capital cost lever, not merely a scheduling parameter.

7.3 Balance Target — Q_{balance}

At this stage, the vessel geometry and insulation specification have not yet been finalised. The framework applies a calibrated placeholder estimate:

$$Q_{\text{balance}} = k \cdot m_{\text{product}} \cdot C_{p,\text{liquid}} \cdot T_{\text{operating}} \quad (6)$$

Table 1. Insulation tier reference values for Q_{balance} estimation

Insulation Tier	Description	k (per hour)	k (per second)
Well Insulated	Modern mineral wool or polyurethane foam cladding, fully sealed and weather-protected	2%	5.56×10^{-6}
Lightly Insulated	Basic fibreglass wrap or partial cladding	5%	1.39×10^{-5}
Uninsulated	Bare metal vessel with no thermal cladding	12%	3.33×10^{-5}

Step 2: Thermofluid Selection and Property Characterisation

The selection of the primary thermal transport medium is one of the most consequential decisions in the design process. Every subsequent calculation—hydraulic resistance, pump sizing, heat transfer coefficient estimation, coil surface area, and thermal battery volume—is directly conditioned by the thermophysical properties of this fluid. A poor thermofluid selection cannot be corrected by oversizing the coil or upgrading the pump.

7.4 Selection Criteria

The thermofluid must satisfy the operational, chemical, regulatory, and economic constraints of the specific installation. The following criteria govern the selection decision and must each be evaluated before a fluid is confirmed.

Operating Temperature Range — The thermofluid must remain in a stable single liquid phase throughout its entire operating cycle — from cold standby conditions through to peak operating temperature. The fluid’s lower operating limit must comfortably exceed its freezing or pour point under the worst-case ambient conditions of the installation site. Its upper operating limit must remain below its boiling point at system operating pressure, and below the fluid’s rated thermal degradation threshold with an adequate margin. Proximity to either boundary introduces phase instability, vapour lock risk, and accelerated fluid degradation.

Viscosity Profile — Dynamic viscosity governs the hydraulic resistance of the circulation loop, the Reynolds number regime within the coil, and the internal convective

heat transfer coefficient. Fluids with high viscosity at operating temperature produce elevated pressure drops, unfavourable laminar flow conditions, and reduced heat transfer performance. The viscosity profile across the full operating temperature range must be reviewed, not merely the value at peak temperature. For systems that undergo cold starts from ambient conditions, the viscosity at the lowest expected startup temperature determines the worst-case hydraulic loading on the pump.

Thermal Stability — Thermofluids subjected to prolonged high-temperature operation are susceptible to oxidative degradation, thermal cracking, and polymerisation. These degradation mechanisms increase fluid viscosity over time, reduce thermal conductivity, deposit insulating fouling layers on internal pipe surfaces, and in severe cases generate corrosive breakdown products. The designer must confirm that the selected fluid's rated continuous operating temperature comfortably exceeds the system's peak operating requirement, and must establish a fluid monitoring and replacement schedule appropriate to the application.

Material Compatibility — The thermofluid must be chemically compatible with all wetted materials in the circulation loop — including the internal coil pipe alloy, pump casing and seal materials, expansion vessel lining, and any inline instrumentation wetted surfaces. Incompatibility manifests as accelerated corrosion, seal degradation, or metallic contamination of the fluid, each of which compromises system integrity and fluid thermal performance over time. The pipe material confirmed in Step 4 must be cross-checked against the thermofluid selection at that stage.

Regulatory and Safety Classification — In food processing, pharmaceutical, and potable water adjacent applications, the thermofluid must carry an appropriate food-grade or regulatory certification. In enclosed or poorly ventilated plant environments, flash point and vapour toxicity ratings must be evaluated against local safety codes. The designer must confirm that the selected fluid's safety classification is consistent with the installation environment before proceeding.

Economic Considerations — Fluid cost encompasses not only initial procurement but also the volume required to prime the full circulation loop, ongoing top-up requirements due to evaporative or leakage losses, disposal costs at end of service life, and the cost implications of the fluid monitoring programme. Higher-performance specialty fluids frequently offer a favourable lifecycle cost relative to cheaper alternatives that degrade rapidly or impose higher maintenance burdens.

7.5 Common Thermofluid Classifications

Industrial Treated Water

Applicable range: 0–100 °C (atmospheric pressure). Highest specific heat capacity of any common thermofluid. Preferred for low-to-moderate temperature applications where freeze protection is not required.

Ethylene Glycol–Water Solutions

Applicable range: –20 °C to 120 °C. The 50% volumetric mixture balances freeze protection against hydraulic performance. Standard selection for outdoor installations in temperate and cold climates.

High-Temperature Synthetic Thermal Oils

Applicable range: 50–300 °C or above. Standard for process temperatures exceeding the safe pressurised water ceiling. Higher viscosity at low temperatures imposes severe hydraulic loading during cold starts.

Food-Grade Thermal Oils

Mandatory selection for direct food contact or food-adjacent processing environments. Performance broadly comparable to synthetic thermal oils of equivalent viscosity grade.

7.6 Thermophysical Property Characterisation

The four primary properties required are: density ρ (kg/m³); specific heat capacity C_p (J/kg·K); dynamic viscosity μ (mPa·s); and thermal conductivity k (W/m·K). All are extracted from Appendix B at the bulk mean operating temperature:

$$T_{\text{bulk}} = \frac{T_{\text{supply}} - T_{\text{return}}}{2} \quad (7)$$

For properties between tabulated reference points, linear interpolation is applied:

$$P(T) = P_1 + \frac{T - T_1}{T_2 - T_1} \times (P_2 - P_1) \quad (8)$$

7.6.1 Conservative Viscosity Selection

Dynamic viscosity must be evaluated at the worst-case cold-start condition in addition to the bulk mean operating temperature. The worst-case viscosity μ_{cold} is used in the overall heat transfer coefficient calculation (Step 6). Failing to account for this in high-viscosity thermofluid classes results in a coil that is adequate at steady state but thermally insufficient during startup.

7.7 Downstream Constraints Flagged by Thermofluid Selection

The thermofluid selected in this step introduces constraints that are resolved at later stages of the framework. The designer should note the following dependencies before proceeding.

Velocity Ceiling (v_{max}) — The maximum allowable fluid velocity within the internal coil is determined by the combination of thermofluid properties and pipe material. This ceiling is evaluated and confirmed in Step 4 once the pipe schedule and material are selected. The thermofluid choice influences this ceiling through its density, viscosity, and chemical compatibility with candidate pipe materials. No numerical velocity limit is assigned at this stage.

Pipe Material Compatibility — The chemical compatibility of the selected thermofluid with candidate pipe materials must be formally verified in Step 4. The designer should carry a preliminary note of any known incompatibilities identified during the selection process in Section 2.1 into that step. Material compatibility data is compiled in Appendix C.

Pump Cavitation Risk — The thermofluid’s vapour pressure at its maximum operating temperature determines the available Net Positive Suction Head margin of the circulation pump. Fluids operating close to their vapour pressure ceiling impose stricter constraints on pump inlet conditions and system pressurisation requirements. This is evaluated during pump selection in Step 5.

Step 2 Output Summary

- Selected thermofluid and regulatory classification
- Confirmed operating temperature range
- Thermophysical properties ρ , C_p , μ , k at bulk mean operating temperature
- Worst-case cold-start viscosity μ_{cold} for conservative heat transfer sizing
- Preliminary note of pipe material compatibility constraints and cavitation risk factors

Step 3: Conceptual Vessel Geometry and Baseline Dimensional Framework

Step 3 establishes the physical spatial envelope of the primary process vessel. The vessel dimensions define the geometric canvas within which the internal coil assembly will be packaged, and also define the external surface area through which environmental heat loss occurs.

7.8 Vessel Geometry Selection

Two cap profiles are supported within the VCH framework.

7.8.1 Conical Caps

Standard selection for atmospheric or near-atmospheric pressure vessels where complete gravity drainage is operationally required. The converging geometry directs residual fluid toward the centreline drain point. Simpler and less expensive to fabricate than hemi-ellipsoidal profiles but structurally unsuited to elevated internal pressures without reinforcement rings.

7.8.2 Hemi-Ellipsoidal Caps

Standard selection for pressurised process vessels. The ellipsoidal profile distributes internal pressure stresses uniformly, making them preferred for closed-loop systems operating above atmospheric pressure and any installation governed by pressure vessel design codes.

7.9 Aspect Ratio and Dimensional Relationships

The standard aspect ratio between the cylindrical shell height h_1 and the cap height h_2 is maintained as:

$$\frac{h_2}{h_1} = \frac{1}{4} \Rightarrow h_1 = 4h_2 \quad (9)$$

The overall vessel height (h_{total}) depends on the selected cap configuration (single versus dual). Practical geometric limits are universally applied across all configurations: a minimum internal diameter of 600 mm and an overall height-to-diameter ratio strictly bounded between $0.8 \leq h_{\text{total}}/d \leq 4.0$.

7.10 Baseline Volume Calculations

The volumetric capacity (V_{tank}) is determined by the specific geometric combination of the cylindrical shell and the selected end caps. The equations below detail four standard vessel configurations. By setting $V_{\text{tank}} = V_{\text{target}}$ and substituting $h_1 = 4h_2$, the designer can analytically solve for the unknown spatial dimensions (h_2 , d , h_1 , and h_{total}).

7.10.1 Dual Conical Caps (Top and Bottom)

For a vessel featuring symmetrical conical caps at both ends, the total overall height is $h_{\text{total}} = h_1 + 2h_2 = 6h_2$, and the internal volume is:

$$V_{\text{tank}} = \frac{\pi d^2}{4} \left(h_1 + \frac{2h_2}{3} \right) \quad (10)$$

7.10.2 Single Conical Cap (Bottom Only, Flat Top)

For an atmospheric tank with a flat top and a conical bottom (standard for gravity-drained batch processes), the total overall height is $h_{\text{total}} = h_1 + h_2 = 5h_2$, and the internal volume is:

$$V_{\text{tank}} = \frac{\pi d^2}{4} \left(h_1 + \frac{h_2}{3} \right) \quad (11)$$

7.10.3 Dual Hemi-Ellipsoidal Caps (Top and Bottom)

For a pressure-rated vessel featuring hemi-ellipsoidal caps at both ends, the total overall height is $h_{\text{total}} = h_1 + 2h_2 = 6h_2$. Accounting for the standard 2 : 1 semi-elliptical volume profile, the internal volume is:

$$V_{\text{tank}} = \frac{\pi d^2}{4} h_1 + \frac{\pi d^2 h_2}{3} = \frac{\pi d^2}{4} \left(h_1 + \frac{4h_2}{3} \right) \quad (12)$$

7.10.4 Single Hemi-Ellipsoidal Cap (Bottom Only, Flat Top)

For a non-pressurised vessel with a flat top and a hemi-ellipsoidal bottom cap, the total overall height is $h_{\text{total}} = h_1 + h_2 = 5h_2$, and the internal volume is:

$$V_{\text{tank}} = \frac{\pi d^2}{4} h_1 + \frac{\pi d^2 h_2}{6} = \frac{\pi d^2}{4} \left(h_1 + \frac{2h_2}{3} \right) \quad (13)$$

7.11 Aspect Ratio Flexibility and Engineering Judgement

The standard aspect ratio $h_2/h_1 = 1/4$ is the recommended default and will produce a practical geometry for the majority of industrial applications. However, the designer retains the latitude to deviate from this ratio where site-specific constraints — such as a restricted ceiling height, a constrained floor footprint, or the need to match an existing vessel family — make the standard ratio impractical.

Any deviation from the standard aspect ratio must be documented with a clear engineering rationale, and its downstream implications must be assessed:

Reduction in h_1 Relative to d — A reduction in h_1 relative to d reduces the available cylindrical shell height for coil winding and will directly constrain the maximum achievable coil length in Step 4. If the reduction is significant, the designer should perform a preliminary check at this stage to confirm that adequate coil length remains achievable before committing to the modified geometry.

Increase in h_1 Relative to d — An increase in h_1 relative to d increases the available coil winding height but may introduce thermal stratification concerns in high-viscosity process fluids and will increase structural shell loading. In tall, narrow vessels processing viscous fluids without mechanical agitation, the assumption of uniform bulk temperature made elsewhere in this framework becomes less conservative and should be noted.

7.12 External Surface Area Calculation

The total external surface area (A_{total}) establishes the thermal boundary layer and is critical for quantifying steady-state environmental heat losses during Step 9. It is compiled by summing the main cylindrical shell area with the specific geometric end-cap selections.

The external surface area of the main cylindrical shell is invariant across all configurations:

$$A_{\text{shell}} = \pi d h_1 \quad (14)$$

For vessels utilising a single formed cap at the bottom, the surface area of the atmospheric flat top must be included to ensure heat loss calculations are rigorously bounded:

$$A_{\text{top, flat}} = \frac{\pi d^2}{4} \quad (15)$$

7.12.1 Conical Cap Surface Area

The external surface area for a single conical cap is geometrically defined as:

$$A_{\text{cap, cone}} = \frac{\pi d}{2} \sqrt{\left(\frac{d}{2}\right)^2 + h_2^2} \quad (16)$$

7.12.2 Hemi-Ellipsoidal Cap Surface Area

The external surface area for a single hemi-ellipsoidal cap (representing exactly half of an oblate spheroid) is mathematically approximated by:

$$A_{\text{cap, ellipse}} = \pi \left(\frac{d}{2}\right)^2 \left[1 + \frac{1 - e^2}{e} \operatorname{arctanh}(e)\right], \quad e = \sqrt{1 - \left(\frac{2h_2}{d}\right)^2} \quad (17)$$

7.12.3 Total External Surface Area Assembly

Depending on the specific architectural selection mapped in the previous section, the total surface area A_{total} is assembled using one of the following conditional formulas:

- **Dual Caps (Top and Bottom):**

$$A_{\text{total}} = A_{\text{shell}} + 2A_{\text{cap}} \quad (18)$$

- **Single Cap (Formed Bottom, Flat Top):**

$$A_{\text{total}} = A_{\text{shell}} + A_{\text{cap}} + A_{\text{top, flat}} \quad (19)$$

Step 3 Output Summary

- Selected cap profile and engineering rationale (Dual vs. Single configurations)
- Baseline vessel dimensions: d , h_1 , h_2 , h_{total}
- Baseline internal volume V_{tank} confirming $V_{\text{tank}} \geq V_{\text{target}}$
- Total external surface area A_{total} explicitly defined for thermal driver sizing (Step 9)
- Confirmation of practical aspect ratio bounds

Step 4: Nominal Pipe Size Selection and Maximum Coil Configuration

Step 4 formally assigns the maximum allowable fluid velocity as a hard numerical ceiling and sizes the internal thermal coil to its maximum physically achievable configuration within the vessel envelope. The maximum coil configuration is not the final coil specification—that is determined in Step 7. Step 4 establishes the physical ceiling within which that optimisation operates.

7.13 Material Selection Criteria

The pipe material must satisfy the chemical compatibility requirement established in Step 2 and must meet the thermal, mechanical, and hygienic demands of the application. The material evaluation matrix in Appendix C provides a comprehensive assessment of candidate materials across the following engineering criteria:

Thermal Conductivity — Higher conductivity reduces the conductive resistance term in the overall heat transfer coefficient, directly improving coil thermal performance.

Corrosion and Oxidation Resistance — Critical for long service life in chemically active or high-temperature environments.

Surface Finish — Smooth internal surfaces minimise hydraulic friction losses and resist fouling accumulation; smooth external surfaces maximise convective heat transfer to the process fluid.

Fabrication Characteristics — Weldability, bendability, and availability in helical coil configurations.

Regulatory Compliance — Food-grade certification where required by the application.

Mechanical Durability — Structural integrity and fatigue resistance under cyclic thermal loading.

For the majority of industrial hydronic heating applications involving organic process fluids, austenitic stainless steel to ASTM A312 TP316L is the recommended baseline material. It offers a balance of corrosion resistance, hygienic surface finish, fabrication practicality, and adequate thermal conductivity that is difficult to match at comparable lifecycle cost. Where thermal conductivity is the primary design constraint and fluid velocity can be strictly controlled, copper to ASTM B88 Type L offers substantially higher conductivity at the cost of a significantly more restrictive velocity ceiling.

7.14 Velocity Ceiling Assignment

With the pipe material confirmed, the maximum allowable fluid velocity $v_{\max}$ is formally assigned. This ceiling is determined by the erosion characteristics of the pipe material under the hydrodynamic conditions imposed by the selected thermofluid. It represents the absolute upper boundary for fluid velocity within the coil under all steady-state operating conditions. The circulation pump, Variable Frequency Drive (VFD) configuration, and all automation control logic established in subsequent steps are bounded by this ceiling without exception.

The following velocity ceilings apply within this framework:

Table 2. Velocity ceiling reference values by material-fluid pairing

Pipe Material	Thermofluid Class	$v_{\max}$ (m/s)
Stainless Steel (TP316L)	Water / Glycol Solutions	3.0
Stainless Steel (TP316L)	Thermal Oils	2.0
Carbon Steel (A106 Gr.B)	Water / Glycol Solutions	2.5
Carbon Steel (A106 Gr.B)	Thermal Oils	1.8
Copper (B88 Type L)	All thermofluid classes	1.5

7.15 Nominal Pipe Size Selection

7.15.1 Target Volumetric Flow Rate

The target volumetric flow rate Q_{flow} is the minimum continuous circulation rate required to deliver Q_{design} to the process vessel within the operational window. It is derived from the

design thermal target established in Step 1 and the thermofluid properties characterised in Step 2:

$$Q_{\text{flow}} = \frac{Q_{\text{design}}}{\rho \cdot C_p \cdot \Delta T_{\text{coil}}} \quad (20)$$

Where:

Q_{design} is the fixed design thermal target from Step 1 (W)

ρ is the thermofluid density at bulk operating temperature (kg/m^3)

C_p is the thermofluid specific heat capacity at bulk operating temperature ($\text{J}/\text{kg} \cdot \text{K}$)

ΔT_{coil} is the allowable thermofluid temperature differential across the coil inlet and outlet, capped at 10% of $(T_{\text{operating}} - T_{\text{ambient, worst}})$ as established in the framework overview (K)

This flow rate represents the hydraulic throughput the system must achieve under steady-state operating conditions to satisfy the thermal design target.

This flow rate represents the hydraulic throughput the system must achieve under steady-state operating conditions to satisfy the thermal design target.

7.15.2 Minimum Required Internal Flow Area

$$A_{\text{min}} = \frac{Q_{\text{flow}}}{v_{\text{max}}} \quad (21)$$

The smallest standard NPS schedule with $A_{\text{selected}} \geq A_{\text{min}}$ is selected:

$$v_{\text{actual}} = \frac{Q_{\text{flow}}}{A_{\text{selected}}} \leq v_{\text{max}} \quad (22)$$

7.16 Maximum Coil Geometry

7.16.1 Maximum Coil Centreline Diameter

$$d_{\text{coil}} = d_{\text{tank}} - 0.200 - d_{\text{pipe}} \quad (23)$$

7.16.2 Maximum Theoretical Coil Length

$$p_{\text{min}} = 2 \cdot d_{\text{pipe}} \quad (24)$$

$$N_{\text{max}} = \frac{h_1}{p_{\text{min}}} \quad (25)$$

$$L_{\text{turn}} = \sqrt{(\pi d_{\text{coil}})^2 + p_{\text{min}}^2} \quad (26)$$

$$L_{\text{max}} = N_{\text{max}} \cdot L_{\text{turn}} = \frac{h_1}{p_{\text{min}}} \sqrt{(\pi d_{\text{coil}})^2 + p_{\text{min}}^2} \quad (27)$$

7.17 Packing Efficiency Correction

7.17.1 Packing Efficiency Sub-Factors

The compound packing efficiency factor is the product of five independent sub-factors, each addressing a distinct physical constraint. The designer evaluates each sub-factor against the specific fabrication and operational conditions of the installation and selects the appropriate value from the ranges tabulated below.

Fabrication Quality Factor (η_f) — Accounts for geometric imperfections introduced during pipe bending and helical rolling — including material springback, wall thinning on the outer bend radius, and cross-sectional ovality. These deviations prevent perfect mathematical coil concentricity and require dimensional safety margins to prevent mechanical clashing between adjacent turns during assembly.

Fabrication Quality	η_f
Precision CNC fabrication	0.97–0.99
Industrial standard	0.92–0.96
Low precision/manual	0.85–0.92

Support Structure Factor (η_s) — Accounts for the physical volume consumed by the structural support network required to suspend the coil assembly within the vessel. Under operating conditions, the coil is subjected to its own fluid-filled dead weight, buoyant forces from the surrounding process fluid, and hydrodynamic drag from convective currents. The support skeleton required to resist these loads occupies space and shields portions of the outer pipe perimeter from convective contact with the process fluid.

Support Complexity	η_s
Minimal supports	0.97
Moderate supports	0.93
Heavy support network	0.88

Weld and Cleaning Clearance Factor (η_w) — Accounts for two distinct lifecycle requirements. During fabrication, manual or automated welding torches require a minimum spatial clearance envelope around each joint for torch access, joint execution, and post-weld non-destructive inspection. During operation, the inter-coil spacing must remain sufficient to allow penetration of cleaning solutions — whether manual or automated Clean-In-Place (CIP) spray systems — to prevent fouling accumulation in confined inter-turn spaces.

Fabrication Access	η_w
Automated TIG	0.98
Standard industrial welding	0.94
Difficult field fabrication	0.88

Thermal Expansion Factor (η_t) — Accounts for the cyclic dimensional growth of the pipe assembly during thermal operation. As the system transitions from ambient standby to operating temperature, the metal undergoes linear and volumetric thermal expansion. Without adequate spatial buffers built into the coil geometry, this expansion generates compressive stress vectors along the pipe axis, causing warping, coil misalignment, and stress corrosion cracking at weld junctions over repeated thermal cycles.

Temperature Severity	η_t
Mild ($< 50^\circ\text{C}$)	0.98
Moderate ($50\text{--}120^\circ\text{C}$)	0.94
High thermal cycling	0.88

Dead-Zone Prevention Factor (η_d) — Accounts for the fluid dynamics of the process vessel. A coil wound too densely ceases to function as a permeable heat exchange lattice and instead behaves as a solid barrier, choking the natural convective circulation currents that distribute thermal energy through the bulk process fluid. In high-viscosity fluids, this stagnation causes severe localised overheating immediately within the coil interior and leaves the outer bulk fluid underheated, producing dangerous thermal stratification. The dead-zone prevention factor enforces a minimum open flow path through the coil assembly.

Circulation Quality	η_d
Aggressive forced mixing	0.97
Natural convection assisted	0.90
High viscosity stagnant system	0.80–0.88

Table 3. Compound packing efficiency sub-factors

Factor	Description	Symbol	Range
Fabrication Quality	Geometric imperfections from bending and rolling	η_f	0.85–0.99
Support Structure	Volume consumed by structural support network	η_s	0.88–0.97
Weld & Cleaning Clearance	Torch access and CIP penetration	η_w	0.88–0.98
Thermal Expansion	Cyclic dimensional growth buffer	η_t	0.88–0.98
Dead-Zone Prevention	Minimum open flow path through coil bundle	η_d	0.80–0.97

7.17.2 Compound Packing Efficiency

$$\eta_{\text{pack}} = \eta_f \cdot \eta_s \cdot \eta_w \cdot \eta_t \cdot \eta_d \quad (28)$$

7.17.3 Maximum Practical Coil Length and Surface Area

$$L_{\text{practical}} = L_{\text{max}} \cdot \eta_{\text{pack}} \quad (29)$$

$$A_{\text{practical}} = \pi \cdot d_{\text{pipe}} \cdot L_{\text{practical}} \quad (30)$$

7.18 Packing Efficiency Limits and Constraints

Packing efficiency (η_{pack}) represents the ratio between the actual linear pipe length that can be physically integrated into the vessel versus the theoretical maximum length achievable if the coil turns were wound perfectly tight with zero clearance. It accounts for the unavoidable spatial penalties introduced by fabrication tolerances, structural support skeletons, welding torch access, thermal expansion buffers, and fluid circulation gaps.

Standard Industrial Range — The standard operational envelope spans from 0.85 to 0.98 (85%–98%). High-precision CNC fabrication with minimal structural support hovers near the upper limit (95%+), while heavy-duty, manually welded systems processing highly viscous fluids drop toward the mid-80s.

The 80% Hard Floor Rule — If the compounding of worst-case operational sub-factors drives the calculated packing efficiency below a realistic threshold of 0.80 (80%), the framework discards the calculated value and enforces a hard line floor of 0.80 (80%) for subsequent geometric design steps.

Engineering Note: A calculated efficiency below 80% is a major red flag indicating an overly constrained system. Capping the efficiency at a hard 80% prevents the mathematical models from generating absurdly spaced, un-winnable coil geometries. If a layout genuinely cannot operate safely at an 80% packing density, the designer must upsize the primary vessel dimensions rather than allowing the packing efficiency to degrade any further.

7.19 Iteration Gate — Geometric Feasibility Check

Before proceeding to pump selection, the designer must confirm that the maximum practical coil configuration is geometrically capable of supporting Q_{design} . This is a preliminary check only — the full thermal performance evaluation is conducted in Steps 6 and 7. The purpose here is to catch a fundamental geometric insufficiency early, before pump selection effort is invested in a configuration that cannot work.

A coil configuration is considered geometrically feasible if the maximum practical surface area $A_{\text{practical}}$ is sufficient to deliver Q_{design} under a conservative estimate of the overall heat transfer coefficient U . If:

$$Q_{\text{estimated}} = U_{\text{conservative}} \cdot A_{\text{practical}} \cdot \Delta T_{\text{drive}} \quad (31)$$

is substantially less than Q_{design} , the current pipe diameter is insufficient. The designer must return to Section 4.2, increment the NPS upward by one standard dimension, and repeat all calculations from that point. The larger pipe diameter increases d_{coil} , which increases L_{max} and consequently $L_{\text{practical}}$, expanding the available heat exchange surface area.

Where the geometric feasibility check passes, the designer proceeds to Step 5 with confidence that the coil envelope is capable of supporting the thermal target, subject to confirmation through the full Effectiveness-NTU analysis in Step 6.

Step 4 Output Summary

- Confirmed pipe material, velocity ceiling $v_{\max}$, selected NPS (d_{pipe} , d_{in} , t_w , A_{selected})
- Actual fluid velocity $v_{\text{actual}} \leq v_{\max}$ confirmed
- Maximum coil centreline diameter d_{coil} , minimum pitch $p_{\min}$
- Maximum theoretical coil length $L_{\max}$ and surface area $A_{\max}$
- Compound packing efficiency η_{pack} (all five sub-factors documented)
- Maximum practical coil length $L_{\text{practical}}$ and surface area $A_{\text{practical}}$
- Geometric feasibility check passed

Step 5: Hydraulic Resistance Characterisation and Pump Selection

Step 5 characterises the hydraulic behaviour of the circulation loop and selects a pump capable of driving the thermofluid at a flow rate that satisfies the thermal design target without breaching the velocity ceiling. The steady-state operating condition is defined entirely by the intersection of the system resistance curve and the pump performance curve.

7.20 System Hydraulic Resistance Characterisation

The system resistance curve describes the total pressure loss the circulation pump must overcome as a function of volumetric flow rate. It is composed of two distinct contributions: major losses arising from fluid friction along the pipe length, and minor losses arising from localised flow disturbances at fittings, bends, valves, and manifold connections. Both contributions must be evaluated and summed to construct an accurate system curve.

The system curve is evaluated across a flow rate range spanning from zero to at least 150% of the target volumetric flow rate Q_{flow} established in Step 4. Evaluating the curve across this extended range ensures that the pump-system intersection is fully characterised and that the VFD clamping range can be properly defined. A minimum of ten discrete flow rate points distributed across this range is recommended to produce a smooth, well-defined curve.

7.21 Helical Coil Friction

Major losses represent the pressure energy dissipated by fluid friction along the internal pipe walls over the full practical coil length $L_{\text{practical}}$. In a straight pipe, this is governed directly by the Darcy-Weisbach equation. In a helical coil, the centrifugal force field induced by the pipe curvature generates secondary counter-rotating flow structures — Dean vortices — that increase the effective friction factor relative to a straight pipe of identical length and diameter. This curvature effect must be accounted for explicitly.

7.21.1 Straight Pipe Friction Factor

The baseline Darcy friction factor for straight pipe flow f_{straight} is determined from the Colebrook-White equation, which accounts for both pipe wall roughness and Reynolds number effects under turbulent flow conditions. The Colebrook-White equation is implicit in f_{straight} and is solved numerically using the Newton-Raphson iterative method seeded with the Haaland approximation. The full numerical solution procedure is provided in Appendix A.

The Reynolds number required for this calculation is:

$$Re = \frac{\rho \cdot v \cdot d_{\text{in}}}{\mu} \quad (32)$$

Where:

ρ is the thermofluid density at bulk operating temperature (kg/m^3)

v is the fluid velocity at the evaluation flow rate (m/s)

d_{in} is the internal pipe diameter of the selected NPS (m)

μ is the dynamic viscosity at bulk operating temperature ($\text{Pa} \cdot \text{s}$)

The pipe absolute roughness ε for the selected pipe material is obtained from Appendix C.

7.21.2 Dean Number and Flow Regime Classification

The intensity of the secondary flow field induced by helical curvature is quantified by the dimensionless Dean number:

$$De = Re \cdot \sqrt{\frac{d_{\text{in}}}{d_{\text{coil}}}} \quad (33)$$

Where d_{coil} is the mean centreline diameter of the helical coil (m), equal to d_{coil} as established in Step 4.

The Dean number governs which of Ito's empirical helical friction factor correlations applies. The flow regime is classified as follows:

- $De < 13.5$ — Curvature effects are negligible. The straight pipe friction factor applies directly: $f_{\text{helical}} = f_{\text{straight}}$
- $13.5 \leq De \leq 2000$ — Laminar or transitional regime with significant Dean vortex activity. The laminar helical correction applies.
- $De > 2000$ — Turbulent regime. The turbulent helical correction applies.

7.21.3 Helical Friction Factor Corrections (Ito's Correlations)

For the laminar and transitional regime ($13.5 \leq De \leq 2000$):

$$f_{\text{helical}} = f_{\text{straight}} \cdot \left[\frac{21.5 \cdot De}{(1.56 + \log_{10} De)^{5.73}} \right] \quad (34)$$

For the turbulent regime ($De > 2000$), expressed using the curvature ratio:

$$f_{\text{helical}} = f_{\text{straight}} \cdot \left[Re \cdot \left(\frac{d_{\text{in}}}{d_{\text{coil}}} \right)^2 \right]^{\frac{1}{20}} \quad (35)$$

The full derivation of Ito's correlations and their applicable Reynolds number boundaries are provided in Appendix A.

7.21.4 Major Pressure Loss

The major pressure loss across the full practical coil length is:

$$\Delta P_{\text{major}} = f_{\text{helical}} \cdot \frac{L_{\text{practical}}}{d_{\text{in}}} \cdot \frac{\rho \cdot v^2}{2} \quad (36)$$

Where all variables are evaluated at the flow rate under consideration. This calculation is repeated at each discrete flow rate point across the evaluation range to construct the major loss component of the system curve.

7.22 Minor Losses — Fittings, Valves, and Manifold Connections

Minor losses represent the localised pressure drops induced by flow disturbances at discrete points in the piping network. In extended helical coil systems with multiple inline components, minor losses can contribute between 15% and 25% of total system resistance. Omitting them produces a system curve that is systematically optimistic, causing the actual duty point to occur at a higher flow rate — and therefore higher fluid velocity — than the design predicts.

Minor losses are quantified using the K -factor method, where each fitting or flow disturbance is assigned a dimensionless resistance coefficient K representing its pressure loss as a multiple of the dynamic pressure:

$$\Delta P_{\text{minor}} = K \cdot \frac{\rho \cdot v^2}{2} \quad (37)$$

The total minor pressure loss is the sum of all individual K contributions across the circulation loop evaluated at the characteristic velocity:

$$\Delta P_{\text{minor, total}} = \left(\sum K \right) \frac{\rho \cdot v^2}{2} \quad (38)$$

Reference K -factor values for common hydronic loop components are mentioned in Appendix C. If the exact fitting geometry is not yet determined at this stage of design, the conservative upper bound value should be applied or calculated at the end of design before selecting the pump.

The designer should compile an itemised schedule of all fittings and inline components in the circulation loop and sum their K contributions. Where the loop configuration is not fully detailed at this stage, a blanket minor loss allowance of 20% applied to the major loss value is an acceptable preliminary estimate, to be replaced by an itemised calculation as the detailed design progresses:

$$\Delta P_{\text{minor, allowance}} = 0.20 \cdot \Delta P_{\text{major}} \quad (39)$$

7.22.1 Total System Resistance

The total system pressure loss at any given flow rate is the sum of major and minor contributions:

$$\Delta P_{\text{system}} = \Delta P_{\text{major}} + \Delta P_{\text{minor, total}} \quad (40)$$

For pump selection and system curve construction purposes, pressure loss is commonly expressed as equivalent fluid head H_{system} in metres of fluid column:

$$H_{\text{system}} = \frac{\Delta P_{\text{system}}}{\rho \cdot g} \quad (41)$$

Where g is gravitational acceleration (9.81 m/s^2).

This calculation is performed at each discrete flow rate point across the evaluation range. The resulting dataset of (Q, H_{system}) pairs defines the system resistance curve. As fluid velocity — and therefore flow rate — increases, both major and minor losses increase approximately with the square of velocity, producing the characteristic upward-curving parabolic system resistance profile.

7.22.2 Total System Head

$$H_{\text{system}} = \frac{\Delta P_{\text{major}} + \Delta P_{\text{minor total}}}{\rho g} \quad (42)$$

7.23 VFD Frequency Scaling

When $v_{\text{duty}} > v_{\text{max}}$, the target VFD frequency to clamp velocity to exactly v_{max} :

$$Q_{\text{max},v} = v_{\text{max}} \cdot A_{\text{selected}} \quad (43)$$

$$N_{\text{ratio}} = \frac{Q_{\text{max},v}}{Q_{\text{duty}}} \quad (44)$$

$$f_{\text{VFD}} = f_{\text{nominal}} \cdot N_{\text{ratio}} \quad (45)$$

Pump affinity law verification at reduced speed:

$$H_{\text{pump,scaled}} = H_{\text{pump,nominal}} \cdot N_{\text{ratio}}^2 \geq H_{\text{system}}(Q_{\text{max},v}) \quad (46)$$

Step 5 Output Summary

- System resistance curve (Q, H_{system}) across evaluation range
- Selected pump with performance curve, NPSH data, and materials specification
- Natural duty point $(Q_{\text{duty}}, H_{\text{duty}})$ and velocity compliance status
- VFD frequency f_{VFD} and scaled duty point (where VFD intervention required)
- Confirmed steady-state operating flow rate $Q_{\text{operating}}$ and velocity $v_{\text{operating}} \leq v_{\text{max}}$

Step 6: Thermal Performance Evaluation

Step 6 establishes the maximum thermal power the coil assembly can transmit to the process fluid under both steady-state and cold-start conditions, constituting the first convergence of the hydraulic and thermal design tracks.

7.24 Overall Heat Transfer Coefficient

The overall heat transfer coefficient U is resolved from a series thermal resistance network:

$$\frac{1}{U} = \frac{1}{h_{\text{in}}} + R_{f,\text{in}} + \frac{d_{\text{out}} \ln\left(\frac{d_{\text{out}}}{d_{\text{in}}}\right)}{2k_{\text{pipe}}} + R_{f,\text{out}} + \frac{1}{h_{\text{out}}} \quad (47)$$

7.24.1 Internal Convective Coefficient

$$h_{\text{in}} = \frac{Nu_{\text{in}} \cdot k_{\text{fluid}}}{d_{\text{in}}} \quad (48)$$

Turbulent flow ($Re > 10\,000$) — **Modified Gnielinski:**

$$Nu_{\text{in}} = \frac{\frac{f_{\text{helical}}}{8} (Re - 1000) Pr}{1 + 2.17 \sqrt{\frac{f_{\text{helical}}}{8}} (Pr^{2/3} - 1)} \cdot \left[1 + \left(\frac{d_{\text{in}}}{d_{\text{coil}}} \right)^{0.9} \right] \quad (49)$$

Laminar/transitional flow ($Re \leq 10\,000$) — **Sieder–Tate:**

$$Nu_{\text{in}} = 1.86 \left(\frac{Re Pr d_{\text{in}}}{L_{\text{practical}}} \right)^{1/3} \left(\frac{\mu_{\text{bulk}}}{\mu_{\text{wall}}} \right)^{0.14} \left[1 + 3.6 \left(1 - \frac{d_{\text{in}}}{d_{\text{coil}}} \right) \left(\frac{d_{\text{in}}}{d_{\text{coil}}} \right)^{0.8} \right] \quad (50)$$

7.25 Fouling Resistance Analysis

Fouling accumulation is a dynamic phenomenon governed by the net balance between particulate deposition and fluid shear removal rates. Rather than acting as a static constant, the long-term fouling resistance of a heat transfer surface is dictated by the interaction of several primary physical and operational parameters, including fluid velocity, surface roughness, fluid purity, wall heat flux, and the frequency of operational cycling.

Sizing margins must be scaled relative to these environmental factors. Favourable conditions—characterised by high fluid velocity, smooth surface finishes, high fluid purity, low localized heat flux, and continuous operation—justify the use of minimal fouling allowances. Conversely, unfavourable conditions require an increased design fouling resistance to prevent thermal degradation over the operational service life of the coil.

7.25.1 Internal Fouling (Thermofluid Side)

The internal fouling resistance ($R_{f,\text{in}}$) is primarily influenced by the chemical stability of the heat transfer fluid. High localized surface temperatures act as catalysts for chemical coking, fluid degradation, and mineral precipitation.

Table 4. Internal fouling resistance values ($R_{f,\text{in}}$)

Thermofluid Class	$R_{f,\text{in}}$ ($\text{m}^2\cdot\text{K}/\text{W}$)
Treated utility water — inhibited closed loop	8.8×10^{-5}
Ethylene glycol-water solution — inhibited	1.76×10^{-4}
Synthetic thermal oil — clean service	3.52×10^{-4}
Synthetic thermal oil — aged or degraded	5.28×10^{-4}
Food-grade thermal oil — clean service	3.52×10^{-4}

7.25.2 External Fouling (Process Fluid Side)

External fouling resistance ($R_{f,\text{out}}$) is dictated by the chemical structure of the process fluid and the local hydrodynamics. For highly viscous or organic fluids, the primary drivers are coking—where localized skin temperatures exceed degradation thresholds—and the absence of mechanical shear.

In corporate practice, the external fouling factor is treated as a dynamic variable. Mechanical agitation (via impellers or stirrers) significantly mitigates fouling by stripping away stationary boundary layers and preventing particulate settlement. Designers are justified in applying reduced, optimised fouling allowances for agitated systems; however, under no circumstances should fouling resistances be omitted from the overall coefficient calculation under the assumption of regular maintenance, as initial degradation typically occurs within early operational cycles.

Table 5. External fouling resistance values ($R_{f,\text{out}}$)

Process Fluid Class	$R_{f,\text{out}}$ ($\text{m}^2\cdot\text{K}/\text{W}$)
Light refined vegetable oils	3.52×10^{-4}
Heavy crude vegetable oils — palm, tallow	5.28×10^{-4}
Petroleum distillates — clean service	1.76×10^{-4}
Heavy hydrocarbons — high fouling potential	7.04×10^{-4}
Aqueous process streams — clean	8.8×10^{-5}
Polymer melts and wax products	5.28×10^{-4}

Where the specific process fluid does not appear in these reference tables, the baseline fouling factor should be cross-referenced with conservative empirical data from recognized engineering standards such as *Perry's Chemical Engineers' Handbook*.

7.26 External Convective Heat Transfer

The external convective coefficient h_{out} governs energy transfer from the outer pipe wall to the bulk process fluid. Its evaluation depends on whether the vessel operates under natural convection or mechanical agitation, and — for phase-changing materials — on the thermophysical state of the process fluid at any given instant.

For fluids that transition through a phase change (e.g., solid palm oil slurry melting to liquid), the Prandtl number varies by orders of magnitude across the operating cycle. Because Nusselt correlations depend non-linearly on Pr (typically through $Pr^{1/3}$), a single

bulk property value introduces significant error. The heat transfer lifecycle is therefore evaluated across distinct phase-dependent regimes, and the resulting coefficients combined into an energy-weighted overall coefficient U_{weighted} (Section 7.27).

7.26.1 Natural Convection — Churchill–Chu Correlation

For submerged coils in unagitated vessels, heat transfer from the outer pipe surface is governed by natural convection. Treating each coil turn as a horizontal cylinder, the external Nusselt number is given by the Churchill–Chu correlation:

$$Nu_{\text{out}} = \left[0.6 + \frac{0.387 \cdot Ra^{1/6}}{\left[1 + \left(\frac{0.559}{Pr} \right)^{9/16} \right]^{8/27}} \right]^2 \quad (51)$$

where the Rayleigh number is:

$$Ra = \frac{g \cdot \beta \cdot \Delta T_{\text{drive}} \cdot d_{\text{out}}^3}{\nu \cdot \alpha} \quad (52)$$

with $g = 9.81 \text{ m/s}^2$, β the thermal expansion coefficient of the process fluid (K^{-1}), ΔT_{drive} the temperature difference between the outer wall and the bulk fluid (K), ν the kinematic viscosity (m^2/s), and α the thermal diffusivity (m^2/s) of the process fluid. The derivation of this correlation and the Rayleigh number formulation are provided in Appendix A.

The external heat transfer coefficient follows as:

$$h_{\text{out}} = \frac{Nu_{\text{out}} \cdot k_{\text{process}}}{d_{\text{out}}} \quad (53)$$

The natural convection estimate is retained as the conservative baseline wherever forced mixing cannot be guaranteed throughout the full batch cycle.

7.26.2 Forced Convection — Mechanically Agitated Vessels

Mechanical agitation physically disrupts the thermal boundary layer, rendering buoyancy-driven natural convection negligible. In this regime, the external Nusselt number is governed by the impeller-based correlation:

$$Nu_{\text{out}} = C \cdot Re_{\text{ia}}^a \cdot Pr^{1/3} \cdot \left(\frac{\mu}{\mu_{\text{w}}} \right)^{0.14} \quad (54)$$

where the impeller Reynolds number is:

$$Re_{\text{ia}} = \frac{\rho \cdot N \cdot D^2}{\mu} \quad (55)$$

with ρ the process fluid density (kg/m^3), N the impeller rotational speed (rps), D the impeller diameter (m), and μ the dynamic viscosity at bulk temperature ($\text{Pa}\cdot\text{s}$). The

viscosity ratio $(\mu/\mu_w)^{0.14}$ corrects for the temperature gradient between the bulk fluid and the outer pipe wall.

The dimensionless constants C and a are selected based on agitator geometry and flow regime:

Impeller Type	/	Agitator	Flow Regime		C	a
Flat-Blade (Pitched/Radial)		Turbine	Turbulent ($Re_{ia} > 10,000$)		0.60	0.67
Standard Paddle		Agitator	Transitional / Medium Viscosity		0.54	0.67
Anchor (Swept-wall)		Agitator	Laminar / High Viscosity / Slurry		0.40	0.50
Helical Ribbon		/ Screw	Extreme Viscosity (Non-Newtonian)		0.35	0.50

For solidifying or melting slurries such as palm oil, anchor or paddle agitators are standard practice. The conservative laminar exponent ($a = 0.50$) is recommended in these regimes. Once Nu_{out} is determined, h_{out} and the total external convective resistance are computed as:

$$h_{out} = \frac{Nu_{out} \cdot k_{process}}{d_{out}}, \quad R_{conv,out} = \frac{1}{h_{out} \cdot A_{out}} \quad (56)$$

7.27 Energy-Weighted Overall Coefficient for Phase-Change Systems

For process fluids undergoing a phase transition, h_{out} — and consequently U — differs substantially between the solid and liquid states. The framework computes U independently for each phase using the phase-specific external convection coefficients, then combines them into a single energy-weighted coefficient:

$$U_{\text{weighted}} = \varphi_{\text{solid}} \cdot U_{\text{solid}} + \varphi_{\text{liquid}} \cdot U_{\text{liquid}} \quad (57)$$

where φ_{solid} and φ_{liquid} are the energy allocation fractions satisfying $\varphi_{\text{solid}} + \varphi_{\text{liquid}} = 1$.

7.28 Thermal Performance via the Effectiveness–NTU Method

Thermal performance is evaluated using the Effectiveness–NTU (ε -NTU) method. This formulation is preferred over the direct LMTD approach because it does not require prior knowledge of the thermofluid outlet temperature, which is an output of the heat exchange analysis rather than an independently specified boundary condition.

7.28.1 Governing Assumptions

The heating fluid circulating through the coil has a finite thermal capacity rate $C_h = \dot{m}_h c_{p,h}$. The process fluid batch, owing to its comparatively large thermal mass and the short residence time of the coil fluid, behaves as an isothermal reservoir at any given instant. This is mathematically equivalent to a heat exchanger configuration in which the capacity-rate ratio approaches zero:

$$C_r = \frac{C_{\min}}{C_{\max}} \rightarrow 0 \quad (58)$$

analogous to an evaporator or condenser in conventional heat exchanger analysis. The same condition applies during the latent melting zone, where the process fluid temperature is fixed at T_m and its effective C_p is mathematically infinite. Under this limiting condition, the effectiveness relation reduces to:

$$\varepsilon = 1 - e^{-NTU} \quad (59)$$

with C_h taken as $C_{\min}$ throughout all zones.

7.28.2 Thermal Capacity Rate and NTU

The thermal capacity rate of the circulating thermofluid is:

$$C_{\text{coil}} = \dot{m} \cdot C_p = (\rho \cdot Q_{\text{operating}}) C_p \quad (60)$$

Because $C_{\text{tank}} \rightarrow \infty$, $C_{\min} = C_{\text{coil}}$ throughout, and the maximum thermodynamically achievable heat transfer rate is:

$$Q_{\text{fluid}} = C_{\text{coil}} \cdot (T_{\text{supply}} - T_{\text{tank}}) \quad (61)$$

The Number of Transfer Units for each thermal zone, characterised by its own overall heat transfer coefficient (U_1 for the solid phase, U_2 for the liquid phase), is:

$$NTU_1 = \frac{U_1 A}{C_h}, \quad NTU_2 = \frac{U_2 A}{C_h} \quad (62)$$

with corresponding zone effectivenesses:

$$\varepsilon_1 = 1 - e^{-NTU_1}, \quad \varepsilon_2 = 1 - e^{-NTU_2} \quad (63)$$

Since U , A , and C_h are constant within a given zone, ε_1 and ε_2 remain invariant throughout their respective stages irrespective of the evolving bulk temperature. The full derivation of the ε -NTU method and the $C_r = 0$ simplification are provided in Appendix A.

7.28.3 Zone-wise Heat Transfer and Processing Time

The complete batch cycle is partitioned into three sequential thermal zones.

Zone 1 — Sensible Heating of the Solid Phase ($T_i \rightarrow T_m$) An energy balance on the processing material yields:

$$M c_{p,\text{solid}} \frac{dT_b}{dt} = \varepsilon_1 C_h (T_{h,\text{in}} - T_b) \quad (64)$$

which integrates to give the time required to raise the bulk temperature to the melting point:

$$t_1 = \frac{M c_{p,\text{solid}}}{\varepsilon_1 C_h} \ln \left(\frac{T_{h,\text{in}} - T_i}{T_{h,\text{in}} - T_m} \right) \quad (65)$$

Zone 2 — Latent Phase Change ($T = T_m$, constant) The bulk temperature remains fixed at T_m , so the temperature driving force and the heat transfer rate are both constant throughout the melting stage:

$$q_2 = \varepsilon_2 C_h (T_{h,\text{in}} - T_m) \quad (66)$$

The melting duration follows directly from the latent energy requirement:

$$t_2 = \frac{M L_f}{q_2} \quad (67)$$

Zone 3 — Sensible Heating of the Liquid Phase ($T_m \rightarrow T_f$) The governing structure is analogous to Zone 1, with liquid-phase properties substituted:

$$t_3 = \frac{M c_{p,\text{liquid}}}{\varepsilon_2 C_h} \ln \left(\frac{T_{h,\text{in}} - T_m}{T_{h,\text{in}} - T_f} \right) \quad (68)$$

Total Process Duration

$$t_{\text{total}} = t_1 + t_2 + t_3 \quad (69)$$

7.28.4 Total Energy Requirement

The total thermal energy required to convert the processing material from the initial solid state at T_i to the final liquid state at T_f is determined solely by the thermodynamic state of the material:

$$Q_{\text{total}} = M c_{p,\text{solid}} (T_m - T_i) + M L_f + M c_{p,\text{liquid}} (T_f - T_m) \quad (70)$$

The role of the ε -NTU formulation is therefore not to determine Q_{total} , which is fixed by the thermodynamic state, but to establish the instantaneous heat transfer rate and the time required to deliver this energy in each zone, given the coil geometry, thermofluid flow rate, and zonal overall heat transfer coefficients.

7.28.5 Maximum Transmissible Thermal Power

The maximum thermal power transmissible by the coil assembly at the confirmed operating condition is:

$$Q_{\text{max}} = Q_{\text{fluid}} \cdot [1 - \exp(-NTU)] = \dot{m} C_p (T_{\text{supply}} - T_{\text{tank}}) \cdot \left[1 - \exp\left(-\frac{U A_{\text{practical}}}{\dot{m} C_p}\right) \right] \quad (71)$$

This expression is evaluated twice: using U_{steady} to establish the nominal steady-state maximum, and using U_{cold} to establish the worst-case cold-start maximum.

7.29 Generator Sizing Ceiling

The external thermal utility must be sized to satisfy the peak instantaneous demand rather than a time-averaged load. Two critical phases bound the peak:

Cold-Start Onset ($t = 0$): The thermal driving force is at its maximum ($T_{h,\text{in}} - T_i$), with the overall coefficient depressed to its worst-case solid-state value U_{cold} :

$$q_{\text{start}} = \varepsilon_{\text{cold}} C_h (T_{h,\text{in}} - T_i) \quad (72)$$

Latent Melting Stage: The driving force narrows to ($T_{h,\text{in}} - T_m$), but U rises to its steady-state maximum U_{steady} as the material liquefies and natural convection or agitation develops:

$$q_2 = \varepsilon_{\text{steady}} C_h (T_{h,\text{in}} - T_m) \quad (73)$$

The required generator thermal output capacity is:

$$Q_{\text{generator}} = \max(q_{\text{start}}, q_2) \cdot F_s \quad (74)$$

where F_s is an engineering factor of safety, typically 1.15–1.20.

7.30 Thermal Adequacy Verification

The maximum transmissible thermal power under both operating conditions is compared against the design thermal target Q_{design} established in Step 1.

Condition A — Thermally Adequate: $Q_{\text{max}} \geq Q_{\text{design}}$ under both steady-state and cold-start conditions. The coil configuration is confirmed adequate. Proceed to Step 7.

Condition B — Cold-Start Constraint: $Q_{\text{max}} \geq Q_{\text{design}}$ under steady-state conditions but not at cold-start. The startup condition is the binding constraint. Mitigation options include elevating the thermofluid supply temperature during startup, specifying a pre-heating arrangement, or increasing coil surface area. Where no adjustment is practicable, the limitation is formally documented and the startup procedure defined accordingly.

Condition C — Thermally Inadequate: $Q_{\text{max}} < Q_{\text{design}}$ under both conditions. The coil configuration is fundamentally insufficient. Return to Step 4, increment the supply NPS by one standard dimension, and repeat Steps 4 through 6. The increased pipe diameter raises d_{coil} , $L_{\text{practical}}$, and A_s , each of which improves Q_{max} .

Step 6 Output Summary

- Steady-state coefficient U_{steady} and cold-start coefficient U_{cold} , with all constituent resistances itemised
- Energy-weighted coefficient U_{weighted} for phase-change systems
- Thermal capacity rate C_{coil} and maximum potential heat transfer Q_{fluid}
- Zone-wise NTU values (NTU_1 , NTU_2) and effectivenesses (ε_1 , ε_2)
- Zone-wise processing times (t_1 , t_2 , t_3) and total batch duration t_{total}
- Maximum transmissible power Q_{max} under steady-state and cold-start conditions
- Generator sizing ceiling $Q_{\text{generator}}$
- Confirmation that $Q_{\text{max}} \geq Q_{\text{design}}$ under both evaluation conditions

Step 7: Coil Length Determination

Step 7 determines the minimum coil length required to satisfy the design thermal target Q_{design} established in Step 1, given the pipe diameter selected in Step 4, the overall heat transfer coefficient U from Step 6, and the thermofluid capacity rate $C_{\min}$. Rather than iterating through intermediate effectiveness and NTU values, the required length is resolved directly from a closed-form inversion of the ε -NTU relationship.

7.31 Closed-Form Inversion of the ε -NTU Relationship

The transmissible thermal power from Step 6 is:

$$Q_{\max} = \varepsilon \cdot C_{\min} \cdot \Delta T = (1 - e^{-NTU}) C_{\min} \cdot \Delta T = \left(1 - e^{-\frac{UA}{C_{\min}}}\right) C_{\min} \cdot \Delta T \quad (75)$$

Setting $Q_{\max} = Q_{\text{design}}$ and solving for A :

$$e^{-\frac{UA}{C_{\min}}} = 1 - \frac{Q_{\text{design}}}{C_{\min} \Delta T} = \frac{C_{\min} \Delta T - Q_{\text{design}}}{C_{\min} \Delta T} \quad (76)$$

$$A_{\text{required}} = \ln\left(\frac{C_{\min} \Delta T}{C_{\min} \Delta T - Q_{\text{design}}}\right) \cdot \frac{C_{\min}}{U} \quad (77)$$

Substituting the geometric identity $A = \pi d_{\text{pipe}} L$:

$$L_{\text{required}} = \ln\left(\frac{C_{\min} \Delta T}{C_{\min} \Delta T - Q_{\text{design}}}\right) \cdot \frac{C_{\min}}{U \cdot \pi d_{\text{pipe}}} \quad (78)$$

where:

Q_{design} is the safety-margined design heat duty from Step 1 (W)

$C_{\min} = \dot{m} C_p$ is the thermofluid thermal capacity rate (W/K)

$\Delta T = T_{\text{supply}} - T_{\text{tank}}$ is the worst-case driving temperature difference, evaluated at cold-start and fouled conditions (K)

U is the overall heat transfer coefficient at the governing condition (W/m²·K)

d_{pipe} is the pipe outer diameter for the selected NPS (m)

Equation (78) is exact for all NTU regimes and requires no iterative solution. Two physical relationships are immediately evident from its structure: L_{required} scales inversely with d_{pipe} , confirming that a larger bore pipe directly reduces the coil length required for the same thermal duty; and the logarithmic argument is always greater than unity provided $Q_{\text{design}} < C_{\min} \Delta T$, which must hold by construction since Q_{design} cannot exceed the thermodynamic ceiling.

7.32 Governing Condition for ΔT and U

Equation (78) is evaluated under the most conservative combination of operating conditions: the cold-start driving temperature difference (maximum ΔT , minimum bulk fluid temperature) paired with the cold-start overall heat transfer coefficient U_{cold} (solid-phase external convection, fully fouled surfaces). This pairing produces the largest L_{required} and ensures the optimised coil satisfies Q_{design} under degraded conditions, not merely under ideal steady-state operation.

For phase-change systems, U_{weighted} from Section 7.27 is substituted in place of U_{cold} where the energy-weighted coefficient is the more conservative value.

7.33 Geometric Feasibility Check

The resolved length must satisfy the physical constraint established in Step 4:

$$L_{\text{required}} \leq L_{\text{practical}} \quad (79)$$

If this condition is violated, the vessel geometry cannot accommodate the required heat transfer area at the selected NPS. The design returns to Step 4, the supply NPS is incremented by one standard dimension, and Steps 4 through 7 are repeated. As d_{pipe} increases, L_{required} decreases (via the $1/d_{\text{pipe}}$ dependence in Equation (78)) while $L_{\text{practical}}$ simultaneously increases due to the expanded vessel clearances — both effects act to satisfy the feasibility condition.

7.34 Optimised Helical Geometry

With L_{required} confirmed feasible, the buildable helical geometry is recovered by back-calculating the coil pitch directly from the helical arc-length relation. The standard arc-length equation can be rearranged to solve for the coil pitch (p_{coil}) algebraically as an explicit function:

$$p_{\text{coil}} = \frac{\pi \cdot d_{\text{coil}}}{\sqrt{\left(\frac{L_{\text{required}}}{h_1}\right)^2 - 1}} \quad (80)$$

The resulting pitch must satisfy the established fabrication bounds:

$$p_{\text{min}} = 2 d_{\text{pipe}} \leq p_{\text{coil}} \leq \frac{h_1}{3} \quad (81)$$

The lower bound prevents coil turn contact and ensures adequate external fluid flow access to the pipe surfaces; the upper bound prevents an excessively stretched pitch that would reduce the effective winding height utilisation below one-third of the available vessel height h_1 .

Dynamic Geometric Adjustment Because the pitch equation is explicitly defined, the coil dimensions can be evaluated dynamically. If the calculated p_{coil} violates the fabrication bounds, the designer can hold L_{required} constant and actively adjust either the

coil diameter (d_{coil}) or the allowable winding height (h_1) independently. Modifying one of these spatial parameters while fixing the other pulls the resulting pitch back into physical manufacturing compliance without altering the thermal capacity of the coil.

The final physical turn count and actual winding height follow as:

$$N_{\text{coil}} = \frac{h_1}{p_{\text{coil}}}, \quad h_{\text{coil}} = N_{\text{coil}} \cdot p_{\text{coil}}, \quad h_{\text{clearance}} = h_1 - h_{\text{coil}} \geq 50 \text{ mm} \quad (82)$$

7.35 Hydraulic Review at Optimised Length

Reducing the coil from $L_{\text{practical}}$ to L_{required} decreases the total frictional resistance of the circuit. The system curve is therefore recomputed at L_{required} and compared against the pump duty point confirmed in Step 5. If the reduction in hydraulic resistance is sufficient to shift the operating point materially — altering the flow rate, velocity, or Reynolds number beyond the bounds assumed in Step 6 — the pump selection is reviewed and a reduced specification considered. Velocity compliance at the updated flow condition is confirmed before the design is advanced.

Step 7 Output Summary

- Required coil length L_{required} from the closed-form inversion of the ε -NTU relationship
- Feasibility confirmation: $L_{\text{required}} \leq L_{\text{practical}}$
- Optimised helical pitch p_{coil} within $[p_{\text{min}}, h_1/3]$
- Final turn count N_{coil} , winding height h_{coil} , end clearance $h_{\text{clearance}} \geq 50 \text{ mm}$
- Confirmed buildable coil specification: $L_{\text{required}}, d_{\text{coil}}, p_{\text{coil}}, N_{\text{coil}}, h_{\text{coil}}$
- Updated system curve at L_{required} , velocity compliance confirmed, pump review outcome

Step 8: Volumetric Displacement Compensation and Final Vessel Dimensioning

When a helical coil assembly is submerged inside a process vessel, it physically occupies volume that would otherwise be available to the process fluid. A vessel fabricated to its baseline internal dimensions will therefore deliver less usable process capacity than the client specified. Step 8 quantifies and corrects this displacement.

7.36 Coil Assembly Displacement Volume

$$V_{\text{coil}} = \frac{\pi d_{\text{pipe}}^2}{4} \cdot L_{\text{coil}} \quad (83)$$

$$V_{\text{ancillary}} = k_{\text{anc}} \cdot V_{\text{coil}} \quad (84)$$

$$V_{\text{displacement}} = V_{\text{coil}} \cdot (1 + k_{\text{anc}}) \quad (85)$$

Table 6. Ancillary displacement factor k_{anc}

Support Complexity	k_{anc}
Minimal support — lightweight coil, low process fluid density	0.08
Moderate support — standard industrial configuration	0.12
Heavy support network — large coil, high-density process fluid	0.18

7.37 Dimensional Adjustment

$$V_{\text{gross}} = V_{\text{target}} + V_{\text{displacement}} \quad (86)$$

Option A — Vertical Shell Extension (Preferred):

$$\Delta h_1 = \frac{V_{\text{displacement}}}{\pi d_{\text{tank}}^2 / 4} \quad (87)$$

$$h_{1,\text{final}} = h_1 + \Delta h_1 \quad (88)$$

Option B — Radial Diameter Expansion: Internal diameter $d_{\text{tank,final}}$ is solved from the volume formula with V_{gross} substituted for V_{target} using the Step 3 procedure.

Step 8 Output Summary

- Primary coil pipe displacement V_{coil} (using *external* pipe diameter)
- Ancillary allowance $V_{\text{ancillary}}$ and total displacement $V_{\text{displacement}}$
- Gross fabrication volume V_{gross}
- Selected adjustment option and final vessel dimensions: d_{tank} , $h_{1,\text{final}}$, h_2 , h_{final}
- Updated external surface area $A_{\text{total,final}}$
- Q_{loss} feedback check satisfied (deviation $< 10\%$)

Step 9: Thermal Driver Sizing

Step 9 retires the Q_{loss} placeholder estimates and replaces them with a rigorous series thermal resistance network calculation using the finalised vessel geometry. It determines the total thermal power capacity required of the primary heating or cooling source.

7.38 Insulation Specification**Table 7.** Common industrial insulation materials

Insulation Material	k_{ins} (W/m·K)	Typical Max. Temp.
Mineral wool — glass fibre	0.033–0.045	250 °C
Mineral wool — rock wool	0.035–0.050	400 °C
Polyurethane foam — rigid	0.022–0.028	120 °C
Cellular glass	0.038–0.055	—
Calcium silicate	0.055–0.075	650 °C
Aerogel blanket	0.012–0.018	—

7.39 Rigorous Thermal Resistance Network

$$R_{\text{total}} = R_{\text{internal}} + R_{\text{vessel wall}} + R_{\text{insulation}} + R_{\text{external}} \quad (89)$$

$$R_{\text{internal}} = \frac{1}{h_{\text{int}} \cdot A_{\text{total,final}}} \quad (90)$$

$$R_{\text{vessel wall}} = \frac{t_{\text{wall}}}{k_{\text{wall}} \cdot A_{\text{total,final}}} \quad (91)$$

$$R_{\text{insulation}} = \frac{L_{\text{ins}}}{k_{\text{ins}} \cdot A_{\text{total,final}}} \quad (92)$$

$$R_{\text{external}} = \frac{1}{h_{\text{ext}} \cdot A_{\text{total,final}}} \quad (93)$$

Steady-state heat loss:

$$Q_{\text{loss}} = \frac{T_{\text{operating}} - T_{\text{worst ambient}}}{R_{\text{total}}} \quad (94)$$

7.40 Total Thermal Driver Capacity

For n vessels operating simultaneously, the thermal driver must be sized to accommodate a strictly defined operating envelope. This range is bounded by the absolute peak demand and the standby minimum, with an optimal nominal operating point situated in between to ensure plant longevity.

The total system capacity envelope is defined by the following driver operating range:

$$Q_{\text{driver,max}} = \sum_{i=1}^n \left(Q_{\text{design}}^{(i)} + Q_{\text{loss}}^{(i)} \right) \quad (95)$$

$$Q_{\text{driver,opt}} = \sum_{i=1}^n \left(1.5 \cdot Q_{\text{batch}}^{(i)} + Q_{\text{loss}}^{(i)} \right) \quad (96)$$

$$Q_{\text{driver,min}} = \sum_{i=1}^n Q_{\text{loss}}^{(i)} \quad (97)$$

Operating Envelope Definitions:

- **Peak Load Limit** ($Q_{\text{driver,max}}$): The absolute geometric maximum thermal power the system can physically draw during worst-case, simultaneous cold-start conditions across all vessels.
- **Optimal Nominal Load** ($Q_{\text{driver,opt}}$): Sized at 50% above the raw process heat requirement (Q_{batch}) plus steady-state losses. This represents the optimal best-case thermal capacity required to confidently maintain production throughput while allowing the central thermal plant to operate at a roughly 50% duty cycle. This deliberate overcapacity extends the lifespan of the firing/heating hardware and provides immediate operational reserve for transient load spikes.
- **Standby Floor** ($Q_{\text{driver,min}}$): The baseline continuous thermal output required purely to offset transmission and environmental heat losses when the vessels are holding at temperature.

Step 9 Output Summary

- Confirmed insulation specification: material, k_{ins} , thickness L_{ins}
- Complete series thermal resistance network with all four components
- Rigorously calculated Q_{loss} for each vessel
- Q_{loss} feedback loop formally resolved (deviation $< 15\%$ or documented return to Step 1)
- Total thermal driver capacity envelope ($Q_{\text{driver,max}}$, $Q_{\text{driver,opt}}$, and $Q_{\text{driver,min}}$)
- Optimal capacity ($Q_{\text{driver,opt}}$) selected for 50% duty cycle plant operation
- Selected commercial thermal driver specification: type, rated capacity, modulation range

Step 10: Thermal Battery Sizing

Step 10 establishes the volumetric capacity and geometric parameters of the thermal battery—the utility fluid reservoir that stores conditioned thermofluid and supplies it continuously to the circulation loop. The tank volume is selected within a rigorously defined operating envelope, strictly bounded by minimum hardware constraints and maximum operational limits.

7.41 Hydronic Priming Volume

The baseline hydronic volume represents the absolute minimum fluid inventory required to physically fill the entire circulation loop. It is the sum of the internal coil volume, the distribution piping network, and all ancillary components:

$$V_{\text{hydronic}} = V_{\text{coil,internal}} + V_{\text{distribution}} + V_{\text{ancillary}} \quad (98)$$

The internal volume of the primary heat exchange coil is calculated using its internal diameter (d_{in}) and total physical length (L_{coil}):

$$V_{\text{coil,internal}} = \frac{\pi d_{\text{in}}^2}{4} \cdot L_{\text{coil}} \quad (99)$$

7.42 Driver-Specific Minimum Volume Constraints

To ensure stable operation and prevent premature equipment degradation, the thermal battery must provide sufficient thermal inertia. The required minimum buffer volume varies depending on the fundamental mechanics of the selected thermal generator.

7.42.1 Electric Immersion Heater Arrays

For electric immersion systems, the volume must provide adequate physical clearance for the heating elements ($V_{\text{clearance}}$) and a thermal flux buffer ($V_{\text{flux buffer}}$). The flux buffer absorbs residual thermal energy in the event of a sudden circulation failure, preventing localized fluid boiling or thermal fluid degradation (cracking) based on the maximum allowable temperature rise rate $(dT/dt)_{\text{limit}}$:

$$V_{\text{electric}} = V_{\text{clearance}} + V_{\text{flux buffer}}, \quad V_{\text{flux buffer}} = \frac{Q_{\text{element}}}{\rho C_p \left(\frac{dT}{dt}\right)_{\text{limit}}} \quad (100)$$

7.42.2 Fired Thermal Fluid Heaters

Fired heaters utilizing natural gas or diesel burners cannot modulate down to zero and require a minimum burner runtime ($t_{\text{min runtime}}$) to ensure complete combustion and prevent damaging short-cycling. The thermal battery must be large enough to absorb the minimum burner output ($Q_{\text{min burn}}$) across the allowable control deadband ($\Delta T_{\text{deadband}}$):

$$V_{\text{fired}} = \frac{Q_{\text{min burn}} \cdot t_{\text{min runtime}}}{\rho C_p \Delta T_{\text{deadband}}} \quad (101)$$

7.42.3 Vapour Compression Chillers

For chilled systems, vapour compression compressors incorporate anti-recycle timers ($\tau_{\text{anti-recycle}}$) to protect internal motor windings from overheating due to excessive start-stop cycles. The tank volume must sustain the process load during this mandatory off-cycle:

$$V_{\text{chiller}} = \frac{Q_{\text{min chiller}} \cdot \tau_{\text{anti-recycle}}}{\rho C_p \Delta T_{\text{deadband}}} \quad (102)$$

7.42.4 Overall Minimum Volume Floor

The definitive minimum volume constraint (V_{min}) for the system is the mathematical upper bound of the requisite hydronic priming volume and the specific hardware protection limit:

$$V_{\text{min}} = \max(V_{\text{hydronic}}, V_{\text{driver specific}}) \quad (103)$$

7.43 Maximum Volume Constraint

While large thermal batteries provide excellent stability, an excessively oversized tank increases spatial footprint, capital cost, and initial startup times. The maximum practical volume (V_{max}) is constrained by the peak thermal power (Q_{peak}) and the usable temperature differential (ΔT_{allow}):

$$V_{\text{max}} = \frac{Q_{\text{peak}}}{\rho C_p \Delta T_{\text{allow}} \cdot \eta_{\text{usable}}} \quad (104)$$

where $\Delta T_{\text{allow}} = T_{\text{operating}} - T_{\text{min allowed}}$ and the usable volume fraction is typically defined as $\eta_{\text{usable}} = 0.80$ to account for thermal stratification and incomplete mixing.

7.44 System Stability Assessment

The dynamic responsiveness of the thermal loop is characterized by its thermal time constant (S_t). This metric quantifies the thermal inertia of the battery relative to the peak process load, dictating the required sophistication of the control hardware:

$$S_t = \frac{\rho \cdot V_{\text{battery}} \cdot C_p}{Q_{\text{peak}}} \quad (105)$$

Table 8. Thermal time constant reference bands

S_t Range	System Behaviour	Control Hardware Implication
$S_t > 1800 \text{ s}$	High inertia — very stable	Standard PID, conventional VFD/SSR
$600 < S_t \leq 1800 \text{ s}$	Moderate inertia — stable	Careful PID tuning, responsive VFD ramp rates
$180 < S_t \leq 600 \text{ s}$	Low inertia — reactive	Fast-acting SCR controllers mandatory
$S_t \leq 180 \text{ s}$	Very low inertia — volatile	Advanced model-predictive control required

7.45 Volume Selection

The final specified thermal battery volume (V_{battery}) must fall strictly within the established boundary constraints:

$$V_{\min} \leq V_{\text{battery}} \leq V_{\max} \quad (106)$$

The specific point selected within this envelope is governed by a balance of available capital expenditure, plant floor footprint limitations, and the dynamic capability of the specified control hardware.

7.46 Thermal Battery Tank Geometry

Once V_{battery} is finalized, the physical tank geometry is derived utilizing the same compound vessel methodology established in Step 3. Applying the standard aspect ratio for vertical cylindrical vessels ($h_{1,\text{bat}} = 4h_{2,\text{bat}}$), the total volume formulation is set equal to V_{battery} to mathematically solve for the conical base height ($h_{2,\text{bat}}$), vessel diameter (d_{bat}), cylindrical height ($h_{1,\text{bat}}$), and total overall height ($h_{\text{total,bat}}$).

The tank design is subject to a minimum internal diameter of 400 mm to allow for adequate sensor placement and maintenance access, with an overall height-to-diameter ratio strictly maintained between:

$$0.8 \leq \frac{h_{\text{total,bat}}}{d_{\text{bat}}} \leq 4.0 \quad (107)$$

Step 10 Output Summary

- Hydronic priming volume V_{hydronic} (all three components individually quantified)
- Driver-specific minimum volume constraint evaluated
- Overall minimum V_{min} and maximum V_{max} with documented ΔT_{allow} and η_{usable}
- Thermal time constant S_t at V_{min} and V_{max} , with control hardware implications
- Selected V_{battery} with documented engineering basis
- Buildable thermal battery tank geometry: d_{bat} , $h_{1,\text{bat}}$, $h_{2,\text{bat}}$, $h_{\text{total,bat}}$

Step 11 : Design Verification and Release

Step 11 is the formal verification gate of the VCH Sizing Framework. It consolidates every governing constraint from Steps 1–10 into a structured audit across four domains, confirming that the complete design is internally consistent, physically realisable, and thermally adequate before release for detailed structural, electrical, and automation engineering. The design is released **only when every check is documented as passed**.

7.47 Thermal Verification Checks

Ref.	Acceptance Criterion	Corrective Action if Failed
T1	Q_{design} identical across all steps	Restore Step 1 value; re-run affected steps
T2	$Q_{\text{total}} = Q_{\text{solid}} + Q_{\text{melt}} + Q_{\text{liquid}}$	Return to Step 1; include all energy components
T3	$T_{\text{supply}} - T_{\text{operating}} \geq 10\text{ }^{\circ}\text{C}$	Increase T_{supply} or select higher-ceiling thermofluid
T4	$\Delta T_{\text{coil}} \leq 0.10(T_{\text{op}} - T_{\text{wa}})$	Return to Step 4; recalculate Q_{flow}
T5	$Q_{\text{max,steady}} \geq Q_{\text{design}}$	Return to Step 6; consider NPS increment
T6	$Q_{\text{max,cold}} \geq Q_{\text{design}}$	Return to Step 6; confirm U_{cold} then NPS increment
T7	$0 < \varepsilon_{\text{target}} < 0.9$	Increase T_{supply} or improve thermofluid
T8	$ Q_{\text{loss,rig}} - Q_{\text{loss,ph}} / Q_{\text{loss,ph}} \leq 15\%$	Return to Step 9; if exceeded, return to Step 1
T9	$Q_{\text{driver,rated}} \geq Q_{\text{total}}$	Return to Step 9; select higher-capacity driver

Table 9. Thermal verification checks

7.48 Hydraulic Verification Checks

Ref.	Acceptance Criterion	Corrective Action if Failed
H1	$v_{\text{operating}} \leq v_{\text{max}}$	Adjust VFD; if insufficient, increment NPS
H2	Nusselt correlation matches confirmed Re regime	Return to Step 6; apply correct correlation
H3	Ito correlation matches confirmed De regime	Return to Step 5; apply correct f_{helical}
H4	$NPSH_a - NPSH_r \geq 0.5 \text{ m}$	Return to Step 5; adjust inlet conditions
H5	$\Delta P_{\text{system}} \leq \Delta P_{\text{seal,rated}}$	Return to Step 5; consider parallel coil circuits
H6	Minor losses included and method documented	Return to Step 5; rebuild system curve
H7	$V_{\text{battery}} \geq V_{\text{hydraulic}}$	Return to Step 10; increase V_{battery}

Table 10. Hydraulic verification checks

7.49 Geometric Verification Checks

Ref.	Acceptance Criterion	Corrective Action if Failed
G1	$V_{\text{gross}} = V_{\text{target}} + V_{\text{disp}}$ within 1%	Recompute dimensions; update Step 8
G2	Radial clearance $\geq 100 \text{ mm}$	Return to Step 4; verify d_{coil} formula
G3	$p_{\text{min}} \leq p_{\text{coil}} \leq h_1/3$	Return to Step 7; apply corrective pitch bounds
G4	$h_{\text{clearance}} \geq 50 \text{ mm}$ at each end	Return to Step 7; reduce N_{coil} by one turn
G5	$L_{\text{coil}} \leq L_{\text{practical}}$	Increment NPS; repeat Steps 4–7
G6	$0.8 \leq h/d \leq 4.0$ for both vessels	Return to Step 3 or Step 10; adjust aspect ratio
G7	$V_{\text{coil}} = \frac{\pi d_{\text{pipe}}^2}{4} L_{\text{coil}}$ (external diameter)	Return to Step 8; recompute with d_{pipe}

Table 11. Geometric verification checks

7.50 Control Stability Verification Checks

Ref.	Acceptance Criterion	Corrective Action if Failed
C1	$V_{\min} \leq V_{\text{battery}} \leq V_{\max}$	Return to Step 10; select within envelope
C2	S_t consistent with control hardware specification	Increase V_{battery} or upgrade control hardware
C3	Driver output spans $Q_{\text{driver},\min}$ to $Q_{\text{driver},\max}$	Return to Step 9; select wider modulation range

Table 12. Control stability verification checks

7.51 Global Cross-Cutting Checks

GL1 — Global Energy Balance

$Q_{\text{driver,rated}} \geq \sum_i (Q_{\text{design}}^{(i)} + Q_{\text{loss}}^{(i)})$, with all component values traceable to their source step without modification.

GL2 — Worst-Case Ambient Temperature Consistency

$T_{\text{worst ambient}}$ is identical in Steps 1, 5, 9, and 10.

GL3 — Thermofluid Property Evaluation Temperature Consistency

ρ , C_p , k_{fluid} in Steps 4, 5, 10 all evaluated at the same T_{bulk} . μ_{cold} used only in Step 6 U_{cold} calculation.

GL4 — Unit System Consistency

All dimensional quantities traceable to SI unit inputs.

7.52 Design Release Statement

When all checks in Sections 11.1–11.4 are documented as passed and the verification summary record is complete, the VCH design is formally released for detailed engineering.

Released Design Package

- Process vessel buildable specification — geometry, volume, cap profile, material (Steps 3 and 8)
- Internal coil assembly specification — L_{coil} , d_{coil} , d_{pipe} , NPS schedule, p_{coil} , N_{coil} , h_{coil} (Steps 4 and 7)
- Circulation pump specification — model, performance curve, VFD frequency range, seal specification (Step 5)
- Overall heat transfer coefficient documentation — U_{steady} , U_{cold} , all resistance components itemised (Step 6)
- Thermal driver specification — type, rated capacity, modulation range (Step 9)
- Thermal battery tank buildable specification — geometry, volume, thermofluid inventory (Step 10)
- Verification summary record — all checks passed and documented (Step 11)

References and Core Standards

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APPENDIX A

MATHEMATICAL DERIVATIONS AND WORKED EXAMPLE

This appendix consolidates all formal derivations, iterative procedures, and supporting mathematics referenced in the main body of the VCH Sizing Framework. Sections 8.1 and 8.2 establish foundational procedures used across multiple steps of the design framework. Sections 8.3 through 8.10 follow the step numbering of the framework itself. All equation numbers within this appendix are self-contained and cross-referenced from the main body by the labels shown in each section heading.

8 Appendix A

8.1 A.A—Linear Interpolation for Thermophysical Property Lookup

Thermophysical properties — density ρ , dynamic viscosity μ , specific heat capacity c_p , thermal conductivity k_f , and volumetric thermal expansion coefficient β — are evaluated at local fluid temperatures by linear interpolation over a discrete property table. This procedure is referenced in Steps 2, 5, 6, and 9 of the main framework.

8.1.1 Procedure

Let $\{T_j, \phi_j\}_{j=1}^M$ denote a tabulated property pair where T_j are monotonically increasing temperature nodes and ϕ_j are the corresponding property values. For a query temperature T^* satisfying $T_j \leq T^* \leq T_{j+1}$, the interpolated property is:

$$\phi(T^*) = \phi_j + \frac{T^* - T_j}{T_{j+1} - T_j} (\phi_{j+1} - \phi_j). \quad (108)$$

The bounding index j is located by a bisection search on the sorted temperature array in $O(\log M)$ comparisons. Extrapolation beyond the table range ($T^* < T_1$ or $T^* > T_M$) must be flagged as an error; the calculation should be aborted and the user required to supply data at the relevant temperature.

8.1.2 Validity and Accuracy Bounds

Equation (108) is valid provided:

1. T^* lies within the tabulated range: $T_1 \leq T^* \leq T_M$.
2. The property ϕ varies smoothly and without phase change across the interval $[T_j, T_{j+1}]$.
3. The node spacing $\Delta T_j = T_{j+1} - T_j$ is sufficiently small that the linear approximation introduces negligible error relative to the physical property variation.

For water and common heat-transfer fluids over industrial operating ranges, linear interpolation between nodes spaced at 10 K or less introduces errors below 0.5% for ρ , c_p , and k_f , and below 2% for μ away from the saturation boundary [1].

Note: Dynamic viscosity μ exhibits the strongest nonlinearity with temperature, particularly for viscous fluids at low temperatures. Where high accuracy is required for μ , a log-linear interpolation of the form $\ln[\mu(T)]$ is recommended in place of (108).

8.2 A.B—Worked Numerical Example

The following end-to-end calculation traces a single representative industrial scenario through all eleven steps of the design framework. All intermediate values are stated explicitly so that a practising engineer may use this section as a verification reference when implementing the framework for the first time. All fluid properties are obtained via (108) applied to the tables in Appendix B.

8.2.1 Design Scenario

- **Working fluid:** Water at coil inlet $T_{\text{in}} = 80^\circ\text{C}$, mass flow rate $\dot{m} = 0.5\text{ kg/s}$.
- **Tank fluid:** Water at bulk temperature $T_\infty = 20^\circ\text{C}$.
- **Design duty:** $Q_{\text{design}} = 25\text{ kW}$.
- **Vessel aspect ratio:** $\lambda = 1.5$, hemispherical heads.
- **Tube geometry:** $d_i = 25\text{ mm}$, $d_o = 28\text{ mm}$, $k_w = 16\text{ W/(m K)}$ (austenitic stainless steel).
- **Coil centreline diameter:** $D_c = 0.30\text{ m}$.
- **Insulation:** $t_{\text{ins}} = 50\text{ mm}$, $k_{\text{ins}} = 0.04\text{ W/(m K)}$ (mineral wool).

8.2.2 Step 1 — Fluid Properties

At film temperature $T_f = (80 + 20)/2 = 50^\circ\text{C}$: $\rho = 988\text{ kg/m}^3$, $\mu = 5.47 \times 10^{-4}\text{ Pa s}$, $k_f = 0.644\text{ W/(m K)}$, $c_p = 4181\text{ J/(kg K)}$, $Pr = 3.55$.

8.2.3 Step 2 — Reynolds Number

$$Re = \frac{4\dot{m}}{\pi d_i \mu} = \frac{4 \times 0.5}{\pi \times 0.025 \times 5.47 \times 10^{-4}} = 46,400 \quad (\text{turbulent}).$$

8.2.4 Step 3 — Inner Nusselt Number

Petukhov friction factor: $f = (0.790 \ln 46400 - 1.64)^{-2} = 0.02093$.

$$Nu_{\text{straight}} = \frac{(0.02093/8)(46400 - 1000) \times 3.55}{1 + 12.7 \sqrt{0.02093/8} (3.55^{2/3} - 1)} = 183.4.$$

Curvature ratio $\delta = 0.028/0.30 = 0.0933$. Correction: $C_{\text{curv}} = 1 + 3.6(1 - 0.0933)(0.0933)^{0.8} = 1.252$.

$$Nu_{\text{coil}} = 183.4 \times 1.252 = 229.6, \quad h_i = \frac{229.6 \times 0.644}{0.025} = 5916\text{ W/(m}^2\text{ K)}.$$

8.2.5 Step 4 — Outer Natural Convection

At $\Delta T = 60$ K and film properties of water at 50°C : $Ra_{d_o} = g\beta\Delta T d_o^3/(\nu\alpha) = 4.27 \times 10^4$.

$$Nu_{\text{outer}} = \left[0.60 + \frac{0.387 \times (4.27 \times 10^4)^{1/6}}{[1 + (0.559/3.55)^{9/16}]^{8/27}} \right]^2 = 9.87, \quad h_o = \frac{9.87 \times 0.644}{0.028} = 226.8 \text{ W}/(\text{m}^2 \text{ K}).$$

8.2.6 Step 5 — Overall U

$$\begin{aligned} \frac{1}{U} &= \frac{r_o}{h_i r_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + \frac{1}{h_o} \\ &= \frac{0.014}{5916 \times 0.0125} + \frac{0.014 \ln(1.120)}{16} + \frac{1}{226.8} \\ &= 1.890 \times 10^{-4} + 9.83 \times 10^{-5} + 4.409 \times 10^{-3} = 4.726 \times 10^{-3} \text{ m}^2 \text{ K}/\text{W}, \\ U &= 211.6 \text{ W}/(\text{m}^2 \text{ K}). \end{aligned}$$

8.2.7 Step 6 — Required Coil Length

$C_{\min} = 0.5 \times 4181 = 2090.5 \text{ W}/\text{K}$, $\Delta T = 80 - 20 = 60$ K.

$$L_{\text{required}} = \ln\left(\frac{2090.5 \times 60}{2090.5 \times 60 - 25000}\right) \times \frac{2090.5}{211.6 \times \pi \times 0.028} = 25.02 \text{ m}.$$

8.2.8 Step 7 — Coil Pitch and Turn Count

With $h_1 = 0.80$ m and $D_c = 0.30$ m, the closed-form pitch from (198):

$$p_{\text{coil}} = \frac{\pi \times 0.30}{\sqrt{(25.02/0.80)^2 - 1}} = 30.1 \text{ mm}.$$

Turn arc length: $\ell = \sqrt{(\pi \times 0.30)^2 + 0.0301^2} = 0.9426$ m. $N_t = \lfloor 25.02/0.9426 \rfloor = 26$ turns (floor, to ensure fit within h_1). Winding height: $h_{\text{wind}} = 26 \times 0.0301 = 0.783$ m. End clearance: $h_{\text{clearance}} = 0.800 - 0.783 = 17$ mm; since this is below the 50 mm minimum, D_c is incremented and the Newton–Raphson procedure of Section 8.7.3 is applied.

8.2.9 Step 8 — Displacement Compensation

$$V_{\text{disp}} = \frac{\pi \times 0.028^2}{4} \times 25.02 = 0.01541 \text{ m}^3 = 15.4 \text{ L}.$$

8.2.10 Step 9 — Steady-State Heat Loss

With $t_w = 3$ mm, $k_s = 16 \text{ W}/(\text{m K})$, $D_i = 0.50$ m, $L_s = 0.80$ m:

$$R_{\text{wall}} = \frac{\ln(1.012)}{2\pi \times 16 \times 0.80} = 1.47 \times 10^{-4} \text{ K}/\text{W}, \quad R_{\text{ins}} = \frac{\ln(1.20)}{2\pi \times 0.04 \times 0.80} = 0.7272 \text{ K}/\text{W}.$$

Table 13. Worked Example — Summary of Key Results

Step	Quantity	Symbol	Value
2	Reynolds number	Re	46 400
3	Inner heat transfer coeff.	h_i	5916 W/(m ² K)
4	Outer heat transfer coeff.	h_o	226.8 W/(m ² K)
5	Overall heat transfer coeff.	U	211.6 W/(m ² K)
6	Required coil length	L_{required}	25.02 m
7	Helix pitch	p_{coil}	30.1 mm
8	Coil displacement volume	V_{disp}	15.4 L
9	Steady-state heat loss	$\dot{Q}_{\text{loss}}$	82.5 W
10	Thermal battery max. volume	V_{max}	2189 L

Neglecting inner and outer convective resistances (small relative to insulation): $R_{\text{tot}} \approx 0.7274 \text{ K/W}$, $\dot{Q}_{\text{loss}} = (80 - 20)/0.7274 = 82.5 \text{ W}$.

8.2.11 Step 10 — Thermal Battery Sizing

$\dot{Q}_{\text{total}} = 82.5 + 25000 = 25\,083 \text{ W}$, $t_{\text{batch}} = 3600 \text{ s}$, $\Delta T_{\text{store}} = T_{\text{max}} - T_{\text{min}} = 10 \text{ K}$:

$$V_{\text{max}} = \frac{25083 \times 3600}{988 \times 4181 \times 10} = 2.189 \text{ m}^3 = 2189 \text{ L}.$$

Thermal time constant: $\tau_{\text{th}} = 988 \times 2.189 \times 4181 \times 0.7274 = 6.574 \times 10^6 \text{ s} \approx 76 \text{ days}$, confirming excellent thermal autonomy for this vessel size.

8.3 A.C—Vessel Geometry Derivations (Step 3)

8.3.1 Algebraic Solution for Vessel Dimensions — Conical-Capped Vessel

Geometry Definition

The vessel comprises a vertical cylinder of internal diameter d and cylindrical wall height h_2 , capped at the base by a right circular cone of half-angle θ . The aspect ratio is defined as:

$$\alpha = \frac{h_2}{d}, \quad \alpha > 0. \quad (109)$$

The conical cap height follows from the half-angle:

$$h_c = \frac{d}{2} \tan \theta. \quad (110)$$

Volume Formula and Cubic Solution

The total internal volume is the sum of the cylindrical and conical contributions:

$$V = \frac{\pi d^2}{4} h_2 + \frac{\pi d^2}{12} h_c. \quad (111)$$

Substituting (109) and (110) into (111) and collecting terms:

$$V = \frac{\pi d^3}{24} (6\alpha + \tan \theta). \quad (112)$$

Setting $V = V_{\text{target}}$ and solving for d :

$$d = \left(\frac{24 V_{\text{target}}}{\pi (6\alpha + \tan \theta)} \right)^{1/3}. \quad (113)$$

The cylindrical wall height and conical cap height follow immediately:

$$h_2 = \alpha d, \quad (114)$$

$$h_c = \frac{d}{2} \tan \theta. \quad (115)$$

Note: Equation (113) is a closed-form cube root — no iterative solution is required for the conical-capped geometry. This distinguishes it from the hemi-ellipsoidal case in Section 8.3.2, where the cap volume coefficient differs.

8.3.2 Algebraic Solution for Vessel Dimensions — Hemi-Ellipsoidal-Capped Vessel

Geometry Definition

The base cap is a hemi-ellipsoid with equatorial radius $d/2$ and polar semi-axis b . The ratio $\beta = 2b/d$ defines the cap slenderness. A standard shallow (2:1 ellipsoidal) cap uses $\beta = 0.25$; a hemisphere corresponds to $\beta = 1$.

Volume Formula and Cubic Solution

The volume of a hemi-ellipsoid of equatorial radius $d/2$ and polar semi-axis $b = \beta d/2$ is:

$$V_{\text{cap}} = \frac{\pi d^3 \beta}{12}. \quad (116)$$

Total vessel volume:

$$V = \frac{\pi d^2}{4} (\alpha d) + \frac{\pi d^3 \beta}{12} = \frac{\pi d^3}{12} (3\alpha + \beta). \quad (117)$$

Explicit solution for d :

$$d = \left(\frac{12 V_{\text{target}}}{\pi (3\alpha + \beta)} \right)^{1/3}, \quad (118)$$

with $h_2 = \alpha d$ and cap polar semi-axis $b = \beta d/2$.

8.3.3 Conical Cap Lateral Surface Area from First Principles

A right circular cone of base radius $r = d/2$ and half-angle θ has slant height:

$$\ell_s = \frac{r}{\sin \theta} = \frac{d}{2 \sin \theta}. \quad (119)$$

The lateral surface area is obtained by unrolling the cone into a flat sector of radius ℓ_s and arc length $2\pi r$:

$$A_{\text{cone}} = \pi r \ell_s = \frac{\pi d^2}{4 \sin \theta}. \quad (120)$$

Note: Equation (120) excludes the base circle area πr^2 , which is not an external surface in a closed vessel. The cylindrical wall lateral area is $A_{\text{cyl}} = \pi d h_2$.

8.3.4 Hemi-Ellipsoidal Cap Surface Area and Working Approximation

8.3.5 Exact Formula via Elliptic Integral

For a hemi-ellipsoid with equatorial radius $a = d/2$ and polar semi-axis b , defining eccentricity $e = \sqrt{1 - (b/a)^2}$ for the oblate case ($b < a$):

$$A_{\text{hemi,exact}} = 2\pi a^2 \left(1 + \frac{1 - e^2}{e} \operatorname{arctanh}(e) \right). \quad (121)$$

For the prolate case ($b > a$), with $e = \sqrt{1 - (a/b)^2}$:

$$A_{\text{hemi,prolate}} = 2\pi a^2 \left(1 + \frac{b}{ae} \arcsin(e) \right). \quad (122)$$

8.3.6 Knud Thomsen Working Approximation

For engineering calculations the Knud Thomsen approximation is used with $p = 1.6075$ [12]:

$$A_{\text{hemi,approx}} \approx 2\pi \left(\frac{2(d/2)^p b^p + (d/2)^{2p}}{3} \right)^{1/p}. \quad (123)$$

This approximation has a maximum relative error of 1.061% across all ellipsoid axis ratios [12], well within the fabrication tolerance of industrial pressure vessels (typically $\pm 1.5\%$ on wall thickness).

8.3.7 Volume Formulas for Standard Vessel Cap Geometries

The total internal vessel volume is the sum of the cylindrical shell volume and the cap volume. For a vertical cylinder of internal diameter d and straight-shell height h_1 , with a single base cap of depth h_2 , the four standard cap profiles yield the following closed-form volume expressions.

Hemispherical Cap ($h_2 = d/2$) A hemisphere of diameter d has volume $\frac{\pi d^3}{12}$. With $h_2 = d/2$:

$$V = \frac{\pi d^2}{4} h_1 + \frac{\pi}{6} \left(\frac{d}{2} \right)^3 \times 2 = \frac{\pi d^2}{4} \left(h_1 + \frac{2h_2}{3} \right). \quad (124)$$

Conical Cap A right circular cone of base diameter d and height h_2 has volume $V_{\text{cone}} = \frac{1}{3} \cdot \frac{\pi d^2}{4} \cdot h_2$. The total vessel volume is:

$$V = \frac{\pi d^2}{4} h_1 + \frac{\pi d^2 h_2}{12} = \frac{\pi d^2}{4} \left(h_1 + \frac{h_2}{3} \right). \quad (125)$$

Full Ellipsoidal Cap (Semi-Ellipsoid, $a = d/2$, $b = h_2$) A full hemi-ellipsoid with equatorial radius $a = d/2$ and polar semi-axis $b = h_2$ has volume $V_{\text{hemi}} = \frac{2}{3}\pi a^2 b = \frac{\pi d^2 h_2}{6}$. The total vessel volume is:

$$V = \frac{\pi d^2}{4} h_1 + \frac{\pi d^2 h_2}{6} = \frac{\pi d^2}{4} \left(h_1 + \frac{2h_2}{3} \right). \quad (126)$$

Note: Equations (124) and (126) are structurally identical. A hemispherical cap is the special case of a hemi-ellipsoid where the polar semi-axis equals the equatorial radius: $h_2 = d/2$. The coincidence of form confirms dimensional consistency.

Spherical Cap (Partial Sphere of Radius R , Height h_2) A spherical cap of height h_2 cut from a sphere of radius $R = (d^2 + 4h_2^2)/(8h_2)$ has exact volume:

$$V_{\text{cap}} = \frac{\pi h_2}{6} \left(\frac{3d^2}{4} + h_2^2 \right). \quad (127)$$

For the shallow-cap approximation ($h_2 \ll d$), retaining only the leading term:

$$V \approx \frac{\pi d^2}{4} h_1 + \frac{\pi d^2 h_2}{3} = \frac{\pi d^2}{4} \left(h_1 + \frac{4h_2}{3} \right). \quad (128)$$

8.3.8 Cap Surface Areas and External Surface Correction for Wall Thickness

Conical Cap Lateral Surface Area A right circular cone of base diameter d and vertical height h_2 has slant height $\ell_s = \sqrt{(d/2)^2 + h_2^2}$. The lateral surface area (excluding the base circle, which is not an external surface in a closed vessel) is:

$$A_{\text{cap,cone}} = \frac{\pi d}{2} \sqrt{\left(\frac{d}{2}\right)^2 + h_2^2}. \quad (129)$$

Ellipsoidal Cap Surface Area For a hemi-ellipsoid with equatorial radius $a = d/2$ and polar semi-axis $b = h_2$, defining the oblate eccentricity ($h_2 < d/2$):

$$e = \sqrt{1 - \left(\frac{2h_2}{d}\right)^2}, \quad (130)$$

the exact lateral surface area is:

$$A_{\text{cap,ellipse}} = \pi \left(\frac{d}{2}\right)^2 \left[1 + \frac{1 - e^2}{e} \operatorname{arctanh}(e) \right]. \quad (131)$$

For the prolate case ($h_2 > d/2$), define $e = \sqrt{1 - (d/2h_2)^2}$ and replace $\operatorname{arctanh}(e)$ with $\operatorname{arcsin}(e)/e$ in (131).

Wall Thickness Correction for External Surface Area The expressions above are referred to the *internal* surface. For heat loss calculations in Step 9, the relevant area is the *external* surface of the vessel shell, which accounts for the wall thickness t_w . The external diameter and cap dimensions are:

$$d_{\text{ext}} = d + 2t_w, \quad h_{2,\text{ext}} = h_2 + t_w. \quad (132)$$

The external cylindrical shell area is:

$$A_{\text{shell,ext}} = \pi d_{\text{ext}} h_1. \quad (133)$$

Cap areas are recomputed by substituting d_{ext} and $h_{2,\text{ext}}$ into (129) or (131) as appropriate. The total external surface area for the thermal resistance network is:

$$A_{\text{total,ext}} = A_{\text{shell,ext}} + A_{\text{cap,ext}}. \quad (134)$$

For most industrial process vessels with $t_w/d < 0.05$, the wall thickness correction to the cap area is below 2% and may be neglected in preliminary design. It must be retained in the rigorous Step 9 heat loss calculation.

8.4 A.D—NPS Selection and Maximum Coil Configuration (Step 4)

8.4.1 Target Volumetric Flow Rate from Thermal Demand

The heating fluid must deliver thermal power $\dot{Q}_{\text{design}}$ to the process. For a single-phase fluid with no phase change, the sensible heat equation is:

$$\dot{Q}_{\text{design}} = \dot{m}_{\text{hot}} c_{p,\text{hot}} \Delta T_{\text{coil}}, \quad (135)$$

where $\Delta T_{\text{coil}} = T_{\text{hot,in}} - T_{\text{hot,out}}$ is the hot-side temperature drop. Solving for mass flow rate and converting to volumetric flow rate using density ρ_{hot} at the mean coil temperature:

$$\dot{V} = \frac{\dot{m}_{\text{hot}}}{\rho_{\text{hot}}} = \frac{\dot{Q}_{\text{design}}}{\rho_{\text{hot}} c_{p,\text{hot}} \Delta T_{\text{coil}}}. \quad (136)$$

All fluid properties are evaluated at $T_{\text{mean}} = (T_{\text{hot,in}} + T_{\text{hot,out}})/2$ using (108).

8.4.2 Minimum Internal Flow Area from Velocity Ceiling Constraint

The continuity equation relates volumetric flow rate to mean velocity u :

$$\dot{V} = u \frac{\pi d_i^2}{4}. \quad (137)$$

Enforcing $u \leq u_{\text{max}}$, the minimum required internal flow area is:

$$A_{\text{flow,min}} = \frac{\dot{V}}{u_{\text{max}}}, \quad (138)$$

and the corresponding minimum internal diameter:

$$d_{i,\text{min}} = \sqrt{\frac{4 \dot{V}}{\pi u_{\text{max}}}}. \quad (139)$$

The NPS schedule is selected as the smallest standard size with $d_i \geq d_{i,\min}$. The actual mean velocity is back-calculated as:

$$u = \frac{4 \dot{V}}{\pi d_i^2} \leq u_{\max}. \quad (140)$$

8.4.3 Helical Coil Arc Length from Parametric Integration

Parametric Definition

A helical coil of N turns, centreline diameter D_c , and axial pitch p is described by:

$$x(\varphi) = \frac{D_c}{2} \cos \varphi, \quad (141)$$

$$y(\varphi) = \frac{D_c}{2} \sin \varphi, \quad (142)$$

$$z(\varphi) = \frac{p}{2\pi} \varphi, \quad (143)$$

where $\varphi \in [0, 2\pi N]$ is the azimuthal parameter.

Arc Length Integration

The arc length element is:

$$ds = \sqrt{\left(\frac{dx}{d\varphi}\right)^2 + \left(\frac{dy}{d\varphi}\right)^2 + \left(\frac{dz}{d\varphi}\right)^2} d\varphi. \quad (144)$$

Substituting the derivatives of (141)–(143) and applying $\sin^2 \varphi + \cos^2 \varphi = 1$:

$$ds = \sqrt{\frac{D_c^2}{4} + \frac{p^2}{4\pi^2}} d\varphi. \quad (145)$$

Since the integrand is constant in φ , integrating over N complete turns:

$$L_{\text{coil}} = N \sqrt{(\pi D_c)^2 + p^2}. \quad (146)$$

Note: Equation (146) uses the coil *centreline* diameter $D_c = d_{\text{tank}} - 0.200 - d_o$, where the 0.200 m term accounts for the mandatory 100 mm radial clearance on each side. All dimensions must be in consistent units of metres.

8.4.4 Compound Packing Efficiency and Practical Coil Length

The geometric maximum coil length from (146) assumes ideal continuous winding with no manufacturing or operational constraints. The practical maximum is obtained by applying the compound packing efficiency factor η_{pack} , which is the product of five independent sub-factors, each addressing a distinct physical constraint:

$$\eta_{\text{pack}} = \eta_f \cdot \eta_s \cdot \eta_w \cdot \eta_t \cdot \eta_d. \quad (147)$$

Reference values for each sub-factor are tabulated in the main framework (Step 4, Tables 1 through 5). Typical compound values across standard industrial coil designs fall in the range $\eta_{\text{pack}} \in [0.55, 0.75]$.

The practical maximum coil length is:

$$\boxed{L_{\text{practical}} = L_{\text{max}} \times \eta_{\text{pack}},} \quad (148)$$

where L_{max} is evaluated from (146) at the minimum structural pitch $p_{\text{min}} = 2d_o$ and maximum number of turns $N_{\text{max}} = \lfloor h_1/p_{\text{min}} \rfloor$.

8.5 A.E—Hydraulic Resistance and Pump Selection (Step 5)

8.5.1 Reynolds Number for Internal Pipe Flow

$$\boxed{Re = \frac{\rho u d_i}{\mu} = \frac{4 \dot{m}}{\pi d_i \mu}.} \quad (149)$$

All fluid properties are evaluated at bulk mean temperature via (108). Flow regime classification for straight pipe: $Re < 2300$ laminar; $2300 \leq Re \leq 4000$ transitional; $Re > 4000$ turbulent. These boundaries are modified for helical geometry by the Dean number (Section 8.5.3).

8.5.2 Colebrook-White Equation and Newton-Raphson Solution

Colebrook-White Equation

For turbulent flow in a pipe of internal roughness ε :

$$\frac{1}{\sqrt{f_s}} = -2 \log_{10} \left(\frac{\varepsilon/d_i}{3.7} + \frac{2.51}{Re \sqrt{f_s}} \right), \quad (150)$$

where f_s denotes the straight-pipe Darcy friction factor. For laminar flow ($Re < 2300$):

$$f_s = \frac{64}{Re}. \quad (151)$$

Haaland Seed Estimate

$$f_0 = \left(-1.8 \log_{10} \left[\left(\frac{\varepsilon/d_i}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \right)^{-2}. \quad (152)$$

Newton-Raphson Iterative Procedure

Define $x = f_s^{-1/2}$ and the residual function $F(x) = 0$:

$$F(x) = x + 2 \log_{10} \left(\frac{\varepsilon/d_i}{3.7} + \frac{2.51 x}{Re} \right), \quad (153)$$

with derivative:

$$F'(x) = 1 + \frac{2}{\ln 10} \cdot \frac{2.51/Re}{\frac{\varepsilon/d_i}{3.7} + \frac{2.51 x}{Re}}. \quad (154)$$

The Newton-Raphson update at iteration k :

$$x^{(k+1)} = x^{(k)} - \frac{F(x^{(k)})}{F'(x^{(k)})}. \quad (155)$$

Initialise with $x^{(0)} = f_0^{-1/2}$ from (152). Terminate when $|x^{(k+1)} - x^{(k)}| < 10^{-8}$. Recover friction factor: $f_s = 1/(x^{(k+1)})^2$. Convergence is achieved in three or fewer iterations across the full turbulent regime [7].

8.5.3 Dean Number Definition and Flow Regime Classification

The Dean number quantifies the ratio of centrifugal to viscous forces in helical pipe flow [8]:

$$De = Re \sqrt{\frac{d_i}{D_c}}. \quad (156)$$

Curvature stabilises the boundary layer, delaying the laminar-to-turbulent transition. The critical Dean number below which flow remains laminar despite $Re > 2300$ is $De_{\text{crit}} = 11.6$ [9], corresponding to:

$$Re_{\text{crit,coil}} = 11.6 \sqrt{\frac{D_c}{d_i}}. \quad (157)$$

Flow regime classification for helical coils:

- $De < 11.6$: laminar — use Sieder-Tate (Section 8.6.3).
- $11.6 \leq De \leq 100$: transitional — interpolate between laminar and turbulent Ito correlations.
- $De > 100$: turbulent — use modified Gnielinski (Section 8.6.2).

8.5.4 Ito's Helical Friction Factor Correlations

Laminar Regime ($De < 11.6$)

$$f_{\text{hel,lam}} = \frac{64}{Re} [1 + 0.033(\log_{10} De)^4]. \quad (158)$$

Turbulent Regime ($De > 100$)

$$f_{\text{hel,turb}} = 0.304 Re^{-0.25} \left(\frac{D_c}{d_i} \right)^{-0.1}. \quad (159)$$

Transitional Regime ($11.6 \leq De \leq 100$)

Linear interpolation in De :

$$f_{\text{hel,trans}} = f_{\text{hel,lam}}|_{De=11.6} + \frac{De - 11.6}{88.4} (f_{\text{hel,turb}}|_{De=100} - f_{\text{hel,lam}}|_{De=11.6}). \quad (160)$$

8.5.5 Darcy-Weisbach Equation: Derivation and Application to Helical Coil Geometry

Derivation from the Mechanical Energy Equation

For steady, incompressible, fully developed internal flow, the mechanical energy equation between two cross-sections separated by length L reduces to:

$$\frac{\Delta P}{\rho} = h_f g, \quad (161)$$

where h_f is the head loss due to wall friction. Dimensional analysis of the friction force on a control volume of length dx and internal diameter d_i gives the wall shear stress $\tau_w = f_s \rho u^2 / 8$, where f_s is the Darcy friction factor. Integrating over the full pipe length and converting to pressure units:

$$\Delta P_{\text{major}} = f_s \frac{L}{d_i} \frac{\rho u^2}{2}, \quad (162)$$

which is the Darcy-Weisbach equation. The factor $\rho u^2 / 2$ is the dynamic pressure of the flow. The Darcy friction factor f_s is four times the Fanning friction factor and is determined from the Colebrook-White equation (Section 8.5.2) for turbulent flow and from $f_s = 64 / Re$ for laminar flow.

Application to Helical Coil Geometry

For a helical coil, the straight-pipe friction factor f_s is replaced by the Ito helical friction factor f_{hel} from Section 8.5.4, which accounts for the additional pressure loss arising from centrifugal secondary flow in the coil cross-section:

$$\Delta P_{\text{major}} = f_{\text{hel}} \frac{L_{\text{coil}}}{d_i} \frac{\rho u^2}{2}, \quad (163)$$

where L_{coil} is the total coil arc length from (146) and $f_{\text{hel}} \geq f_s$ for all Dean numbers, confirming that helical geometry always incurs higher frictional resistance than an equivalent straight pipe at the same flow conditions.

8.5.6 Minor Losses, Total System Resistance, and Head Conversion

K-Factor Method for Minor Losses

Each fitting contributes:

$$\Delta P_{\text{minor},j} = K_j \frac{\rho u^2}{2}. \quad (164)$$

Total system pressure drop:

$$\Delta P_{\text{system}} = \left(f_{\text{hel}} \frac{L_{\text{coil}}}{d_i} + \sum_{j=1}^M K_j \right) \frac{\rho u^2}{2}. \quad (165)$$

Conversion to Equivalent Head

$$H_{\text{system}} = \frac{\Delta P_{\text{system}}}{\rho g}, \quad (166)$$

where $g = 9.81 \text{ m/s}^2$.

8.5.7 Pump Affinity Laws and VFD Frequency Scaling

For a geometrically similar pump operating at rotational speed n , the affinity laws follow from invariance of the dimensionless flow, head, and power coefficients [3]:

$$\frac{\dot{V}_2}{\dot{V}_1} = \frac{n_2}{n_1}, \quad (167)$$

$$\frac{H_2}{H_1} = \left(\frac{n_2}{n_1} \right)^2, \quad (168)$$

$$\frac{P_2}{P_1} = \left(\frac{n_2}{n_1} \right)^3. \quad (169)$$

A Variable Frequency Drive (VFD) scales pump speed proportionally to supply frequency f_{VFD} . The target VFD frequency required to achieve velocity-compliant flow rate $\dot{V}_{\text{design}}$ is:

$$f_{\text{VFD}} = f_{\text{rated}} \frac{\dot{V}_{\text{design}}}{\dot{V}_{\text{rated}}}. \quad (170)$$

The scaled duty head is verified as:

$$H_{\text{duty}} = H_{\text{rated}} \left(\frac{f_{\text{VFD}}}{f_{\text{rated}}} \right)^2 \geq H_{\text{system}}. \quad (171)$$

If (171) is not satisfied, a larger NPS or pump frame is selected and the procedure repeats from Section 8.4.2.

8.6 A.F—Thermal Performance Evaluation (Step 6)

8.6.1 Series Thermal Resistance Network and Overall Heat Transfer Coefficient

For a helical coil of inner radius r_i , outer radius r_o , wall conductivity k_w , and length L , the overall heat transfer coefficient referred to the outer surface is obtained from the series resistance network (inner convection, cylindrical wall conduction, outer convection):

$$\frac{1}{U} = \frac{r_o}{h_i r_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + \frac{1}{h_o}. \quad (172)$$

Including fouling resistances $R_{f,i}$ and $R_{f,o}$ on the inner and outer surfaces respectively:

$$\frac{1}{U} = \frac{r_o}{h_i r_i} + R_{f,i} \frac{r_o}{r_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + R_{f,o} + \frac{1}{h_o}. \quad (173)$$

8.6.2 Modified Gnielinski Correlation for Turbulent Helical Coil Flow

Baseline Gnielinski Correlation

The Gnielinski correlation for a smooth straight tube provides the Nusselt number for turbulent forced convection in the range $3,000 \lesssim Re \lesssim 5 \times 10^6$ and $0.5 \lesssim Pr \lesssim 2,000$:

$$Nu_{\text{straight}} = \frac{(f_s/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f_s/8}(Pr^{2/3} - 1)}, \quad (174)$$

where f_s is obtained from the Petukhov relation $f_s = (0.790 \ln Re - 1.64)^{-2}$. The subtraction of 1000 from Re in the numerator corrects for the transitional regime entry length effect.

Curvature Correction for Helical Geometry

Secondary Dean-flow vortices in a helically coiled tube augment convective heat transfer relative to the straight-tube baseline. The Manlapaz-Churchill correction [10] quantifies this enhancement as a multiplicative factor dependent on the curvature ratio $\delta = d_i/D_c$:

$$Nu_{\text{coil}} = \frac{(f_{\text{hel}}/8)(Re - 1000)Pr}{1 + 2.17\sqrt{f_{\text{hel}}/8}(Pr^{2/3} - 1)} \cdot \left[1 + \left(\frac{d_i}{D_c} \right)^{0.9} \right], \quad (175)$$

where f_{hel} is the Ito helical friction factor. The bracketed curvature term is always greater than unity, confirming heat transfer enhancement over the straight-tube case. The internal convective coefficient follows as:

$$h_{\text{in}} = \frac{Nu_{\text{coil}} k_f}{d_i}. \quad (176)$$

8.6.3 Sieder-Tate Correlation for Laminar and Transitional Helical Coil Flow

For laminar and transitional flow ($De < 11.6$, or equivalently Graetz number $Gz = Re Pr d_i/L_{\text{coil}} > 10$), the Sieder-Tate correlation is applied with the Manlapaz-Churchill curvature correction [10]:

$$Nu_{\text{coil}} = 1.86 \left(\frac{Re Pr d_i}{L_{\text{coil}}} \right)^{1/3} \left(\frac{\mu_b}{\mu_w} \right)^{0.14} \left[1 + 3.6 \left(1 - \frac{d_i}{D_c} \right) \left(\frac{d_i}{D_c} \right)^{0.8} \right]. \quad (177)$$

The first grouped term is the classical Graetz-Leveque thermal entry length factor. The viscosity ratio $(\mu_b/\mu_w)^{0.14}$ corrects for the temperature dependence of viscosity across the boundary layer: μ_b is evaluated at the bulk fluid temperature and μ_w at the estimated wall temperature $T_w \approx (T_{\text{supply}} + T_{\text{tank}})/2$. The final bracketed term is the curvature correction. For low-viscosity fluids or small temperature differences, the viscosity correction factor is typically within 2% of unity and may be set to 1.0 in preliminary calculations.

8.6.4 Prandtl Number Definition and Role in Convective Correlations

$$Pr = \frac{\mu c_p}{k_f} = \frac{\nu}{\alpha_{\text{th}}}, \quad (178)$$

where $\nu = \mu/\rho$ is kinematic viscosity and $\alpha_{\text{th}} = k_f/(\rho c_p)$ is thermal diffusivity. The Prandtl number governs the relative thickness of the hydrodynamic and thermal boundary layers: $\delta_{\text{th}}/\delta_{\text{hydro}} \approx Pr^{-1/3}$.

8.6.5 Churchill-Chu Correlation: Physical Basis, Rayleigh Number, and External Convective Coefficient

Rayleigh Number

The Rayleigh number governs the intensity of natural convection by quantifying the ratio of buoyancy-driven to diffusion-limited transport. It is the product of the Grashof number Gr and the Prandtl number Pr :

$$Ra_{d_o} = Gr_{d_o} \cdot Pr = \frac{g \beta (T_w - T_\infty) d_o^3}{\nu^2} \cdot \frac{\nu}{\alpha_{th}} = \frac{g \beta \Delta T_{drive} d_o^3}{\nu \alpha_{th}}, \quad (179)$$

where $g = 9.81 \text{ m/s}^2$, β is the volumetric thermal expansion coefficient of the process fluid evaluated at the film temperature $T_f = (T_w + T_\infty)/2 \text{ (K}^{-1}\text{)}$, $\Delta T_{drive} = T_w - T_\infty$ is the wall-to-bulk temperature difference (K), d_o is the external pipe diameter serving as the characteristic length (m), ν is the kinematic viscosity (m^2/s), and $\alpha_{th} = k_f/(\rho c_p)$ is the thermal diffusivity (m^2/s).

For ideal gases, $\beta = 1/T_f \text{ (K}^{-1}\text{, absolute temperature)}$. For liquids, β must be obtained from tabulated property data via the relation $\beta = -(\partial\rho/\partial T)_P/\rho$.

Churchill-Chu Correlation

Treating each turn of the helical coil as a long horizontal isothermal cylinder submerged in the bulk process fluid, the Churchill-Chu correlation [11] gives the average external Nusselt number, valid across the full range $Ra_{d_o} \lesssim 10^{12}$:

$$Nu_{out} = \left[0.60 + \frac{0.387 Ra_{d_o}^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right]^2. \quad (180)$$

The constant 0.60 represents the conduction limit at $Ra \rightarrow 0$, while the $Ra^{1/6}$ term captures the convective enhancement. The grouping in the denominator accounts for the Prandtl-number dependence of the thermal boundary layer thickness relative to the velocity boundary layer. The external convective coefficient is:

$$h_{out} = \frac{Nu_{out} k_{process}}{d_o}, \quad (181)$$

where $k_{process}$ is the thermal conductivity of the process fluid evaluated at the film temperature.

8.6.6 Impeller-Based Forced Convection Correlation for Mechanically Agitated Vessels

For submerged coils in mechanically agitated vessels, the conventional pipe-flow Reynolds number is replaced by the Impeller Reynolds Number, which characterises the turbulence and shear imposed by the rotating agitator:

$$Re_{ia} = \frac{\rho N D^2}{\mu}, \quad (182)$$

where ρ is the process fluid density (kg/m^3), N is the impeller rotational speed (rev/s), D is the impeller diameter (m), and μ is the dynamic viscosity at bulk temperature ($\text{Pa}\cdot\text{s}$).

The external Nusselt number is:

$$Nu_{\text{out}} = C \cdot Re_{\text{ia}}^a \cdot Pr^{1/3} \cdot \left(\frac{\mu}{\mu_w} \right)^{0.14}, \quad (183)$$

where C and a are geometry- and regime-dependent constants (tabulated in Step 6 of the main framework), and μ_w is the viscosity evaluated at the outer pipe wall temperature $T_{w,\text{out}}$.

The $Pr^{1/3}$ dependence reflects the classical forced convection scaling of thermal boundary layer thickness, while the viscosity ratio $(\mu/\mu_w)^{0.14}$ corrects for the viscosity gradient between the bulk fluid and the pipe wall. The external convective coefficient follows as:

$$h_{\text{out}} = \frac{Nu_{\text{out}} k_{\text{process}}}{d_o}. \quad (184)$$

Note: In agitated vessels Ra is rendered negligible and (180) must not be used. The two external convection regimes are mutually exclusive: natural convection governs unagitated operation; the impeller correlation governs agitated operation.

8.6.7 Logarithmic Mean Temperature Difference — Derivation and Arithmetic Mean Approximation

Derivation

For a coil immersed in a well-mixed tank at constant temperature T_∞ , the energy balance on a differential element $dA = \pi d_o dx$ gives $d\theta/\theta = -(U\pi d_o/\dot{m}c_p) dx$, where $\theta = T_f - T_\infty$. Integrating over the full coil length:

$$q = U A_o \Delta T_{\text{lm}}, \quad \Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}, \quad (185)$$

where $\Delta T_1 = T_{\text{in}} - T_\infty$ and $\Delta T_2 = T_{\text{out}} - T_\infty$.

Arithmetic Mean Approximation Validity

The arithmetic mean $\Delta T_{\text{am}} = (\Delta T_1 + \Delta T_2)/2$ approximates ΔT_{lm} within 5% when the ratio $R = \Delta T_1/\Delta T_2$ satisfies $0.54 \lesssim R \lesssim 1.86$, i.e. when the two endpoint temperature differences differ by less than a factor of approximately 1.86.

8.6.8 Effectiveness-NTU Method: Full Derivation for $C_r = 0$

For a submerged coil in a large well-mixed tank, the tank thermal capacity rate $C_{\text{tank}} \rightarrow \infty$, giving capacity ratio $C_r = C_{\text{min}}/C_{\text{max}} = 0$. The energy balance on the coil stream reduces to the ODE $d\theta/dA = -(U/C_{\text{min}})\theta$, where $\theta = T_{\text{min}} - T_\infty$. Integrating:

$$\ln\left(\frac{\theta_{\text{out}}}{\theta_{\text{in}}}\right) = -NTU, \quad NTU = \frac{U A_o}{C_{\text{min}}}. \quad (186)$$

The heat exchanger effectiveness:

$$\varepsilon = \frac{q}{q_{\text{max}}} = \frac{C_{\text{min}}(T_{\text{min,out}} - T_{\text{min,in}})}{C_{\text{min}}(T_\infty - T_{\text{min,in}})}, \quad (187)$$

simplifies, after substituting the integrated ODE result, to:

$$\boxed{\varepsilon = 1 - e^{-NTU}, \quad (C_r = 0).} \quad (188)$$

8.6.9 Maximum Transmissible Thermal Power

$$q_{\max} = C_{\min} (T_{h,\text{in}} - T_{\infty}) = \dot{m} c_p (T_{\text{in}} - T_{\infty}). \quad (189)$$

Combining with (188):

$$\boxed{q = (1 - e^{-UA_o/C_{\min}}) C_{\min} (T_{\text{in}} - T_{\infty}).} \quad (190)$$

This expression encapsulates coil geometry (through A_o), thermal resistances (through U from (172) or (173)), flow conditions (through $C_{\min}$), and driving potential (through $T_{\text{in}} - T_{\infty}$). It is evaluated twice in Step 6: once using U_{steady} for nominal performance, and once using U_{cold} (worst-case cold-start viscosity) for conservative sizing.

8.7 A.G—Coil Length Determination (Step 7)

8.7.1 Closed-Form Inversion of the ε -NTU Relationship for Required Coil Length

Starting Point

From the ε -NTU result (190), the actual heat transfer rate delivered by the coil is:

$$Q = \left(1 - e^{-\frac{UA}{C_{\min}}}\right) C_{\min} \Delta T, \quad (191)$$

where $A = \pi d_{\text{pipe}} L$ is the external heat transfer surface area and $\Delta T = T_{\text{supply}} - T_{\text{tank}}$ is the driving temperature difference.

Derivation

Setting $Q = Q_{\text{design}}$ and rearranging for the exponential term:

$$e^{-\frac{UA}{C_{\min}}} = 1 - \frac{Q_{\text{design}}}{C_{\min} \Delta T} = \frac{C_{\min} \Delta T - Q_{\text{design}}}{C_{\min} \Delta T}. \quad (192)$$

Taking the natural logarithm of both sides and solving for A :

$$A_{\text{required}} = \ln\left(\frac{C_{\min} \Delta T}{C_{\min} \Delta T - Q_{\text{design}}}\right) \cdot \frac{C_{\min}}{U}. \quad (193)$$

Substituting $A = \pi d_{\text{pipe}} L$ and solving for L :

$$\boxed{L_{\text{required}} = \ln\left(\frac{C_{\min} \Delta T}{C_{\min} \Delta T - Q_{\text{design}}}\right) \cdot \frac{C_{\min}}{U \cdot \pi d_{\text{pipe}}}.} \quad (194)$$

This result is exact for all NTU regimes and requires no iterative solution. The argument of the logarithm is always greater than unity provided $Q_{\text{design}} < C_{\min} \Delta T$, which must hold

by the second law of thermodynamics. Two structural properties are immediately evident: $L_{\text{required}} \propto 1/d_{\text{pipe}}$, confirming that a larger bore pipe reduces the required coil length for the same thermal duty; and $L_{\text{required}} \propto 1/U$, so improved heat transfer conditions similarly reduce the coil length requirement.

Equation (194) is evaluated using U_{cold} (worst-case, fully fouled, cold-start coefficient) to produce the conservative upper bound on L_{required} .

8.7.2 Geometric Feasibility Check and Helical Pitch Closed-Form Solution

The resolved length must satisfy the physical constraint from Step 4:

$$L_{\text{required}} \leq L_{\text{practical}}. \quad (195)$$

If (195) is violated, the NPS is incremented upward and Steps 4 through 7 are repeated.

Closed-Form Pitch Solution

With L_{required} confirmed feasible, the helical pitch is recovered by inverting the arc-length relation (146) with $N = h_1/p$:

$$\frac{h_1}{p} \sqrt{(\pi d_{\text{coil}})^2 + p^2} = L_{\text{required}}. \quad (196)$$

Squaring both sides and collecting p^2 terms:

$$p^2 (L_{\text{required}}^2 - h_1^2) = h_1^2 (\pi d_{\text{coil}})^2. \quad (197)$$

Solving explicitly:

$$p_{\text{coil}} = \frac{\pi d_{\text{coil}}}{\sqrt{\left(\frac{L_{\text{required}}}{h_1}\right)^2 - 1}}, \quad (198)$$

valid when $L_{\text{required}} > h_1$, which is satisfied in all physically meaningful coil configurations. The pitch must satisfy the fabrication bounds:

$$p_{\text{min}} = 2 d_{\text{pipe}} \leq p_{\text{coil}} \leq \frac{h_1}{3}. \quad (199)$$

When additional geometric constraints couple p to d_{coil} — for instance, when the coil diameter must be adjusted after displacement compensation in Step 8 — the Newton-Raphson procedure of Section 8.7.3 is applied.

8.7.3 Newton-Raphson Iterative Procedure for Helical Pitch

When the closed-form solution of (198) cannot be applied directly, the transcendental equation (196) is solved iteratively. Defining the residual $F(p)$ and its derivative:

$$F(p) = \frac{h_1}{p} \sqrt{(\pi d_{\text{coil}})^2 + p^2} - L_{\text{required}}, \quad (200)$$

$$F'(p) = \frac{-h_1 (\pi d_{\text{coil}})^2}{p^2 \sqrt{(\pi d_{\text{coil}})^2 + p^2}}. \quad (201)$$

The Newton-Raphson procedure proceeds as follows:

1. Initialise $p^{(0)} = d_{\text{pipe}} (1 + \delta_{\text{min}})$, where δ_{min} is the minimum inter-turn gap fraction.
2. At iteration k , compute:

$$\Delta p = -\frac{F(p^{(k)})}{F'(p^{(k)})}, \quad p^{(k+1)} = p^{(k)} + \Delta p.$$

3. Enforce the lower bound: if $p^{(k+1)} < d_{\text{pipe}} (1 + \delta_{\text{min}})$, reset $p^{(k+1)} = d_{\text{pipe}} (1 + \delta_{\text{min}})$.
4. Terminate when $|\Delta p| < \tau = 10^{-6} \text{ m}$. Recover $p_{\text{coil}} = p^{(k+1)}$.

Convergence is quadratic, typically requiring fewer than ten iterations. If $F'(p) \rightarrow 0$ near $p = \pi d_{\text{coil}}$, switch to bisection on $[d_{\text{pipe}}(1 + \delta_{\text{min}}), h_1/2]$.

8.7.4 Turn Count, Winding Height, and End Clearance Formulations

Turn arc length per revolution:

$$\ell_{\text{turn}} = \sqrt{(\pi d_{\text{coil}})^2 + p_{\text{coil}}^2}. \quad (202)$$

Number of turns, rounded down to ensure the coil fits within h_1 :

$$N_t = \left\lfloor \frac{h_1}{p_{\text{coil}}} \right\rfloor. \quad (203)$$

Actual winding height:

$$h_{\text{wind}} = N_t p_{\text{coil}}. \quad (204)$$

End clearance at each extremity:

$$h_{\text{clearance}} = h_1 - h_{\text{wind}} \geq 50 \text{ mm}. \quad (205)$$

If (205) is violated, N_t is reduced by one and the thermal adequacy check is repeated with the reduced surface area.

8.8 A.H—Volumetric Displacement Compensation (Step 8)

8.8.1 Primary Coil Pipe Displacement Volume

The coil displaces volume equal to the full external cross-section of the tube, since the tube interior (thermofluid) is inaccessible to the process fluid. Using the *external* diameter d_o :

$$V_{\text{disp}} = \frac{\pi d_o^2}{4} L_{\text{coil}}. \quad (206)$$

Using d_i instead of d_o would underestimate the displacement by the tube wall volume and is incorrect.

8.8.2 Gross Fabrication Volume Target

$$\boxed{V_{\text{gross}} = V_{\text{target}} + V_{\text{disp}} + V_{\text{ancillary}}}, \quad (207)$$

where $V_{\text{ancillary}} = k_{\text{anc}} \times V_{\text{disp}}$ with k_{anc} from the support complexity table in Step 8.

8.8.3 Option A — Additional Shell Height for Vertical Extension

With vessel diameter d held constant, the additional cylindrical height required to recover $V_{\text{gross}} - V_{\text{target}}$:

$$\boxed{\Delta h = \frac{4(V_{\text{disp}} + V_{\text{ancillary}})}{\pi d^2}, \quad h_{1,\text{final}} = h_1 + \Delta h.} \quad (208)$$

8.8.4 Option B — Adjusted Vessel Diameter for Radial Expansion

With shell height h_1 held constant, the adjusted diameter is obtained by substituting V_{gross} for V_{target} in the Step 3 cubic solution. For the conical-capped case from (113):

$$\boxed{d_{\text{new}} = \left(\frac{24 V_{\text{gross}}}{\pi(6\alpha + \tan \theta)} \right)^{1/3}}. \quad (209)$$

For the hemi-ellipsoidal case, substitute V_{gross} into (118).

8.8.5 Updated External Surface Area After Dimensional Adjustment

Option A (diameter d fixed, new height $h_{1,\text{final}}$)

$$A_{\text{shell,A}} = \pi d h_{1,\text{final}}, \quad A_{\text{total,A}} = A_{\text{shell,A}} + A_{\text{caps}}. \quad (210)$$

A_{caps} is evaluated using (120) or (123) as appropriate and is unchanged from Step 3.

Option B (height h_1 fixed, new diameter d_{new})

Both shell and cap areas must be recomputed with d_{new} in place of d , using the surface area formulas from Section 8.3.3 or 8.3.4.

8.9 A.I—Thermal Driver Sizing (Step 9)

8.9.1 Internal Natural Convection Resistance at the Vessel Boundary

The inner vessel wall is treated as a vertical flat plate or large-diameter horizontal cylinder. Applying the Churchill-Chu correlation (180) with characteristic length equal to the vessel inner diameter D_i and $\Delta T = T_{\text{bulk}} - T_{\text{wall,i}}$:

$$h_{\text{nc,i}} = \frac{Nu_{D_i} k_f}{D_i}, \quad (211)$$

and the corresponding resistance referred to inner surface area A_i :

$$R_{\text{conv,i}} = \frac{1}{h_{\text{nc,i}} A_i}. \quad (212)$$

For conservative preliminary estimation, $h_{nc,i}$ may be taken as 50–200 W/(m² K) for low-to-moderate viscosity process fluids and 10–50 W/(m² K) for high-viscosity organic fluids.

8.9.2 Vessel Wall Conductive Resistance — Flat Wall and Cylindrical Formulations

Cylindrical Formulation

$$R_{\text{wall,cyl}} = \frac{\ln(r_o/r_i)}{2\pi k_s L_s}. \quad (213)$$

Flat Wall Approximation

$$R_{\text{wall,flat}} = \frac{t_w}{k_s A_{\text{wall}}}. \quad (214)$$

The flat-wall approximation introduces less than 5% error when $t_w/r_i < 0.1$ (thin-walled condition).

8.9.3 Insulation Conductive Resistance — Flat Wall and Cylindrical Formulations

Cylindrical Formulation

$$R_{\text{ins,cyl}} = \frac{\ln(r_{\text{ins}}/r_o)}{2\pi k_{\text{ins}} L_s}, \quad (215)$$

where $r_{\text{ins}} = r_o + t_{\text{ins}}$.

Flat Wall Approximation

$$R_{\text{ins,flat}} = \frac{t_{\text{ins}}}{k_{\text{ins}} A_{\text{ext}}}, \quad (216)$$

where A_{ext} is from Section 8.8.5. The cylindrical form should be preferred when insulation thickness is not negligible relative to the vessel radius.

8.9.4 External Convective Resistance Formulation

$$R_{\text{conv,o}} = \frac{1}{h_{\text{ext}} A_{\text{ins}}}, \quad (217)$$

where A_{ins} is the outer insulation surface area and h_{ext} is estimated from the Churchill-Chu correlation applied to the outer insulation geometry, or taken from the reference values in Step 9 of the main framework.

8.9.5 Total Series Resistance Network and Steady-State Heat Loss Rate

$$R_{\text{tot}} = R_{\text{conv,i}} + R_{\text{wall}} + R_{\text{ins}} + R_{\text{conv,o}}. \quad (218)$$

$$\dot{Q}_{\text{loss}} = \frac{T_{\text{bulk}} - T_{\text{worst,amb}}}{R_{\text{tot}}}. \quad (219)$$

$T_{\text{worst,amb}}$ is the worst-case ambient temperature established in Step 1 — coldest recorded for heating applications, hottest for cooling.

8.9.6 Multi-Vessel Total Thermal Driver Capacity

For N_v vessels, each with heat loss $\dot{Q}_{\text{loss},j}$ and process heat demand $Q_{\text{design},j}$:

$$\dot{Q}_{\text{driver,total}} = \sum_{j=1}^{N_v} (\dot{Q}_{\text{loss},j} + Q_{\text{design},j}). \quad (220)$$

A diversity factor $f_d \leq 1$ may be applied if simultaneous peak demand across all vessels is not required by the operating schedule.

8.9.7 Driver Turndown Ratio and Heating Element Count Formulations

Minimum number of elements to cover peak demand at rated output per element $\dot{Q}_{\text{rated}}$:

$$N_{e,\text{max}} = \left\lceil \frac{\dot{Q}_{\text{driver,total}}}{\dot{Q}_{\text{rated}}} \right\rceil. \quad (221)$$

Minimum number of elements to sustain standing losses without cycling:

$$N_{e,\text{min}} = \left\lceil \frac{\sum_{j=1}^{N_v} \dot{Q}_{\text{loss},j}}{\dot{Q}_{\text{rated}}} \right\rceil. \quad (222)$$

If the minimum stable driver output at turndown exceeds $\sum_j \dot{Q}_{\text{loss},j}$, the driver will cycle at low demand; $N_{e,\text{max}}$ should be increased or a smaller modulating element added.

8.10 A.J—Thermal Battery Sizing (Step 10)

8.10.1 Hydronic Priming Volume

$$V_{\text{prime}} = V_{\text{pipe}} + V_{\text{coil,int}} + V_{\text{exp}}, \quad (223)$$

where, using the *internal* pipe diameter d_i :

$$V_{\text{coil,int}} = \frac{\pi d_i^2}{4} L_{\text{coil}}, \quad (224)$$

$$V_{\text{pipe}} = \sum_{k=1}^m \frac{\pi d_{i,k}^2}{4} \ell_k, \quad (225)$$

$$V_{\text{exp}} = f_{\text{exp}} (V_{\text{pipe}} + V_{\text{coil,int}}), \quad f_{\text{exp}} \approx 0.04\text{--}0.08. \quad (226)$$

Note the use of d_i here, in contrast to d_o in the displacement calculation (206). The priming volume is the fluid payload inside the loop, not the volume the coil displaces in the process vessel.

8.10.2 Electric Element Flux Buffer Volume Formulation

$$V_{\text{buffer,e}} = V_{\text{clearance}} + V_{\text{flux}}, \quad V_{\text{flux}} = \frac{\dot{Q}_e}{\rho c_p (dT/dt)_{\text{limit}}}, \quad (227)$$

where $(dT/dt)_{\text{limit}}$ is the maximum allowable temperature ramp rate the SCR or VFD control system can track without triggering element high-temperature safety interlocks (K/s), and $V_{\text{clearance}}$ is the volume required to keep all active elements fully submerged at minimum standby fluid level.

8.10.3 Fired Heater Minimum Volume from Burner Turndown Constraints

To prevent the fluid temperature from rising through the controller deadband ΔT_{db} faster than the burner can throttle down during its minimum runtime $t_{\text{min,run}}$ at minimum stable output $\dot{Q}_{\text{min}} = \dot{Q}_{\text{rated}}/TR$:

$$V_{\text{min,fired}} = \frac{\dot{Q}_{\text{min}} t_{\text{min,run}}}{\rho c_p \Delta T_{\text{db}}}. \quad (228)$$

8.10.4 Chiller Minimum Volume from Compressor Anti-Recycle Constraints

The fluid temperature must not exceed $T_{\text{set}} + \Delta T_{\text{db}}$ during the compressor anti-recycle lockout period τ_{arc} :

$$V_{\text{min,chiller}} = \frac{\dot{Q}_{\text{loss}} \tau_{\text{arc}}}{\rho c_p \Delta T_{\text{db}}}. \quad (229)$$

8.10.5 Maximum Volume Derivation from First-Law Transient Energy Balance

Governing ODE

Applying the first law to a well-mixed thermal battery of volume V with no driver input ($\dot{Q}_{\text{source}} = 0$):

$$\rho V c_p \frac{dT}{dt} = -\dot{Q}_{\text{loss}}(T) - \dot{Q}_{\text{proc}}, \quad (230)$$

where $\dot{Q}_{\text{loss}}(T) = (T - T_{\text{worst,amb}})/R_{\text{tot}}$ and $\dot{Q}_{\text{proc}} = Q_{\text{design}}/t_{\text{batch}}$ is the constant peak extraction rate.

Integration and Autonomy Boundary

Under the conservative assumption $T \approx T_{\text{set}}$, the total energy extracted during one batch cycle t_{batch} is:

$$\Delta E = (\dot{Q}_{\text{loss}} + \dot{Q}_{\text{proc}}) t_{\text{batch}}. \quad (231)$$

This must be provided entirely by cooling the battery through the allowable temperature drawdown $\Delta T_{\text{allow}} = T_{\text{max}} - T_{\text{min}}$:

$$\Delta E = \rho V c_p \Delta T_{\text{allow}}. \quad (232)$$

Equating (231) and (232):

$$V_{\max} = \frac{(\dot{Q}_{\text{loss}} + \dot{Q}_{\text{proc}}) t_{\text{batch}}}{\rho c_p \Delta T_{\text{allow}} \eta_{\text{usable}}}, \quad (233)$$

where $\eta_{\text{usable}} = 0.80$ is the usable thermal fraction accounting for stratification and standing losses.

8.10.6 Thermal Time Constant Formulation

Setting $\dot{Q}_{\text{proc}} = 0$ in (230) and linearising the loss term, the natural cooling response is exponential with time constant:

$$\tau_{\text{th}} = \rho V c_p R_{\text{tot}}. \quad (234)$$

A larger volume, better insulation (higher R_{tot}), and higher fluid thermal mass all increase thermal autonomy as expected.

8.10.7 Thermal Battery Tank Geometric Dimensioning Procedure

The selected volume $V_{\text{batt}} \in [V_{\min}, V_{\max}]$ is converted to physical dimensions by applying the Step 3 cubic solution with V_{batt} in place of V_{target} and the same aspect ratio α and cap profile selected for the process vessel.

For the conical-capped case from (113):

$$d_{\text{batt}} = \left(\frac{24 V_{\text{batt}}}{\pi(6\alpha + \tan \theta)} \right)^{1/3}, \quad h_{2,\text{batt}} = \alpha d_{\text{batt}}. \quad (235)$$

For the hemi-ellipsoidal case, substitute into (118). The aspect ratio sanity bounds of Step 3 apply: $h_{\text{total,batt}}/d_{\text{batt}} \in [0.8, 4.0]$. A minimum internal diameter of 400 mm is recommended for thermal battery tanks to ensure adequate element submersion and maintenance access.

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APPENDIX B

THERMOPHYSICAL PROPERTY DATASETS AND REFERENCE MATERIAL

This appendix consolidates all baseline thermodynamic matrices, geometric line profiles, and lifecycle safety tolerances referenced across the core design framework. The tabulated data blocks listed herein provide the explicit property pairs required to execute the first-order linear interpolation algorithms detailed in Section A.A. Sections B.A through B.D serve as the unified data bank for optimising calculations in Steps 2, 4, 5, 6, 8, and 9.

9 Appendix B

9.1 B.A—Aqueous and Industrial Transport Fluid Data Matrices

The following datasets represent the baseline thermodynamic properties of standard process and hydronic heat-transfer fluids. Designers must look up parameters at local bulk mean temperatures using first-order linear splines. All property values are evaluated at atmospheric pressure unless otherwise stated.

Pressurised systems. For pressurised closed-loop systems, minor corrections to density and specific heat may be applied per the fluid manufacturer’s pressurised property charts; however, for the operating envelopes covered by this framework, the atmospheric-baseline values introduce negligible error for sizing purposes.

9.1.1 Pure Water Base Fluid (Process/Utility Standard)

Table 14 indexes properties for distilled/purified water lines across atmospheric heating intervals, primarily utilised for process batch calculations and clean utility runs. Extended to 100°C at atmospheric pressure; pressurised systems may use water as a transport fluid beyond this limit with appropriate saturation-margin analysis.

Table 14. Thermophysical Properties of Pure Water (0°C to 100°C)

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
0	999.8	4217	1.787	1.787	0.561	13.44
20	998.2	4182	1.002	1.004	0.598	7.01
40	992.2	4179	0.653	0.658	0.631	4.32
60	983.2	4185	0.467	0.475	0.654	2.99
80	971.8	4196	0.354	0.364	0.670	2.22
100	958.4	4216	0.282	0.294	0.679	1.75

9.1.2 30% Propylene Glycol Aqueous Solution (Closed Loop Anti-Freeze Option)

Table 15 provides metrics for commercial-grade inhibited propylene glycol mixtures at 30% by volume. Freeze point approximately −13°C. This fluid profile is common in HVAC

and industrial process loops due to its low toxicity and GRAS (Generally Recognised As Safe) regulatory classification.

Table 15. Thermophysical Properties of 30% Propylene Glycol Solution by Volume

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
0	1042.0	3710	4.150	3.983	0.382	40.31
30	1035.0	3780	1.820	1.758	0.395	17.43
60	1029.0	3840	0.940	0.913	0.407	8.87
90	1025.0	3890	0.520	0.507	0.415	4.87
120	1014.0	3950	0.340	0.335	0.421	3.19

9.1.3 50% Propylene Glycol Aqueous Solution (Heavy Freeze-Protection Standard)

Table 16 provides the industry-standard 50% by volume concentration, delivering a freeze point of approximately -33°C . This is the most commonly specified concentration for outdoor or unheated plant installations in temperate and cold climates.

Cold-start hydraulics. The viscosity penalty at low temperatures is significant; cold-start hydraulic loading must be evaluated against pump curve data.

Table 16. Thermophysical Properties of 50% Propylene Glycol Solution by Volume

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
-20	1058.0	3570	22.400	21.172	0.358	223.2
0	1052.0	3620	8.600	8.175	0.366	85.0
20	1044.0	3670	3.800	3.640	0.374	37.3
40	1036.0	3730	2.050	1.979	0.381	20.1
60	1026.0	3790	1.210	1.179	0.387	11.8
80	1015.0	3850	0.780	0.769	0.391	7.7
100	1002.0	3910	0.530	0.529	0.394	5.3

9.1.4 40% Ethylene Glycol Aqueous Solution (High-Efficiency HVAC/Chilling Loop)

Table 17 catalogs industrial ethylene glycol parameters at 40% by volume, with a freeze point of approximately -24°C . Optimised for heavy-duty process chilling runs.

Toxicity. Ethylene glycol is toxic and requires appropriate secondary containment in food-adjacent environments.

9.1.5 50% Ethylene Glycol Aqueous Solution (Maximum Standard Freeze Protection)

Table 18 documents the 50% by volume ethylene glycol concentration, delivering a freeze point of approximately -37°C . This is the maximum concentration recommended for standard hydronic service; higher concentrations offer diminishing freeze-point depression with increasing viscosity penalty. Widely stocked as a pre-mixed inhibited product.

Table 17. Thermophysical Properties of 40% Ethylene Glycol Solution by Volume

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
−20	1078.0	3380	9.420	8.738	0.401	79.43
10	1063.0	3480	2.860	2.691	0.418	23.81
40	1049.0	3580	1.340	1.277	0.429	11.18
70	1033.0	3680	0.790	0.765	0.438	6.64
100	1016.0	3790	0.510	0.502	0.443	4.36

Table 18. Thermophysical Properties of 50% Ethylene Glycol Solution by Volume

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
−30	1089.0	3260	27.600	25.344	0.384	234.4
−10	1079.0	3310	7.400	6.858	0.393	62.4
10	1069.0	3360	3.280	3.068	0.401	27.5
30	1057.0	3420	1.870	1.769	0.408	15.7
60	1039.0	3510	0.980	0.943	0.416	8.3
90	1019.0	3610	0.590	0.579	0.421	5.1

9.1.6 Therminol 66 (Synthetic Organic High-Temperature Thermal Oil Standard)

Table 19 documents the behaviour of Therminol 66, a synthetic organic heat-transfer liquid manufactured by Eastman. Applied when hot-side temperatures must exceed 150°C to 345°C without generating extreme steam pressures.

Cold-start viscosity correction. Note the extreme Prandtl number at cold-start conditions (50°C): the cold-start viscosity correction for U_{cold} in Step 6 is mandatory for this fluid class.

Table 19. Thermophysical Properties of Therminol 66 Liquid Heat Transfer Fluid

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
50	981.0	1660	58.200	59.327	0.116	832.74
100	953.0	1845	6.800	7.135	0.112	111.96
150	923.0	2030	1.950	2.113	0.108	36.65
200	893.0	2215	0.880	0.985	0.104	18.74
250	861.0	2400	0.490	0.569	0.101	11.64
300	828.0	2585	0.310	0.374	0.097	8.26
345	795.0	2760	0.210	0.264	0.093	6.23

9.1.7 Therminol VP-1 / Dowtherm A (Diphenyl Oxide Eutectic, High-Performance Industrial Standard)

Table 20 covers Therminol VP-1 and its equivalent Dowtherm A (both are diphenyl/diphenyl oxide eutectic blends with near-identical property profiles). Operating range 12°C to 400°C. This is the primary benchmark fluid for solar thermal, concentrated chemical processing, and high-temperature reactor jacket applications.

Vapour pressure / food contact. Vapour pressure is non-negligible above 200°C; system pressurisation must be confirmed against saturation data. Do not use in food-contact environments.

Table 20. Thermophysical Properties of Therminol VP-1 / Dowtherm A

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	1060.0	1550	3.990	3.764	0.138	44.81
50	1039.0	1600	1.880	1.809	0.135	22.27
100	1004.0	1680	0.940	0.936	0.130	12.15
150	968.0	1760	0.560	0.579	0.125	7.88
200	930.0	1850	0.370	0.398	0.119	5.75
250	891.0	1950	0.260	0.292	0.113	4.49
300	849.0	2060	0.190	0.224	0.107	3.66
350	805.0	2180	0.145	0.180	0.101	3.13
400	757.0	2310	0.112	0.148	0.095	2.73

9.1.8 Caloria HT 43 (Mineral-Based Thermal Oil, Low-to-Medium Temperature)

Table 21 documents Caloria HT 43 (ExxonMobil), a highly refined mineral oil thermal fluid. Operating range -10°C to 170°C . Lower procurement cost than synthetic alternatives makes it attractive for budget-constrained installations with moderate temperature requirements.

Oxidation stability. Oxidation stability is inferior to synthetic fluids; fluid monitoring and periodic replacement schedules are mandatory above 140°C . Not food-grade.

Table 21. Thermophysical Properties of Caloria HT 43 Mineral Thermal Oil

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	869.0	1950	43.500	50.058	0.134	633.2
40	857.0	2000	18.200	21.237	0.131	278.0
60	845.0	2060	8.600	10.178	0.129	137.4
80	833.0	2120	4.500	5.402	0.127	75.1
100	820.0	2180	2.600	3.171	0.125	45.3
130	800.0	2270	1.250	1.563	0.122	23.3
160	780.0	2360	0.680	0.872	0.119	13.5

9.1.9 Mobiltherm 603 (Mineral-Based Thermal Oil, Standard Industrial Grade)

Table 22 covers Mobiltherm 603 (ExxonMobil Mobil), a widely distributed mineral thermal oil for continuous service up to 315°C . One of the most commonly stocked thermal oils globally, making it the practical default for installations where supply chain reliability is prioritised over peak thermal performance.

Fire safety. Flash point $\approx 180^{\circ}\text{C}$; fire safety classification must be confirmed against local plant codes.

Table 22. Thermophysical Properties of Mobiltherm 603 Thermal Oil

Temp. $T (^{\circ}\text{C})$	Density $\rho (\text{kg/m}^3)$	Sp. Heat $c_p (\text{J/kg}\cdot\text{K})$	Dyn. Viscosity $\mu (\times 10^{-3} \text{ Pa}\cdot\text{s})$	Kin. Viscosity $\nu (\times 10^{-6} \text{ m}^2/\text{s})$	Thermal Cond. $k (\text{W/m}\cdot\text{K})$	Prandtl Pr
40	868.0	1980	16.800	19.355	0.133	250.2
80	844.0	2080	4.750	5.629	0.129	76.6
120	820.0	2190	1.920	2.341	0.125	33.6
160	795.0	2300	0.970	1.220	0.121	18.4
200	769.0	2420	0.570	0.741	0.117	11.8
250	736.0	2570	0.330	0.448	0.112	7.6
300	702.0	2720	0.210	0.299	0.107	5.3

9.1.10 Paratherm NF (Food-Safe Synthetic Thermal Oil, NSF HX-1 Registered)

Table 23 documents Paratherm NF, a synthetic paraffinic thermal fluid carrying NSF HX-1 registration for incidental food contact. Operating range -20°C to 260°C . This is the preferred selection for food processing, pharmaceutical manufacturing, and confectionery applications where regulatory compliance requires a certified food-safe fluid but the operating temperature exceeds the practical range of glycol-water solutions.

Service profile. Thermally stable with low odour and low toxicity.

Table 23. Thermophysical Properties of Paratherm NF Food-Safe Thermal Oil

Temp. $T (^{\circ}\text{C})$	Density $\rho (\text{kg/m}^3)$	Sp. Heat $c_p (\text{J/kg}\cdot\text{K})$	Dyn. Viscosity $\mu (\times 10^{-3} \text{ Pa}\cdot\text{s})$	Kin. Viscosity $\nu (\times 10^{-6} \text{ m}^2/\text{s})$	Thermal Cond. $k (\text{W/m}\cdot\text{K})$	Prandtl Pr
20	840.0	2040	10.700	12.738	0.140	155.9
50	823.0	2100	4.100	4.983	0.137	62.8
80	806.0	2170	1.980	2.457	0.134	32.1
120	781.0	2270	0.880	1.127	0.130	15.4
160	756.0	2370	0.470	0.622	0.126	8.8
200	729.0	2470	0.290	0.398	0.122	5.9
240	701.0	2580	0.195	0.278	0.118	4.3

9.1.11 Syltherm 800 (Silicone-Based Thermal Fluid, Wide-Range Specialty)

Table 24 covers Syltherm 800 (Dow), a polydimethylsiloxane-based (silicone) thermal fluid rated for continuous service from -40°C to 260°C , with intermittent use to 315°C . Outstanding low-temperature fluidity (low pour point, moderate viscosity at -40°C) makes it the dominant selection for pharmaceutical freeze-dry jacket loops, semiconductor temperature conditioning, and wide-range process cycling.

Cost and compatibility. Higher cost than hydrocarbon alternatives; silicone fluid contamination is incompatible with certain downstream processes and coating operations.

Table 24. Thermophysical Properties of Syltherm 800 Silicone Thermal Fluid

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
−40	1006.0	1380	21.000	20.874	0.132	219.5
−20	990.0	1420	8.700	8.788	0.130	95.1
0	975.0	1460	4.200	4.308	0.128	47.9
40	943.0	1540	1.600	1.697	0.124	19.9
80	910.0	1620	0.800	0.879	0.120	10.8
120	877.0	1710	0.470	0.536	0.116	6.9
160	843.0	1800	0.310	0.368	0.112	5.0
200	808.0	1900	0.220	0.272	0.108	3.9
260	756.0	2050	0.145	0.192	0.102	2.9

9.1.12 50% Glycerin Aqueous Solution (Sanitary Food / Pharmaceutical Standard)

Table 25 provides metrics for refined aqueous glycerol loops at 50% by mass. This choice is common in sanitary process facilities due to its bio-safe, non-toxic profile under accidental product-contact failure scenarios.

Laminar flow risk. The notably high viscosity demands careful pump selection and hydraulic loop analysis; laminar flow conditions are likely at lower operating temperatures and must be evaluated against the Nusselt number correlations appropriate for the flow regime.

Table 25. Thermophysical Properties of 50% Glycerin Solution by Mass

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
10	1134.0	3220	7.950	7.011	0.415	61.68
30	1124.0	3300	3.820	3.399	0.428	29.45
50	1113.0	3390	2.180	1.959	0.439	16.84
70	1101.0	3480	1.390	1.262	0.446	10.85
90	1088.0	3570	0.960	0.882	0.450	7.62

9.1.13 Food-Grade Vegetable Thermal Oil (Confectionery / Edible Oil Processing)

Table 26 provides representative thermophysical data for refined food-grade vegetable oil (palm/sunflower/canola blend), applicable to confectionery, chocolate processing, edible oil refining, and dairy fat handling installations where direct product contact is a credible failure mode.

Data basis. Properties are presented for a representative refined palm-sunflower blend; actual values vary with specific oil composition and should be verified against the supplier's data sheet for the procurement batch.

Operating ceiling. Operating ceiling approximately 180°C; oxidation degradation accelerates above 150°C and fluid replacement intervals must be shortened accordingly.

Table 26. Thermophysical Properties of Food-Grade Vegetable Thermal Oil

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
40	910.0	1950	36.000	39.560	0.168	417.9
60	898.0	1990	16.200	18.040	0.166	194.4
80	886.0	2030	8.500	9.594	0.164	105.2
100	873.0	2080	4.900	5.613	0.162	62.9
120	860.0	2130	3.100	3.605	0.160	41.3
150	840.0	2200	1.700	2.024	0.157	23.8

9.1.14 D'Limonene-Based Thermal Fluid (Bio-Derived, Food-Adjacent Specialty)

Table 27 documents thermal fluid properties for high-purity d-limonene, a bio-derived citrus terpene increasingly specified in food-adjacent installations due to its natural origin, low toxicity, and pleasant odour profile. Operating range approximately 0°C to 175°C. While thermophysical performance is modest relative to synthetic thermal oils, regulatory acceptance in food-contact risk environments and lower environmental impact make it a commercially viable niche selection.

Fire classification. Flash point $\approx$ 49°C; fire classification is Class IC flammable liquid and strict hot-work permit controls apply.

Table 27. Thermophysical Properties of D'Limonene Bio-Derived Thermal Fluid

Temp. T (°C)	Density ρ (kg/m ³)	Sp. Heat c_p (J/kg·K)	Dyn. Viscosity μ ($\times 10^{-3}$ Pa·s)	Kin. Viscosity ν ($\times 10^{-6}$ m ² /s)	Thermal Cond. k (W/m·K)	Prandtl Pr
20	841.0	1860	0.920	1.094	0.143	11.97
40	828.0	1910	0.730	0.882	0.140	9.97
60	814.0	1960	0.590	0.725	0.138	8.38
80	800.0	2010	0.490	0.613	0.135	7.29
100	786.0	2060	0.410	0.522	0.133	6.35
140	756.0	2170	0.300	0.397	0.128	5.09

9.2 B.B—Solid Material and Insulation Thermal Properties

Conductive resistances (R_{wall} and $R_{insulation}$) across Steps 6 and 9 rely on core material transport metrics. Table 28 establishes wall conductivities, while Table 29 provides insulation property targets evaluated at 20°C atmospheric reference settings.

Table 28. Thermal Conductivities of Standard Structural Metal Alloys

Metallurgical Material Specification	Thermal Conductivity k (W/m·K)
Carbon Steel (AISI 1020 / Standard Line)	50.0
Stainless Steel 304 (Austenitic Standard)	16.2
Stainless Steel 316 (High-Corrosion Resist)	15.0
Nickel Alloy 625 (Inconel Core Variant)	9.8
Copper (Drawn Tube Standard C12200)	390.0
Titanium (Grade 2 Unalloyed Core)	21.9

Table 29. Thermal Properties of Industrial Shell Insulation Claddings

Insulation Material Type	Thermal Conductivity k (W/m·K)	Max Temp Rating
Rigid Polyurethane Foam (PUF)	0.024	120
Mineral Wool / Rockwool Board	0.038	650
Fiberglass Blanket Insulation	0.040	540
Expanded Polystyrene (EPS)	0.035	75
Calcium Silicate Structural Sheet	0.055	900

9.3 B.C—Industrial Design Boundary Allowances

To ensure the sized machinery retains performance margins across multi-year operational cycles, fluid friction and thermal performance elements must be penalised using surface degradation metrics.

9.3.1 Absolute Pipe Material Surface Roughness Values

Table 30 specifies the absolute physical surface roughness (ε) used as the core input seed parameter for the Colebrook-White iterative hydraulic head loss calculations (Step 5).

Table 30. Absolute Surface Roughness Parameters (ε)

Piping Material Specification Profile	Absolute Roughness ε (mm)
Drawn Plastic Piping (PVC / CPVC / PVDF)	0.0015
Drawn Copper Tubing (Clean Commercial)	0.0015
Commercial Stainless Steel Line (New/Clean)	0.0150
Commercial Carbon Steel / Wrought Iron	0.0450
Galvanized Iron / Zinc Coated Steel	0.1500
Heavily Corroded / Scaled Cast Iron Shell	1.5000

9.3.2 Standard Industrial Fluid Fouling Heat Transfer Resistances

Table 31 logs the standard thermal fouling resistance penalties (R_{f_in} , R_{f_out}). These terms apply localised heat-blocking penalties to the series thermal network calculation (Step 6) to model scaling and process deposition across the lifespan of the equipment.

Table 31. Standard Industrial Fouling Thermal Resistance Penalties

Fluid Circuit Operating Environment	Fouling Resistance R_f ($\text{m}^2 \cdot \text{K} / \text{W}$)
Distilled Water / Demineralised Loop Fluid	0.0001
Treated Boiler Feedwater (Closed Utility Run)	0.0002
Clean Inhibited Glycol Aqueous Solutions	0.0002
City Tap Water (Standard Municipal Intake)	0.0003
River Water / Untreated Cooling Tower Fluid	0.0005
Light Organic Process Fractions / Light Oils	0.0004
Synthetic Thermal Oils (Clean Service)	0.0002
Mineral Thermal Oils (Standard Service)	0.0003
Silicone Thermal Fluids (Clean Service)	0.0001
Heavy Crude Fractions / Slurry Process Fluids	0.0010

APPENDIX C

COMMERCIAL PIPING STANDARDS AND MATERIAL PROFILES

This appendix consolidates the mechanical, geometric, and thermomechanical data required to execute the Nominal Pipe Size (NPS) selection and structural material confirmation outlined in Step 4 of the framework. It includes standardised ASME dimensional boundaries for fluid flow cross-sections and comprehensive material evaluation matrices to ensure thermodynamic efficiency, structural integrity, and chemical compatibility across the operational lifespan of the hydronic system.

10 Appendix C

10.1 C.A—ASME B36.10M and B36.19M Pipe Dimensional Data

To safely constrain internal fluid velocities below the erosion ceiling ($v \leq v_{max}$), the theoretical minimum flow area calculated in Step 4 must be matched to a discrete, commercially available pipe profile. Table 32 lists standard dimensions for Welded and Seamless Wrought Steel (ASME B36.10M) and Stainless Steel (ASME B36.19M) pipe across typical industrial schedules.

Table 32. Standard Commercial Pipe Dimensions (NPS 1/2" to 8")

Nominal Size NPS (inches)	Outer Diameter d_{outer} (mm)	Schedule Designator	Wall Thickness t_{wall} (mm)	Inner Diameter d_{in} (mm)	Internal Flow Area A_{in} ($\times 10^{-3}$ m ²)
1/2"	21.34	Sch 10S	2.11	17.12	0.230
	21.34	Sch 40	2.77	15.80	0.196
	21.34	Sch 80	3.73	13.88	0.151
1"	33.40	Sch 10S	2.77	27.86	0.610
	33.40	Sch 40	3.38	26.64	0.557
	33.40	Sch 80	4.55	24.30	0.464
1-1/2"	48.26	Sch 10S	2.77	42.72	1.433
	48.26	Sch 40	3.68	40.90	1.314
	48.26	Sch 80	5.08	38.10	1.140
2"	60.33	Sch 10S	2.77	54.79	2.358
	60.33	Sch 40	3.91	52.51	2.165
	60.33	Sch 80	5.54	49.25	1.905
3"	88.90	Sch 10S	3.05	82.80	5.384
	88.90	Sch 40	5.49	77.92	4.768
	88.90	Sch 80	7.62	73.66	4.261
4"	114.30	Sch 10S	3.05	108.20	9.195
	114.30	Sch 40	6.02	102.26	8.213
	114.30	Sch 80	8.56	97.18	7.417
6"	168.28	Sch 10S	3.40	161.48	20.480
	168.28	Sch 40	7.11	154.06	18.641
	168.28	Sch 80	10.97	146.34	16.819
8"	219.08	Sch 10S	3.76	211.56	35.153
	219.08	Sch 40	8.18	202.72	32.276
	219.08	Sch 80	12.70	193.68	29.462

10.2 C.B—Thermomechanical Material Profiles

Selecting the correct material for the vessel shell and internal helical coil directly impacts the Overall Heat Transfer Coefficient (U) derived in Step 6, as well as the maximum structural limits of the system. Table 33 indexes the baseline mechanical and thermal properties of standard industrial materials.

Table 33. Thermomechanical Properties of Commercial Structural & Piping Materials

Material Specification (Common Designation)	Density ρ (kg/m ³)	Thermal Cond. k (W/m · K)	Yield Strength S_y (MPa)	Max Operating Temp (°C)	Relative Cost Factor
<i>Ferrous Alloys (Standard Vessel & Structural Shells)</i>					
Carbon Steel (AISI 1020)	7850	50.0	250	425	1.0x
Carbon Steel (A106 Grade B)	7850	48.0	240	400	1.1x
Cast Iron (Gray, Class 30)	7150	45.0	130	350	0.8x
<i>Stainless Alloys (Sanitary & Corrosive Environments)</i>					
Stainless Steel (304 / 304L)	8000	16.2	215	870	3.5x
Stainless Steel (316 / 316L)	8000	15.0	205	870	4.5x
Duplex Stainless (2205)	7800	19.0	450	300	6.0x
<i>Non-Ferrous & Specialty Metals (High Thermal Transfer)</i>					
Copper (Drawn, C12200)	8940	390.0	100	200	4.0x
Aluminum (6061-T6)	2700	167.0	275	150	2.0x
Titanium (Grade 2 Unalloyed)	4510	21.9	275	315	12.0x
Inconel (Alloy 625)	8440	9.8	460	980	15.0x
<i>Industrial Polymers (Low-Temp Chilling & Utility)</i>					
PVC (Polyvinyl Chloride)	1380	0.19	40	60	0.5x
CPVC (Chlorinated PVC)	1550	0.14	55	93	0.8x
PEX (Cross-linked Polyethylene)	940	0.40	15	95	0.7x
PTFE (Teflon / Lined Piping)	2200	0.25	25	260	8.0x

10.3 C.C—Material Compatibility and Selection Criteria Matrix

To prevent galvanic corrosion, chemical degradation, or thermal failure, the selected piping and vessel materials must be validated against the primary process fluid and the hydronic heat-transfer fluid. Table 34 provides a selection matrix mapping standard materials to their appropriate operational environments.

Reading the matrix. **EXCELLENT** denotes a standard industry application with a service lifespan greater than 15 years. **GOOD** indicates acceptable service subject to the noted condition. **POOR** indicates rapid degradation, galvanic potential, or unacceptable process contamination, and should be avoided without the qualifying mitigation shown.

Table 34. Material Suitability Matrix for Hydronic Process Fluids

Hydronic / Process Fluid	Carbon Steel	SS 304 / 316	Copper	CPVC / PEX
Pure Water (Deionized / RO)	POOR (Rusts)	EXCELLENT	GOOD	EXCELLENT
Treated Boiler Feedwater (Hot)	EXCELLENT	EXCELLENT	GOOD	POOR (Temp Limits)
Propylene / Ethylene Glycols	GOOD (w/ Inhibitor)	EXCELLENT	EXCELLENT	GOOD (Low Temp)
Therminol 66 (Thermal Oil)	EXCELLENT	EXCELLENT	POOR	UNSUITABLE
Glycerin Solutions (Sanitary)	POOR (Taints)	EXCELLENT	POOR	POOR
Brine / High-Chloride Chilling	POOR (Pitting)	POOR (Requires 316)	GOOD	EXCELLENT

10.4 C.D—Pipe Fittings and Connections

Undersizing risk. Each fitting installed within the hydronic circuit introduces a localised pressure drop that, if neglected, will cause the pump selection to be undersized and the actual flow velocity to fall below design intent. The hydraulic resistance characterisation performed in Step 5 of the framework requires that all fittings be accounted for through their equivalent pipe length contributions.

Table 35 provides standard equivalent length values (L_{eq}) for common wrought and forged fittings conforming to ASME B16.9 (butt-weld), ASME B16.11 (socket-weld and threaded), and ASME B16.5 (flanges). These values are expressed as a ratio to nominal pipe diameter (L_{eq}/d_{in}) and are valid for fully turbulent flow conditions consistent with coil operating Reynolds numbers.

Table 35. Equivalent Length Ratios for Standard Industrial Fittings (ASME B16.9 / B16.11)

Fitting Type	Connection	Equivalent Length Ratio L_{eq}/d_{in}		
	Standard	NPS $\leq 1"$	NPS 1.5"–3"	NPS 4"–8"
<i>Elbows</i>				
90° Standard Elbow	B16.9 BW	30	30	30
90° Long-Radius Elbow	B16.9 BW	16	16	16
45° Standard Elbow	B16.9 BW	16	16	16
<i>Tees</i>				
Tee, Flow-Through (Run)	B16.9 BW	20	20	20
Tee, Flow-Through (Branch)	B16.9 BW	60	60	60
<i>Reducers & Couplings</i>				
Concentric Reducer	B16.9 BW	5	5	5
Eccentric Reducer	B16.9 BW	5	5	5
Full-Coupling / Union	B16.11 SW	2	2	2
<i>Valves (Fully Open)</i>				
Gate Valve	B16.10	8	8	8
Globe Valve	B16.10	340	340	340
Ball Valve (Full Port)	B16.34	3	3	3
Butterfly Valve	B16.34	45	45	45
Check Valve (Swing)	B16.10	100	100	100
<i>Flanged Connections</i>				
Weld-Neck Flange (pair)	B16.5	2	2	2
Slip-On Flange (pair)	B16.5	2	2	2

Note: BW = Butt-Weld; SW = Socket-Weld. Values based on Crane TP-410 turbulent flow data.

For laminar flow ($Re < 2300$), apply the laminar correction factor $K_L = K \cdot (Re/1000)$.

10.5 C.E—Welding and Joining Standards

All pressure-retaining joints in the hydronic vessel and coil assembly must conform to a recognised fabrication code. The applicable standard is determined by the operating pressure class, material group, and service fluid. Table 36 maps common joint configurations to their governing ASME and AWS codes and specifies the minimum examination requirements that must be satisfied before the design verification gate in Step 11.

Sign-off prerequisite. Compliance with the examination requirements listed below is a prerequisite for the formal sign-off on the completed design package.

Table 36. Welding and Joining Standards for Hydronic Pressure Vessels and Coil Assemblies

Joint Application	Governing Code	Weld Process	Filler / Electrode	Elec.	Min. Examination Req.
Carbon Steel Vessel Shell (Butt Weld)	ASME Sec. IX / ASME VIII Div. 1	SMAW, GTAW, GMAW	E7018 / ER70S-6		Visual + RT or UT per UW-11
SS 304/316 Vessel Shell	ASME Sec. IX	GTAW (Root), GMAW (Fill)	ER308L / ER316L		Visual + 100% PT on root pass
Coil-to-Header Connection	ASME B31.3 (Process)	GTAW (orbital preferred)	ER316L / ER308L		Visual + 5% RT spot check
Coil Butt Joints (Same Material)	ASME B31.3	GTAW	Match metal	base	Visual + Hydrostatic proof test
Nozzle / Stub-In (Vessel Shell)	ASME VIII Div. 1 UW-16	SMAW / GTAW	Match metal	base	Visual + MT or PT (full)
Dissimilar Metal (CS to SS)	ASME Sec. IX / AWS D1.6	GTAW (butter layer)	ENiCrFe-3 (Inconel)	(In-	Visual + 100% RT
Flange Face (Slip-On / Weld-Neck)	ASME B16.5 / Sec. IX	SMAW / GTAW	Match metal	base	Visual + hydrostatic leak test
Low-Pressure Utility (< 150 psig)	AWS D1.1 (Structural)	SMAW / FCAW	E7018 / E71T-1		Visual per AWS D1.1 Table 6.1

RT = Radiographic Testing; UT = Ultrasonic Testing; PT = Liquid Penetrant Testing; MT = Magnetic Particle Testing.

All welders must hold current ASME Section IX or AWS QC1 qualification for the applicable process and material group.

10.6 C.F—Gasket and Seal Material Compatibility

Slow seepage risk. Selection errors at this stage are a common root cause of slow seepage failures that are not captured during short-duration hydrostatic proof tests. Flanged connections and inspection ports within the hydronic circuit require a gasket or seal whose material is independently verified against both the process fluid and the operating temperature range.

Table 37 maps standard gasket and elastomeric seal materials to their effective operational envelopes.

De-rating for thermal cycling. The temperature ratings listed represent continuous service limits; cyclic thermal shock conditions typical of batch processing duty should be assessed with a 15% de-rating applied to the upper temperature boundary.

All food-contact or sanitary applications must use FDA-compliant (21 CFR 177) grades of PTFE, EPDM, or Silicone.

Spiral-wound gaskets must be specified with inner and outer guide rings per ASME B16.20.

10.7 C.G—Insulation Materials and Thermal Properties

The environmental heat loss term Q_{loss} quantified in Step 5 of the framework is governed by the total thermal resistance of the vessel boundary layer, which includes the insulation jacket as the dominant resistance element. The selection of insulation type directly affects both the steady-state balance load and the payback period on the installed thermal capacity. Table 38 catalogues the thermal and mechanical properties of industrial insulation systems appropriate for hydronic vessel jacketing across the operating temperature range of the framework.

Mean-temperature values. The effective thermal conductivity values (k) listed are mean-temperature values; for precision loss calculations, the temperature-dependent $k(T)$ curve from the manufacturer's datasheet should be used to integrate across the wall temperature gradient.

For the VCH loss model, minimum recommended insulation thickness targets $R_{ins} \geq 1.0 \text{ m}^2 \cdot \text{K}/\text{W}$ for heated vessels.

Insulation systems in food-processing or pharmaceutical environments must comply with local hygiene codes (e.g. EHEDG, 3-A Sanitary Standards).

Table 37. Gasket and Seal Material Compatibility for Hydronic Process Fluids

Gasket / Seal Material	Max Temp. (°C)	Max Press. (bar)	Compatible Fluids	Incompatible / Caution
PTFE (Virgin / Expanded)	260	70	Water, glycols, glycerin, acids, thermal oils, brine	Strong oxidisers; PTFE cold-flows under bolt load — use spiral-wound type above 40 bar
Spiral-Wound SS/PTFE	450	250	Steam, hot water, glycols, thermal oils	Not for cryogenic; requires flat-face flange
Nitrile (NBR)	120	35	Water, glycols, mineral oils, brine	Aromatic hydrocarbons; thermal oils (Therminol) cause swelling
EPDM (Ethylene Propylene)	150	20	Hot water, steam (low press.), glycols, dilute acids	Petroleum-based oils and solvents; Therminol series
Viton (FKM Fluoroelastomer)	200	35	Thermal oils, glycols, brine, mild acids, hot water	Steam above 150°C; ketones; low-molecular-weight esters
Graphite (Flexible Foil)	450	200	Hot water, steam, thermal oils, glycols, most process fluids	Strong oxidising acids (e.g. HNO ₃); not food-grade without encapsulation
Neoprene (CR)	100	20	Water, glycols, brine, mild alkalis	Aromatic solvents; thermal oils; strong oxidisers
Silicone (VMQ)	230	15	Hot water (low press.), steam (low press.), food-grade glycerin	Petroleum oils; acids; steam above 150°C

10.8 C.H—Coil Fabrication Tolerances and Bend Radius Data

The coil geometry optimisation performed in Step 7 produces a theoretical helix diameter, coil pitch, and number of turns. These ideal dimensions must be mapped to physically realisable fabrication parameters before the coil can be ordered or manufactured. Two constraints govern this translation: the minimum bend radius that a given pipe size and material can sustain without wall thinning or ovality beyond acceptable limits, and the compound packing efficiency factor η_{pack} that accounts for the systematic loss of usable coil turns introduced by fabrication tolerances, structural supports, weld access clearances, thermal expansion gaps, and dead-zone prevention offsets at the top and bottom of the vessel.

Table 39 provides the minimum centreline bend radius (R_{min}) expressed as a multiple of nominal outer diameter (D) for mandrel bending of seamless and welded tube, consistent

Table 38. Thermal and Mechanical Properties of Industrial Vessel Insulation Materials

Insulation Material	Thermal Cond.	Density	Service Temp.	Relative	Typical Application
	k (W/m·K)	ρ (kg/m ³)	Range (°C)	Cost	
<i>Rigid Foam Systems</i>					
Polyurethane Foam (PUR)	0.022–0.028	30–60	–50 to +120	1.0x	Cold tanks, chilling vessels
Polyisocyanurate (PIR)	0.022–0.026	32–48	–50 to +150	1.2x	Heated vessels, pipework
Extruded Polystyrene (XPS)	0.030–0.038	25–45	–50 to +75	0.7x	Low-temp utility; moisture-resistant
<i>Mineral Fibre Systems</i>					
Glass Wool (Wired Mat)	0.032–0.044	10–48	Up to +450	0.6x	General industrial pipework
Rock Wool (Stone Wool)	0.033–0.045	40–200	Up to +750	0.8x	High-temp vessels and steam lines
Calcium Silicate Board	0.055–0.085	190–300	Up to +1000	1.5x	Very high-temp, structural duty
<i>Flexible & Specialty Systems</i>					
Elastomeric Foam (Armaflex)	0.033–0.040	60–80	–50 to +105	1.8x	Cold pipework; self-sealing joints
Aerogel Blanket	0.013–0.018	100–200	–200 to +650	8.0x	Space-constrained high-value duty
Cellular Glass (Foamglas)	0.038–0.055	100–130	–265 to +430	3.5x	Cryogenic and high-humidity service
<i>Cladding and Jacketing (Outer Weather Barrier)</i>					
Aluminium Sheet (0.6–1.0 mm)	N/A	2700	–200 to +200	1.0x	Standard outdoor and indoor jacketing
Stainless Sheet (0.5–0.8 mm)	N/A	8000	All ranges	2.5x	Corrosive or sanitary environments
PVC Jacketing (Vapour Barrier)	N/A	1380	Up to +60	0.5x	Indoor low-temp chilling only

with ASME B16.49 (factory bends) and EN 13480-4 (industrial piping). Table 40 quantifies the individual tolerance deductions that compound into the η_{pack} correction applied in Step 7.

Radii below these limits risk wall thinning > 12.5% or ovality > 8% per ASME B16.49 acceptance criteria. For tight-pitch helical coils wound on a mandrel, confirm that the helix diameter $D_{helix} \geq 2 \times R_{min}$.

η_{pack} is applied as: $N_{active} = \lfloor N_{theoretical} \times \eta_{pack} \rfloor$, where $N_{theoretical}$ is derived from the Step 7 optimisation.

For large-diameter vessels ($ID > 1.5$ m) with multiple coil banks, each bank should carry an independent η_{pack} assessment.

Table 39. Minimum Centreline Bend Radius for Helical Coil Fabrication

Pipe Material	Fabrication Method	Min. Centreline Radius R_{min} (multiples of D_{outer})			
		NPS 1/2"–1"	NPS 1.5"–2"	NPS 3"–4"	NPS 6"–8"
Carbon Steel (Seamless)	Cold Mandrel Bend	1.5 D	2.0 D	3.0 D	5.0 D
Carbon Steel (Seamless)	Hot Induction Bend	1.5 D	1.5 D	2.0 D	3.0 D
Stainless 304/316 (Seamless)	Cold Mandrel Bend	2.0 D	2.5 D	3.5 D	5.0 D
Stainless 304/316 (Seamless)	Hot Induction Bend	1.5 D	2.0 D	2.5 D	3.5 D
Copper (Drawn, C12200)	Cold Spring Bending	1.5 D	2.0 D	3.0 D	N/A
Titanium Grade 2	Cold Mandrel Bend	3.0 D	3.5 D	5.0 D	N/A

Table 40. Coil Packing Efficiency Deduction Factors Comprising η_{pack}

Deduction Source		Physical Basis	Typical Deduction	Notes
Structural Support Legs		Radial supports interrupt coil continuity at ≥ 3 points	2–4 turns	Fixed loss per coil
Weld Access Clearance (Top)		Minimum 50–75 mm clearance from vessel top head to first active turn	1–2 turns	Increases with NPS
Weld Access Clearance (Bottom)		Minimum 50–75 mm clearance from vessel base to last active turn	1–2 turns	Increases with NPS
Thermal Expansion Gap		Axial growth of helical coil at $\Delta T_{operating}$	0.5–1 turn	Per 10 m coil length
Dead-Zone Prevention (Top)		Stagnant fluid zone above coil top must be minimised	1 turn	Heating applications
Dead-Zone Prevention (Bottom)		Stagnant fluid zone below coil bottom must be minimised	1 turn	Cooling applications
Coil Pitch Rounding		Discrete turn count rounded down to nearest integer	0–1 turn	Deterministic
Typical Composite η_{pack} Range			0.80 – 0.92	

APPENDIX D DISTRIBUTED PROCESS AUTOMATION AND CONTROL ARCHITECTURE

VERSION 1 — ARCHIVED

11 Appendix D (Archived)

Architectural Supersession Notice. This version of Appendix D (Version 1) is retained in full as an archived record of the initial single-loop control architecture developed during the first design iteration of the VCH Sizing Framework. It is preserved deliberately to document the evolution of the control philosophy and to demonstrate the engineering decision trail that led to the revised architecture.

This appendix has been superseded in its entirety by **Appendix F — Revised Control Architecture (Version 2)**, which should be consulted for all current design, commissioning, and implementation purposes. The following sections of this document contain content that conflicts with the Version 2 architecture and must not be used as implementation references:

- **Section D.A** — The four operational modes (Mode I through Mode IV) have been replaced by a revised three-loop mode structure (1.A.1, 1.A.2, B, C) in Appendix F, Section F.A. The Mode IV multi-vessel balancing logic has been retained in a modified form.
- **Section D.B** — The instrumentation table is incomplete relative to the Version 2 I/O schedule. Instruments TE-104 and TE-105 (battery inlet and outlet temperature) are absent. The full 25-point I/O schedule is defined in Appendix F, Section F.B.
- **Section D.C** — The PLC state machine reflects the Version 1 mode structure and does not map to the Version 2 operational states. The updated state transition matrix is provided in Appendix F, Section F.C.
- **Section D.D** — The single master-slave cascade topology described herein has been replaced by three independent control loops with defined controller types (PID, PI, P). Refer to Appendix F, Section F.D.

Sections **D.E through D.H** (HMI/SCADA alarm philosophy, PID tuning parameters, network architecture, and FMEA) remain substantively valid and have been carried forward into Appendix F with targeted amendments to reflect the revised I/O schedule and mode structure. Where discrepancies exist between D.E–D.H and their Appendix F counterparts, the Appendix F version takes precedence.

Superseded by: Appendix F — Revised Control Architecture (Version 2)

Retention

rationale: Engineering decision record / portfolio documentation

Section Conflict — D.A. The four-mode structure defined below (Modes I–IV) has been superseded. Refer to Appendix F, Section F.A for the current operational mode architecture. Do not implement from this section.

11.1 D.A Control Philosophy and Multi-Vessel Operational Modes

To maintain fluid velocity beneath the erosion boundary ($v \leq v_{max}$) while fulfilling transient thermal demands during a batch cycle, the system alternates between four primary operational configurations via automated modulating three-way valves (TCVs) and variable-frequency drive (VFD) circulation loops:

1. **Mode I: Baseline Utility Supply (Direct Charging):** Activated when the total process load is below the steady-state boiler capacity ($Q_{load} \leq Q_{steady}$). The thermal battery is structurally isolated, and the utility line directly handles vessel demands using standard feedback proportional-integral-derivative (PID) tuning.
2. **Mode II: Peak Shaving / Battery-Assist Discharge:** Prompted automatically when a new process batch initiates and thermal demand spikes past utility capacity ($Q_{load} > Q_{steady}$). The system activates a feed-forward control block, modulating a secondary three-way valve to blend high-temperature fluid from the thermal battery into the loop, maintaining a fixed batch heating rate without overloading the utility.
3. **Mode III: Isolated Thermal Battery Recharge:** Triggered during process downtime or low-load intervals. Excess utility energy ($Q_{steady} - Q_{load}$) is diverted into the thermal battery vessel, raising its core internal energy state until upper thermal cutoff boundaries are reached.
4. **Mode IV: Multi-Vessel Load Balancing Mode:** Utilizing a distributed cascade architecture, the master Programmable Logic Controller (PLC) monitors real-time temperature differentials ($\Delta T = T_{in} - T_{out}$) across multiple independent vessels. Flow distribution is dynamically scaled using a prioritize-by-deviation algorithm, shifting hydronic volume to vessels exhibiting the steepest temperature drop.

Section Conflict — D.B. The instrumentation table below is incomplete. Instruments TE-104 and TE-105 are absent. The full 25-point I/O schedule supersedes this table. Refer to Appendix F, Section F.B.

11.2 D.B Instrumentation, Sensors, and Actuator Field Profiles

Executing the velocity constraints and energy tracking requirements defined across the core framework requires a synchronized field instrument network. Table 41 catalogs the sensor characteristics, signal transmission standards, and functional roles mandated within the hydronic loop.

Table 41. Field Instrumentation and Signal Mapping Specifications (Version 1 — Incomplete)

Tag ID	Instrument Description	Sensor Type / Element	Signal Protocol	Functional Framework Role
TE-101	Loop Supply Temperature	PT100 RTD (Class A)	4-20 mA / HART	Monitors main utility output profile
TE-102	Battery Core Temperature	Multipoint Thermocouple	4-20 mA / HART	Computes current state-of-charge (SoC)
TE-103	Vessel Batch Temperature	PT100 RTD (Sanitary)	4-20 mA / HART	Feeds the Step 10 transient integration loop
FE-201	Main Line Velocity Sensor	Electromagnetic Flow	Modbus TCP	Validates velocity constraint ($v \leq 2.0$ m/s)
PE-301	Loop Differential Press.	Piezo-Resistive Cell	4-20 mA / HART	Tracks real-time helical head loss (ΔH_f)
TCV-401	Battery Blending Actuator	3-Way Equal Percentage	4-20 mA Loop	Modulates thermal battery discharge volume
VFD-501	Loop Circulation Pump	AC Induction Inverter	Profinet	Alters pump RPM to maintain hydraulic balance

Section Conflict — D.C. The state machine below maps to Version 1 operational modes only and does not reflect the revised mode structure or three-loop architecture. Do not use as an implementation reference. Refer to Appendix F, Section F.C.

11.3 D.C Programmable Logic Controller (PLC) State Transition Matrix

The system logic is implemented within a deterministic state machine to guarantee predictable transitions during sudden multi-vessel load spikes or unexpected system trips. Table 42 provides the logical state matrix used to construct the PLC ladder logic or structured text routines.

Section Conflict — D.D. The single master-slave cascade topology described below has been replaced by three independent control loops. The architectural rationale for this change is documented in Appendix F, Section F.D. Do not implement from this section.

11.4 D.D Cascaded Control Loop Proportional-Integral-Derivative (PID) Architecture

To prevent hydraulic hunting when multiple control valves modulate simultaneously, a cascaded control topology is established. The primary master controller monitors the process vessel temperature error ($e(t) = T_{set} - T_{vessel}$) and continuously computes a target hydronic flow rate requirement.

The slave controller receives this flow setpoint and modulates pump speed via VFD-501, referencing electromagnetic flowmeter FE-201 as its direct feedback element. By nesting the velocity loop inside the slower thermal loop, the slave loop can intercept instantaneous pressure fluctuations across the helical network, smoothing out actuator actions and eliminating the risk of accidental erosion velocity overshoots in the system.

Table 42. PLC State Machine Transition Sequence and Safety Interlocks (Version 1 — Superseded)

Current State	System Profile	Primary Trigger Input	Trigger	Target State	Output	Mandatory Interlock / Safety Guard
01_STANDBY	System idling; pump operating at minimum bypass speed.	Batch initiation command received from HMI terminal.		02_UTILITY_ONLY		Confirm loop pressure PE-301 is within stable static bounds (> 1.5 bar).
02_UTILITY_	Boiler satisfies loop demand; battery remains isolated.	Process load exceeds utility limit: $Q_{load} > 80$ kW.		03_PEAK_ASSIST		Continuous loop velocity tracking; if $v > 2.0$ m/s, trim VFD-501 immediately.
03_PEAK_ASS	TCV-401 modulates open to inject battery energy.	Batch cycle completes; thermal load drops below threshold.		04_RECHARGE		If battery core temperature drops below process target, isolate battery and flag alarm.
04_RECHARGE	Excess utility energy routed into storage matrix.	Battery multipoint sensor TE-102 hits upper ceiling.		01_STANDBY		Disable recharge loop instantly if vessel high-pressure switch trips.
05_FAIL_SAF	All control valves forced to unpowered safety states.	Fluid velocity limit breach ($v > 2.5$ m/s) or hardware fault.		Safe Shutdown (Lockout)		Bypass all process commands; trigger audible field alarm beacon; dump VFD run signals.

Note — Sections D.E through D.H. The following sections are substantively valid and have been carried forward into Appendix F with targeted amendments. Where instrument tags referenced below (e.g. alarm setpoints for TE-101–TE-103) differ from the expanded Version 2 I/O schedule, the Appendix F versions take precedence. These sections are not flagged as conflicting and may be read as background reference.

11.5 D.E HMI / SCADA Screen Layout and Alarm Philosophy

The human-machine interface (HMI) and supervisory control and data acquisition (SCADA) layer serves as the primary operator window into the hydronic network state. Screen organisation follows ISA-101 (HMI Design and Implementation) principles: each display level presents only the information operationally relevant at that level of abstraction, preventing alarm flooding during transient batch events. The hierarchy comprises three display tiers.

Level 1 — System Overview: A single-screen mimic diagram showing all four oper-

ational modes (Section D.A), the real-time state of TCV-401 and VFD-501, live readings from TE-101, TE-102, TE-103, FE-201, and PE-301, the thermal battery state-of-charge (SoC) bar expressed as a percentage of $Q_{battery,max}$, and the current PLC state tag. No control actions are executable from this screen; it is display-only to prevent inadvertent mode changes.

Level 2 — Loop Detail Screens: One screen per controlled loop. Each screen displays the trend history for the relevant temperature, flow, and pressure tags over a configurable 30-minute rolling window, the active setpoint, PV, and output of the governing PID block, and the current interlock status for that loop.

Level 3 — Maintenance and Calibration: Password-protected. Provides manual override capability for individual actuators, PID parameter adjustment, sensor zero/span calibration records, and accumulated run-hour logs for VFD-501.

Table 43. Alarm Setpoint and Priority Classification Matrix

Tag	Alarm Description	Lo / LoLo Limit	Hi / HiHi Limit	Priority	Required Operator Action
TE-101	Supply temperature out of band	Lo: $T_{set} - 5^{\circ}\text{C}$ LoLo: $T_{set} - 15^{\circ}\text{C}$	Hi: $T_{set} + 10^{\circ}\text{C}$ HiHi: $T_{set} + 25^{\circ}\text{C}$	P2 / P1	Investigate utility source; P1 triggers Mode IV bypass
TE-102	Battery core temperature deviation	Lo: $T_{bat,min}^{\circ}\text{C}$ LoLo: $T_{bat,min} - 10^{\circ}\text{C}$	Hi: $T_{bat,max}^{\circ}\text{C}$ HiHi: $T_{bat,max} + 5^{\circ}\text{C}$	P2 / P1	P1 forces immediate recharge isolation; flag for inspection
TE-103	Vessel batch temperature deviation	Lo: $T_{proc} - 3^{\circ}\text{C}$	Hi: $T_{proc} + 5^{\circ}\text{C}$ HiHi: $T_{proc} + 12^{\circ}\text{C}$	P3 / P2	P2 triggers setpoint review and valve trim audit
FE-201	Loop flow / velocity breach	Lo: 0.3 m/s	Hi: 2.0 m/s HiHi: 2.5 m/s	P2 / P1	P1 trips VFD-501 to minimum; latches O5_FAIL_SAFE
PE-301	Differential pressure loss	LoLo: 0.5 bar	HiHi: 6.0 bar	P1	Isolate loop; inspect coil for fouling or mechanical failure
VFD-501	Drive fault or overtemperature	—	Any drive fault code	P1	Acknowledge; switch to manual bypass pump if installed
TCV-401	Valve position feedback mismatch	Commanded – Actual > 5% for > 10 s		P2	Check pneumatic/electric actuator supply and positioner

Deadband of $\pm 0.5^{\circ}\text{C}$ applied to all temperature alarms to suppress chattering.

HiHi and LoLo thresholds trigger automatic PLC state transitions; Hi and Lo announce only. Rows shaded amber are Priority 1 (P1) — the highest annunciation tier.

11.6 D.F PID Tuning Parameters and Recommended Gain Values

The cascaded control topology described in Section D.D requires independently tuned PID blocks for each loop layer. All PID blocks are implemented in the positional (non-velocity) form with output clamping and anti-windup via integrator back-calculation:

$$u(t) = K_p \left[e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right] \quad (236)$$

Derivative action is applied to the process variable only to prevent derivative kick on setpoint steps.

Table 44. Recommended PID Tuning Parameters by Control Loop (Version 1)

Loop Name		Controlled Variable		Actuator	K_p	T_i (s)	T_d (s)	Notes
Master (Outer)	Thermal	Vessel temperature	batch TE-103	TCV-401 position setpoint	1.8–2.5	180–300	15–30	Slow integrating process; low K_p prevents overshoot on batch start
Slave (Inner)	Hydraulic	Loop flow	velocity FE-201	VFD-501 speed (%)	0.8–1.4	8–20	1–3	Fast loop; T_d active to suppress pressure transient spikes
Battery charge	Dis-	Battery ratio	blend via TCV-401	TCV-401 direct	2.0–3.0	60–120	0	Feed-forward dominant; integral only for steady-state trim
Recharge Loop		Battery temperature	core TE-102	Recharge divert valve	1.2–1.8	240–480	0	Very slow thermal mass; derivative omitted
Multi-Vessel Balancing	Bal-	ΔT per vessel	deviation	Per-vessel VFD trim	0.5–1.0	30–60	0	Proportional-integral only; gains intentionally low

Output limits: Master loop 0–100% valve position; Slave loop 20–100% VFD speed.

11.7 D.G Network Architecture and Communication Protocols

The field instrument and actuator network is organised as a three-layer communication hierarchy. Physical isolation between layers is mandatory to prevent process-level noise from propagating into the supervisory layer and to maintain deterministic scan times on the PLC execution bus.

Layer 1 — Field Bus (Process Level): All analogue field instruments communicate over the 4–20 mA analogue loop with superimposed HART digital overlay. Shielded twisted-pair (STP) cable with drain wire grounded at the marshalling cabinet end only is mandatory for all HART runs.

Layer 2 — Control Bus (PLC Level): FE-201 and VFD-501 communicate over Profinet IO (IEC 61158 / IEC 61784), providing cycle times of 1–4 ms. The master PLC also hosts a Modbus TCP server for inter-PLC data exchange in multi-vessel balancing configurations.

Layer 3 — Supervisory Bus (SCADA Level): The HMI/SCADA server communicates with the master PLC via OPC-UA (IEC 62541) over a dedicated VLAN. SCADA historian sampling is configured at 1-second resolution for all analogue tags.

Table 45. Communication Protocol and Network Layer Specifications

Layer	Protocol	Physical Medium	Topology	Cycle / Poll Rate	Connected Devices
Layer 1	4–20 mA + HART (rev 7)	STP 2-core, 0.5 mm ²	Point-to-point star	Analogue continuous	TE-101/102/103, PE-301, TCV-401
Layer 2	Profinet IO (RT Class 1)	Cat6A STP Ethernet	Line / star hybrid	1–4 ms	FE-201, VFD-501
Layer 2	Modbus TCP	Cat6A STP Ethernet	Star from master PLC	500 ms poll	Satellite vessel PLCs
Layer 3	OPC-UA (IEC 62541)	Cat6A, dedicated VLAN	Star from SCADA server	1 s historian	HMI terminals, historian, engineering workstation

All Ethernet segments must use managed switches with VLAN tagging.

IEC 62443-3-3 Security Level 2 recommended for OPC-UA endpoints.

11.8 D.H Failure Mode and Effects Analysis (FMEA) and Fault Response

A structured failure mode and effects analysis is required to validate that the PLC state machine captures all credible fault conditions. The FMEA is structured per IEC 60812. Each failure mode is assigned a Risk Priority Number (RPN) computed as the product of Severity (S), Occurrence (O), and Detectability (D), each scored 1–10.

Hardware interlock threshold. Modes with $RPN \geq 100$ require a dedicated hardware interlock independent of the PLC software layer. The single qualifying row in Table 46 is shaded amber below.

$RPN = S \times O \times D$. All $RPN \geq 100$ items require hardware interlock independent of PLC software.

Table 46. Failure Mode and Effects Analysis (FMEA) — Version 1

Failure Mode	Root Cause	Effect on System	<i>S</i>	<i>O</i>	<i>D</i>	RPI	Recommended Mitigation
FE-201 signal loss	Cable fault or transmitter failure	Inner PID loses flow feedback; VFD runs open-loop	8	3	4	96	Watchdog timer; fallback to minimum VFD speed on signal loss
TCV-401 stuck open	Actuator spring or positioner failure	Uncontrolled battery discharge; vessel overtemperature	9	2	3	54	Mechanical position indicator; TE-103 HiHi trips battery isolation
TCV-401 stuck closed	Positioner or pneumatic supply failure	Battery cannot discharge; thermal demand unmet	5	2	3	30	TE-103 Lo alarm; manual bypass line with block valve
VFD-501 trip / fault	Overtemperature, overcurrent, or supply fault	Loss of loop circulation; coil stagnation risk	9	3	2	54	VFD fault output hardwired to PLC DI; auto-transition to 05_FAIL_SAFE
Loop over-velocity event	VFD run-away or control loop instability	Erosion of coil inner wall	10	2	2	40	Hardware overspeed relay independent of PLC; trips VFD contactor
TE-102 multi-point failure	Thermocouple open-circuit or moisture ingress	SoC calculation invalid; risk of battery over-charge	7	4	4	112	Critical (RPN ≥ 100): Redundant thermocouple pair; majority-vote logic
PLC processor fault	Hardware fault or watchdog timeout	All automated control outputs frozen	10	1	2	20	Hot-standby PLC in duplex configuration
Modbus TCP link loss	Network switch fault or cable break	Master loses satellite vessel data	6	3	3	54	Satellite PLCs revert to local standalone PID on comms timeout
Loop pressure loss	Coil or pipe leak	Risk of fluid escape and thermal hazard	9	2	2	36	PE-301 LoLo hardwired to master isolation valve
SCADA OPC-UA server loss	Server crash or network VLAN fault	Loss of operator visibility	4	3	2	24	PLC operates independently; local HMI panel maintains Level 1 display

APPENDIX E PLC IMPLEMENTATION AND DISCRETE-TIME CONTROL FORMULATIONS

VERSION 1 — ARCHIVED

12 Appendix E (Archived)

Implementation Supersession Notice. This version of Appendix E (Version 1) is retained in full as an archived record of the initial single-loop PLC implementation developed alongside the Version 1 control architecture of Appendix D. It is preserved to document the engineering decision trail and to serve as a reference for the mathematical foundations that carry forward into the revised implementation.

This appendix has been superseded in its entirety by **Appendix G — Revised PLC Implementation and Control Commissioning (Version 2)**, which should be consulted for all current implementation and commissioning purposes. The following sections contain content that conflicts with or is incomplete relative to the Version 2 three-loop architecture:

- **Section E.D** — The scan interval derivation assumes a single universal execution interval applied to one loop. The Version 2 architecture requires four independent scan intervals derived separately for each loop. The per-loop derivation is provided in Appendix G, Section G.B. Do not implement from this section.
- **Section E.E** — The single ST code block covers one generic PID loop only. The Version 2 architecture requires three separate code blocks — one per controlled loop — each with the correct controller type (PID, PI, or P), the correct actuator tag, and the correct scan interval. The three blocks are provided in Appendix G, Section G.E. Do not implement from this section.

Sections **E.A through E.C** (discrete-time PID algorithms, derivative kick mitigation, anti-windup clamping, and EMA sensor filtering) are mathematically universal and have been carried forward into Appendix G, Section G.A without change. Sections **E.F and E.G** (ladder logic guidelines and hardwired relay fallback) are carried forward into Appendix G, Sections G.G and G.H respectively, with minor amendments to reflect the three-loop structure. Section **E.H** (watchdog and fail-safe) is carried forward into Appendix G, Section G.I with one amendment: TCV-402 is added to the safe-state output enforcement block, which was absent in Version 1 because TCV-402 did not exist in the Version 1 architecture. Where any discrepancy exists between E.A–E.C, E.F–E.H and their Appendix G counterparts, the Appendix G version takes precedence.

Superseded by: Appendix G — Revised PLC Implementation and Control Commissioning

(Version 2) Retention rationale: Engineering decision record / mathematical reference

This appendix translates the continuous-time thermal logic established in the sizing framework into the discrete-time difference equations required by industrial microprocessors. The formulations are specifically formatted for Programmable Logic Controllers (PLCs) and Distributed Control Systems (DCS), ensuring safe actuator modulation, electrical noise rejection, and system stability under massive thermal inertia.

Note — Sections E.A through E.C. The following three sections are mathematically universal and have been carried forward into Appendix G, Section G.A without change. They may be read as the authoritative mathematical reference for the discrete-time algorithms underlying all three Version 2 control loops. No conflict with the Version 2 architecture exists in these sections.

12.1 E.A Discrete-Time Proportional-Integral-Derivative (PID) Algorithms

The theoretical continuous-time PID algorithm is defined as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}.$$

Because a PLC samples data at fixed, discrete scan intervals (Δt), this continuous integral is converted algebraically using backward Euler approximations. For hydronic valves and Variable Frequency Drives (VFDs), which require bumpless transfers and smooth actuation, the **Velocity (Incremental) PID Form** is employed. Instead of calculating the absolute output $u(k)$, the PLC calculates the incremental change required for the current scan cycle $\Delta u(k)$:

$$\Delta u(k) = K_p [e(k) - e(k-1)] + K_i \Delta t e(k) + \frac{K_d}{\Delta t} [e(k) - 2e(k-1) + e(k-2)].$$

The final actuator signal sent to field hardware is then accumulated:

$$u(k) = u(k-1) + \Delta u(k).$$

12.2 E.B Derivative Kick Mitigation and Integral Anti-Windup Clamping

Derivative Kick Avoidance. If a plant operator suddenly alters the process setpoint (SP) via the HMI, the error $e(k)$ shifts instantaneously, triggering a large spike in the standard derivative term and forcing the VFD abruptly to maximum frequency. To prevent this, the algorithm is modified to take the derivative of the Process Variable (PV) rather than the error:

$$\text{Derivative Term} = -\frac{K_d}{\Delta t} [PV(k) - 2PV(k-1) + PV(k-2)].$$

Integral Anti-Windup (Conditional Clamping). Large thermal batch processes possess massive inertia, causing the integral term to accumulate indefinitely while the fluid

heats up. To prevent the loop from winding up and overshooting the target, the PLC logic freezes the *incremental* integral contribution $\Delta I(k)$ for the current scan cycle whenever the actuator output has reached its physical saturation boundary. The accumulated output $u(k)$ is not zeroed; only the increment for that cycle is suppressed:

$$\begin{aligned} \text{If } u(k) \geq u_{\max} \text{ and } e(k) > 0: \quad \Delta I(k) &= 0, \\ \text{If } u(k) \leq u_{\min} \text{ and } e(k) < 0: \quad \Delta I(k) &= 0. \end{aligned}$$

This formulation is consistent with the velocity-form ST implementation in Section E.E, where `I_Term` represents the per-scan integral increment $K_i \cdot \Delta t \cdot e(k)$, not the total accumulated sum.

12.3 E.C First-Order Low-Pass Sensor Filtering (EMA)

Industrial fluid environments induce turbulent noise across RTDs (TE-101) and electromagnetic flowmeters (FE-201). Before entering the PID calculation block, raw analog signals are smoothed using a discrete Exponential Moving Average (EMA) low-pass filter:

$$PV_{\text{filtered}}(k) = \alpha \cdot PV_{\text{raw}}(k) + (1 - \alpha) \cdot PV_{\text{filtered}}(k - 1),$$

where the smoothing constant α is derived from the scan rate (Δt) and the desired filter time constant (τ_f):

$$\alpha = \frac{\Delta t}{\tau_f + \Delta t}, \quad 0 < \alpha \leq 1.$$

Section Conflict — E.D. The scan interval derivation below assumes a single universal execution interval applied to one loop, derived from a single reference S_t value. The Version 2 architecture requires four independent scan intervals derived separately for each loop because their time constants differ by orders of magnitude. Applying this single interval to Loops 2, 3, or 4 will produce an unstable hydraulic controller or wasteful CPU oversampling. Do not implement from this section. The per-loop derivation is provided in Appendix G, Section G.B.

12.4 E.D PLC Execution Interval and Hardware Scan Strategy

Executing complex floating-point PID arithmetic on every millisecond PLC hardware cycle needlessly consumes CPU bandwidth and amplifies derivative noise. The control task execution interval (Δt_{scan}) is locked to the physical time constant (S_t) of the thermal battery, validated in Step 11 of the framework. Based on the Nyquist-Shannon constraint for thermal inertia, the logic block is executed at:

$$\Delta t_{\text{scan}} \approx \frac{S_t}{20}.$$

As a representative reference: for a system where Step 10 yields $S_t = 6665 \text{ s}$, this gives $\Delta t_{\text{scan}} \approx 330 \text{ s}$ (5.5 minutes). The divisor of 20 provides a minimum of ten sample points per half-period of the slowest thermal oscillation mode, consistent with standard process control practice. For systems with significantly smaller thermal mass, $S_t/20$ must be re-evaluated and the scan interval reduced accordingly.

Section Conflict — E.E. The ST code block below implements one generic PID loop only. It does not cover Loop 2 (pump flow rate, PI only), Loop 3 (battery recharge, PI only), or the mode-dependent output limit required by Loop 1 in the Version 2 architecture. It also lacks the TCV-402 output, the velocity hard ceiling rung in Loop 2, the battery overtemperature ceiling in Loop 3, and the per-loop heartbeat toggle introduced in Version 2. Do not use this block as the implementation reference. The three separate Version 2 code blocks are provided in Appendix G, Section G.E.

12.5 E.E IEC 61131-3 Structured Text (ST) Implementation

The following standardised ST logic encapsulates the discrete algorithms established above, ready for implementation in Siemens TIA Portal, Beckhoff TwinCAT, or Rockwell Studio 5000 environments.

Listing E.1 — Velocity Form PID with Anti-Windup and EMA Filtering (Version 1 — Single Loop)

```

VAR
    // Inputs and Parameters
    PV_Raw      : REAL; // Raw Analog Sensor Input
    SP          : REAL; // Target Setpoint
    Kp, Ki, Kd  : REAL; // Tuning Parameters
    dt          : REAL; // Scan time in seconds (= St / 20, from Step 10)
    Alpha       : REAL; // Filter Constant (= dt / (tau_f + dt))

    // Memory States (RETAIN: preserved across PLC power cycle)
    PV_Filtered : REAL;
    PV_Prev1    : REAL;
    PV_Prev2    : REAL;
    Error_Prev  : REAL;
    Actuator_Out: REAL;

    // Intermediate Calculation Variables
    Error       : REAL;
    Delta_U     : REAL;
    P_Term      : REAL;
    I_Term      : REAL;
    D_Term      : REAL;
END_VAR

// --- 1. Exponential Moving Average Sensor Filter ---
PV_Filtered := (Alpha * PV_Raw) + ((1.0 - Alpha) * PV_Filtered);

// --- 2. Error Calculation ---
Error := SP - PV_Filtered;

// --- 3. Proportional Term (error-based) ---
P_Term := Kp * (Error - Error_Prev);

```

```

// --- 4. Derivative Term (PV-based: prevents setpoint-step kick) ---
D_Term := -(Kd / dt) * (PV_Filtered - (2.0 * PV_Prev1) + PV_Prev2);

// --- 5. Integral Increment with Anti-Windup Clamping ---
// Suppresses only the per-scan increment; does not zero the accumulator.
IF (Actuator_Out >= 100.0 AND Error > 0.0) OR
   (Actuator_Out <= 0.0 AND Error < 0.0) THEN
    I_Term := 0.0;
ELSE
    I_Term := Ki * dt * Error;
END_IF;

// --- 6. Velocity Form Output Accumulation ---
Delta_U      := P_Term + I_Term + D_Term;
Actuator_Out := Actuator_Out + Delta_U;

// --- 7. Absolute Output Saturation Limits ---
IF Actuator_Out > 100.0 THEN Actuator_Out := 100.0; END_IF;
IF Actuator_Out < 0.0 THEN Actuator_Out := 0.0; END_IF;

// --- 8. Update Memory Registers for Next Scan ---
Error_Prev := Error;
PV_Prev2   := PV_Prev1;
PV_Prev1   := PV_Filtered;

```

Note — Sections E.F through E.H. The following sections are substantively valid and have been carried forward into Appendix G with the amendments noted below. They are not flagged as conflicting and may be read as background reference. Where any discrepancy exists between these sections and their Appendix G counterparts, Appendix G takes precedence.

- **E.F (Ladder Logic):** Carried forward to Appendix G, Section G.G. One amendment: a single global interrupt block for all loops is explicitly prohibited in Version 2 because loop scan rates differ by orders of magnitude.
- **E.G (Relay Fallback):** Carried forward to Appendix G, Section G.H. One amendment: the operational limitations note is extended to explicitly disqualify relay architecture for any installation requiring the four-mode vessel structure of Appendix F.
- **E.H (Watchdog and Fail-Safe):** Carried forward to Appendix G, Section G.I. One amendment: TCV-402 is added to the safe-state output enforcement block in Block 4, which was absent in Version 1 because TCV-402 did not exist in the Version 1 I/O schedule.

12.6 E.F Ladder Logic (LD) Implementation Guidelines

While Structured Text natively supports complex algebraic difference equations, many legacy or visually-oriented plant environments require IEC 61131-3 Ladder Diagram (LD)

architecture. When implementing the VCH control loops in LD, the following structural guidelines apply:

- **Cyclic Execution via Interrupts.** The LD network should reside inside a dedicated cyclic hardware interrupt block (e.g., OB30 in Siemens TIA Portal or a Periodic Task in Rockwell RSLogix). The block execution rate must be statically configured to the Δt_{scan} interval calculated in Section E.D. Internal timers (TON) must not be used to trigger the math block.
- **Mathematical Compute Blocks.** The EMA sensor filter cannot readily be executed with basic add/multiply coils. A dedicated Compute (CPT) instruction block should be used to evaluate the $(1 - \alpha)$ filtering formula in a single rung.
- **Native PID Instruction Utilisation.** Rather than writing the full velocity-form equation manually in LD, the native PID instruction blocks (such as PID_Compact or PIDE) should be used. Block parameters must be explicitly configured to read the *filtered* PV rather than the raw physical analog channel.
- **Boolean Anti-Windup Triggers.** Standard contact structures (XIC/XIO) are used to evaluate actuator saturation. When the PID instruction block output reaches 100% and the error remains positive, a parallel rung branch routes to a MOVE (MOV) instruction that immediately forces the integral sum register to zero.

12.7 E.G Hardwired Relay-Based Control (Non-PLC Architecture)

For smaller, isolated hydronic skids, or as a robust fail-safe override to the primary PLC, a traditional hardwired electromechanical control architecture can be employed without microprocessor intervention.

- **Bang-Bang (On/Off) Control Profiles.** Proportional-Integral modulation is not achievable without a dedicated controller. Continuous analog temperature sensors (RTDs) are replaced with mechanical thermostats or liquid-filled capillary thermal switches, which close when the bulk fluid temperature drops below the physical set-point.
- **Relay Logic and Contactors.** Thermostat switch closure energises a 24 VDC or 120 VAC Master Control Relay (MCR) coil. The MCR subsequently pulls in the heavy-duty main motor contactor, providing three-phase power directly to the primary utility heater or the fixed-speed circulation pump.
- **Hardware Safety Interlocks.** To replicate the PLC's software velocity and flow safeguards, physical flow switches (paddle or thermal dispersion type) and high-limit pressure switches (capsule type) are wired directly in series with the MCR coil. An unexpected drop in flow or a surge in system pressure beyond mechanical limits breaks the series circuit instantly, dropping the contactor and halting the hydronic loop.
- **Operational Limitations.** While hardwired relays are highly robust and immune to software faults, network crashes, and EMI corruption, they suffer from severe thermal hysteresis. Furthermore, relay logic cannot execute dynamic load-balancing or thermal battery peak shaving (Modes II and IV). A hardwired network restricts the hydronic framework exclusively to baseline direct charging and standby states.

12.8 E.H Watchdog Timer and 05_FAIL_SAFE State Implementation

The PLC state machine defined in Appendix D (Section D.C) designates 05_FAIL_SAFE as the terminal safety state, reachable from any operating mode upon detection of a velocity limit breach, hardware fault, or loss of a critical field signal. Reaching this state via software logic alone is insufficient for a safety-critical hydronic loop: a PLC processor fault, scan-cycle overrun, or memory corruption event could prevent the state transition from executing entirely. Two complementary mechanisms are therefore required.

Software Watchdog (PLC Internal). On every scan cycle in which the control block executes normally, the ST block writes a heartbeat token to a dedicated **RETAIN** boolean flag (WD_Heartbeat). A separate watchdog timer task, scheduled at a faster interrupt priority than the main control task, monitors this flag. If the heartbeat is not refreshed within two consecutive watchdog periods, the task unconditionally forces all control outputs to their de-energised fail-safe positions and latches the **FAIL_SAFE_Active** flag, which is hardwired to the field alarm beacon and the VFD run-enable relay.

Hardware Watchdog (Independent of PLC). The VFD run-enable signal (VFD-501 enable input) is routed through an external hardware watchdog relay module, independent of the PLC output card. This relay requires a continuous 1 Hz pulse train from the PLC digital output to remain energised. If the pulse train stops for any reason — including PLC processor halt, output card failure, or loss of 24 VDC control supply — the relay drops out, removing the run-enable signal from the drive and bringing the loop to a controlled deceleration stop.

The ST implementation below covers the software watchdog block and the 05_FAIL_SAFE state transition logic, intended to execute in a high-priority cyclic interrupt task at a fixed 1-second interval, independent of the main control task.

Listing E.2 — Software Watchdog and 05_FAIL_SAFE State Transition Logic (Version 1)

```

VAR
    // Watchdog Coordination (set by main PID block each scan)
    WD_Heartbeat      : BOOL;           // Written TRUE by PID block each
    execution
    WD_Prev           : BOOL;           // Previous-scan heartbeat state
    WD_MissCount       : UINT := 0;     // Consecutive missed heartbeat counter
    WD_TripThreshold   : UINT := 2;     // Trips FAIL_SAFE after N missed scans

    // Velocity Monitoring (read from FE-201 signal chain)
    Velocity_Live      : REAL;          // Current loop velocity in m/s
    Velocity_HiHi      : REAL := 2.5;   // Hard erosion ceiling (Appendix D
    alarm)

    // VFD Hardware Pulse Output (1 Hz square wave to external WD relay)
    HW_WD_Pulse       : BOOL;          // Physical DO to hardware watchdog
    relay
    HW_WD_Toggle       : BOOL;          // Internal toggle state

    // System State Flags
    FAIL_SAFE_Active   : BOOL := FALSE; // Latching; requires manual HMI
    reset
    PLC_State          : UINT;          // Mirrors D.C state machine (05 =
    FAIL)

    // Safe-State Actuator Demand Outputs
    // NOTE: TCV-402 absent in Version 1 --- added in Appendix G, Section
    G.I
    TCV401_Demand      : REAL;          // 0.0 = fully closed (battery
    isolated)
    VFD501_Demand      : REAL;          // 0.0 = stop command to drive
    Alarm_Beacon       : BOOL;          // Hardwired field alarm output
END_VAR

// --- Block 1: Hardware Watchdog Pulse Generator (1 Hz toggle) ---
HW_WD_Toggle := NOT HW_WD_Toggle;
HW_WD_Pulse  := HW_WD_Toggle;

// --- Block 2: Software Heartbeat Monitor ---
IF WD_Heartbeat = WD_Prev THEN
    WD_MissCount := WD_MissCount + 1;
ELSE
    WD_MissCount := 0;
END_IF;
WD_Prev := WD_Heartbeat;

// --- Block 3: FAIL_SAFE Trigger Evaluation ---
IF (WD_MissCount >= WD_TripThreshold) OR

```

```
(Velocity_Live > Velocity_HiHi) THEN
  FAIL_SAFE_Active := TRUE;
END_IF;

// --- Block 4: Safe-State Output Enforcement ---
IF FAIL_SAFE_Active THEN
  PLC_State      := 5;
  TCV401_Demand := 0.0;
  VFD501_Demand := 0.0;
  Alarm_Beacon  := TRUE;
  HW_WD_Pulse   := FALSE;
END_IF;
```

Two implementation requirements must be confirmed during factory acceptance testing (FAT) before commissioning. First, the hardware watchdog relay drop-out time must be verified to be shorter than the VFD deceleration ramp time, ensuring the drive reaches zero speed before any mechanical brake or isolation valve actuates. Second, the `FAIL_SAFE_Active` latch must be confirmed as non-resettable from any automated logic path; reset must require an explicit operator acknowledgement action on the Level 3 HMI screen defined in Appendix D, Section D.E.

APPENDIX F

REVISED CONTROL ARCHITECTURE

This appendix supersedes Appendix D (Version 1) in its entirety and constitutes the current control architecture reference for the VCH Sizing Framework. It should be consulted for all design, commissioning, and implementation purposes. Appendix D is retained as an archived engineering decision record. Appendix G (Revised PLC Implementation and Control Commissioning, Version 2) provides the discrete-time mathematics, code implementation, and commissioning procedures consistent with the architecture defined herein.

This appendix defines the revised system-level control philosophy, instrumentation topology, and operational mode structure governing the hydronic vessel network and centralised thermal battery system. The architecture presented herein corrects and extends the single-loop cascade model of Version 1, replacing it with three independent but physically coupled control loops and a refined four-mode vessel operational structure. The revision was driven by a more rigorous analysis of battery charging and discharging behaviour during simultaneous thermal demand, the introduction of operator-driven alarm response in place of automatic state transitions, and the formalisation of the pump flow rate as an independently controlled variable rather than a subordinate actuator.

13 Appendix F (Revised D)

13.1 F.A Revised Control Philosophy and Operational Modes

13.1.1 F.A.1 Three-Loop Independence Principle

The Version 1 architecture treated the hydronic system as a single cascade loop in which pump speed was a subordinate output of the temperature controller. This formulation was found to be architecturally insufficient for a system in which the pump velocity constraint, the vessel thermal target, and the battery energy state each represent independent physical objectives with independent actuators and independent failure modes.

The revised architecture declares three independent control loops. Each loop has its own setpoint, its own controller, its own actuator, and its own process variable. Independence is architectural — each loop runs its own scan cycle and PID block without receiving setpoint commands from another loop’s output. The loops are physically coupled through the shared hydronic fluid, meaning the output of one loop modifies the disturbance environment of the others, but this coupling is handled as a measurable disturbance rather than a structural dependency.

A fourth loop governs multi-vessel load balancing in distributed installations. This loop operates at the supervisory level and trims individual vessel pump speeds relative to a master deviation signal. It is intentionally non-aggressive and does not interact with the primary three loops at the single-vessel level.

Table 47 defines the four loops, their controller types, and their physical roles.

Table 47. Control Loop Declarations — Version 2

Loop Name		Controller	Process Variable	Actuator	Physical Role
1	Vessel Batch Temperature	PID	TE-103	TCV-401	Maintains vessel fluid at target temperature across all modes
2	Pump Flow Rate	PI	FE-201	VFD-501	Enforces velocity set-point; prevents erosion ceiling breach and stagnation floor violation
3	Battery Recharge	PI	TE-102	TCV-402	Directs surplus driver energy into battery; always permitted to run; set-point is $T_{\text{bat,target}}$
4	Multi-Vessel Balancing	P only	ΔT per vessel	Per-vessel VFD trim	Redistributes flow across vessels by deviation; intentionally lazy to prevent inter-vessel hunting

Loop 4 is active only in multi-vessel installations operating Mode IV. It does not affect single-vessel operation.

13.1.2 F.A.2 Battery Charging and Discharging Simultaneity

A critical correction from Version 1 concerns the mutual exclusion assumption applied to battery charging and discharging. Version 1 implied that these were mutually exclusive states — the battery was either charging (Mode III) or discharging (Mode II), never both simultaneously.

This assumption is physically incorrect for a thermal driver operating below its rated capacity. In normal operation the driver is always running. It simultaneously delivers heat to the vessel through the coil circuit and replenishes the battery through the recharge path governed by Loop 3. The net battery energy state is therefore approximately constant in all modes except one — Mode 1.A.2, where total vessel heat consumption exceeds what the driver alone can supply. Only in this condition is the battery in net energy drain, because the driver is fully committed to the vessel demand and has no surplus to route through Loop 3.

Loop 3 therefore carries no interlock against simultaneous Loop 1 or Loop 2 operation. It runs continuously whenever the driver has surplus capacity, which is the normal condition in all modes except 1.A.2.

13.1.3 F.A.3 Vessel Operational Modes

The vessel batch temperature loop operates across four modes. The mode determines the setpoint, the output limits, and the pump flow setpoint presented to Loops 1 and 2 respectively. The PID block structure does not change between modes — only its boundary conditions change.

Mode 1.A.1 — Heating, Utility Sufficient. The entry condition is that the maximum heat deliverable by the pump-coil circuit at v_{max} does not exceed the driver's allocated

output capacity:

$$\dot{Q}_{\text{pump,max}} \leq \dot{Q}_{\text{allocated}}.$$

The vessel PID setpoint is T_{target} . Loop 2 setpoint is the velocity that satisfies the thermal demand without exceeding v_{max} . The battery is in net neutral — Loop 3 is active and routing surplus driver energy into the battery at the background rate. TCV-401 output upper limit is set to the valve position corresponding to $\dot{Q}_{\text{allocated}}$.

Mode 1.A.2 — Heating, Battery-Assisted. The entry condition is:

$$\dot{Q}_{\text{pump,max}} > \dot{Q}_{\text{allocated}},$$

meaning full pump velocity alone would demand more heat from the driver than the allocated capacity. The battery discharge path opens, and TCV-401 blends supplemented battery fluid into the coil circuit. The driver is fully committed to vessel demand; Loop 3 receives no surplus and the battery enters net drain.

The battery state-of-charge watchdog runs on every scan cycle. When SoC reaches the operator-defined floor threshold $\text{SoC}_{\text{floor}}$, no automatic state transition occurs. Instead the PLC raises a Priority 1 alarm and presents a two-option decision prompt on the Level 1 HMI screen:

- **Option A — Stop Batch:** Graceful shutdown. Loop 1 setpoint is ramped to ambient over a configurable cool-down period. Battery recovers on the next idle recharge cycle.
- **Option B — Continue, Accept Temperature Sacrifice:** The operator accepts reduced thermal performance. TCV-401 output upper limit is reduced, the vessel receives less heat than demanded, and T_{vessel} falls short of T_{target} . The operator owns this decision; the PLC does not override it.

The only automatic hard trip that remains non-negotiable is the velocity ceiling breach transitioning to **FAIL_SAFE** — this is a hardware safety condition, not an operational one, and is never subject to operator override.

Mode B — Holding. The vessel has reached T_{target} . The only thermal demand is the environmental loss term $\dot{Q}_{\text{loss}}$ quantified in Step 9 of the framework. Loop 2 setpoint drops to the minimum velocity required to deliver $\dot{Q}_{\text{loss}}$ through the coil. Loop 1 PID setpoint remains T_{target} ; the integrating nature of the controller naturally throttles TCV-401 toward a low steady-state opening as error approaches zero.

The stagnation floor constraint applies: if $\dot{Q}_{\text{loss}}$ is sufficiently small that it would require $v < v_{\text{min}}$ (0.3 m/s per the FMEA threshold), Loop 2 holds at v_{min} and Loop 1 modulates TCV-401 to prevent overheating. This edge case is most likely to occur in well-insulated vessels at low ambient temperature.

Mode C — Hibernation. The operator defines a hibernation power fraction $\phi \in (0, 1)$ via the HMI. The effective thermal demand becomes:

$$\dot{Q}_{\text{hib}} = \phi \cdot \dot{Q}_{\text{hold}}.$$

This is back-calculated to a reduced temperature setpoint:

$$T_{\text{hib}} = T_{\text{target}} - \Delta T_{\text{hib}},$$

where ΔT_{hib} is derived from $\dot{Q}_{\text{hib}}$ and the vessel thermal resistance network. Loop 2 setpoint scales proportionally downward. The vessel cools to T_{hib} and holds there. Wake-up from hibernation is a manual HMI command that restores T_{target} as the Loop 1 setpoint, and the system re-heats through Mode 1.A.1 or 1.A.2 depending on the $\dot{Q}_{\text{pump,max}}$ evaluation.

Table 48 summarises the mode-dependent boundary conditions applied to Loops 1 and 2.

Table 48. Mode-Dependent Setpoint and Output Limit Summary

Mode	Loop 1 Set-point	Loop 1 Output Limit	Loop 2 Set-point	Battery State
1.A.1	T_{target}	$\leq$ position for $\dot{Q}_{\text{allocated}}$	v satisfying demand, $\leq v_{\text{max}}$	Net neutral; Loop 3 active
1.A.2	T_{target}	Position for $\dot{Q}_{\text{allocated}} + \dot{Q}_{\text{battery}}$	v_{max}	Net drain; SoC watchdog active
B	T_{target}	Unrestricted (PID self-regulates)	v satisfying $\dot{Q}_{\text{loss}}, \geq v_{\text{min}}$	Net neutral; Loop 3 active
C	$T_{\text{hib}} = T_{\text{target}} - \Delta T_{\text{hib}}$	Scaled to $\phi \cdot \dot{Q}_{\text{hold}}$	v scaled to $\phi, \geq v_{\text{min}}$	Net neutral; Loop 3 active

FAIL_SAFE overrides all modes unconditionally on velocity ceiling breach or hardware fault.

13.2 F.B Complete Instrumentation and I/O Schedule

The full instrument and actuator schedule for the Version 2 architecture comprises 25 I/O points across four signal classes. This schedule supersedes the incomplete seven-point table in Appendix D, Section D.B. All field devices must be installed, tagged, and loop-tested against this schedule prior to PLC commissioning.

Analogue Inputs (AI) — 8 Points

Table 49. Analogue Input Schedule

Tag	Instrument	Type / Element	Signal	Functional Role
TE-101	Loop supply temperature	PT100 Class A	RTD 4–20 mA / HART	Monitors driver output temperature; Loop 1 disturbance input
TE-102	Battery core temperature (multipoint)	Type K thermocouple ×3	4–20 mA / HART	SoC proxy via majority-vote logic; Loop 3 process variable
TE-103	Vessel batch temperature	PT100 Sanitary	RTD 4–20 mA / HART	Loop 1 process variable; primary batch quality indicator
TE-104	Battery inlet temperature	PT100	RTD 4–20 mA / HART	Confirms recharge flow entering battery; detects TCV-402 response
TE-105	Battery outlet temperature	PT100	RTD 4–20 mA / HART	Confirms discharge flow leaving battery; validates 1.A.2 supplementation
FE-201	Main loop velocity	Electromagnetic flowmeter	Modbus TCP	Loop 2 process variable; erosion ceiling and stagnation floor enforcement
PE-301	Loop differential pressure	Piezo-resistive cell	4–20 mA / HART	Head loss tracking; leak detection via LoLo threshold
LE-401	Battery SoC (derived)	Calculated from TE-102	Software tag	SoC floor alarm trigger; 1.A.2 watchdog input

TE-104 and TE-105 are new in Version 2. They provide independent confirmation of energy exchange in both directions.

Analogue Outputs (AO) — 3 Points

Table 50. Analogue Output Schedule

Tag	Actuator	Type	Signal	Functional Role
TCV-401	Battery blend / discharge valve	3-way equal percentage	4–20 mA	Loop 1 actuator; modulates battery discharge blend into coil circuit
TCV-402	Recharge divert valve	2-way modulating	4–20 mA	Loop 3 actuator; directs surplus driver energy into battery
VFD-501	Loop circulation pump	AC induction inverter	Profinet	Loop 2 actuator; velocity setpoint from inner PI loop

TCV-402 is new in Version 2. It was absent from the Version 1 I/O schedule.

Digital Inputs (DI) — 8 Points

Table 51. Digital Input Schedule

Tag	Device	Functional Role
HS-601	Emergency stop pushbutton (hard-wired)	Forces FAIL_SAFE unconditionally; bypasses all PLC logic
HS-602	HMI batch start command	Triggers transition from STANDBY to heating mode evaluation
HS-603	HMI hibernate command	Triggers Mode C; ϕ value entered via HMI numeric input
ZS-701	TCV-401 open position confirmation	Confirms valve has physically responded to output demand
ZS-702	TCV-402 open position confirmation	Confirms recharge path is physically open
FS-801	Flow switch, low flow detection	Hardware backup to FE-201 stagnation alarm; wired in parallel
PS-901	High pressure switch (hardwired)	Direct hardware trip to VFD contactor independent of PLC
WD-001	Hardware watchdog relay feedback	Confirms 1 Hz pulse train from D0-103 is being received

Digital Outputs (DO) — 6 Points

Table 52. Digital Output Schedule

Tag	Device	Functional Role
AL-001	Audible field alarm beacon	Energised on SoC floor breach or any Priority 1 alarm
AL-002	Visual alarm beacon (strobe)	Energised on same trigger as AL-001
DO-101	VFD-501 run enable relay	De-energised by hardware watchdog drop-out; removes VFD run permission
DO-102	Driver enable signal	PLC command to driver on/off; interlocked with loop pressure confirmation
DO-103	Hardware watchdog pulse output	1 Hz square wave to external watchdog relay module
DO-104	HMI SoC alarm indicator	Illuminates operator decision prompt on Level 1 screen in Mode 1.A.2

I/O Summary: 8 Analogue Inputs, 3 Analogue Outputs, 8 Digital Inputs, 6 Digital Outputs — **25 points total.**

13.3 F.C Revised PLC State Transition Matrix

The Version 2 state machine maps directly to the four vessel operational modes defined in Section F.A, with **STANDBY** as the idle entry state and **FAIL_SAFE** as the terminal safety latch reachable from any state. The critical departure from Version 1 is the removal of automatic state transitions driven by load threshold comparisons. In Version 2, transitions between heating modes are governed by physical condition evaluation at entry, and the 1.A.2 SoC floor event produces an alarm and operator prompt rather than a forced transition.

FAIL_SAFE remains the sole automatic non-operator-initiated transition and is triggered exclusively by hardware safety conditions — velocity ceiling breach, watchdog timeout, or hardwired emergency stop. It latches unconditionally and requires manual HMI acknowledgement to clear.

Table 53. PLC State Machine Transition Sequence — Version 2

State	System Profile	Entry Condition	Exit Condition	Active Interlocks
STANDBY	All loops at minimum; pump at bypass speed; battery loops inactive.	Power-on de-fault or manual HMI reset after FAIL_SAFE clearance.	HS-602 batch start command received.	Confirm PE-301 > 1.5 bar before permitting transition.
MODE_1A1	Loop 1 drives to T_{target} ; Loop 2 at demand velocity; Loop 3 active recharging.	$\dot{Q}_{\text{pump,max}} \leq \dot{Q}_{\text{allocated}}$ evaluated at batch start.	$T_{\text{vessel}} \geq T_{\text{target}} - \delta \rightarrow \text{MODE_B}$; or condition flips to 1.A.2 on load change.	Continuous velocity monitoring: $v > v_{\text{max}}$ trims VFD-501 immediately.
MODE_1A2	Battery discharge blended via TCV-401; Loop 3 starved of surplus.	$\dot{Q}_{\text{pump,max}} > \dot{Q}_{\text{allocated}}$ evaluated at batch start or detected mid-batch.	$T_{\text{vessel}} \geq T_{\text{target}} - \delta \rightarrow \text{MODE_B}$; SoC $\leq \text{SoC}_{\text{floor}} \rightarrow \text{P1 alarm} + \text{HMI prompt only}$; no forced transition.	SoC watchdog active every scan. D0-104 energised when SoC floor reached.
MODE_B	Loop 1 holds T_{target} ; Loop 2 modulates to offset $\dot{Q}_{\text{loss}}$ only; Loop 3 active.	Arrival at T_{target} from either 1.A.1 or 1.A.2.	HS-603 hibernate command $\rightarrow \text{MODE_C}$; or new batch start $\rightarrow$ re-evaluate 1.A.1 / 1.A.2.	Loop 2 floor: $v \geq v_{\text{min}}$ enforced at all times.
MODE_C	Loop 1 holds T_{hib} ; Loop 2 scaled to ϕ ; Loop 3 active.	HS-603 command with ϕ value confirmed on HMI.	Manual wake command $\rightarrow \text{MODE_B} \rightarrow$ re-heat sequence.	ϕ range check: $0 < \phi < 1$ enforced; out-of-range entry rejected with HMI notification.
FAIL_SAFE	All outputs forced to de-energised positions; alarms latched.	$v > v_{\text{HiHi}}$; watchdog time-out; HS-601 pressed; PS-901 trip.	Manual HMI acknowledgement only, after physical fault clearance confirmed.	D0-103 pulse halted; D0-101 de-energised; AL-001/002 energised; VFD-501 run enable removed.

δ = arrival deadband, recommended 1–2°C. Prevents **MODE_B** entry on momentary overshoot.

13.4 F.D Three-Loop Independent Control Architecture

Loop 1 — Vessel Batch Temperature (PID)

Loop 1 is the primary process control objective of the framework. Its controlled variable is the vessel bulk fluid temperature measured by TE-103. Its actuator is TCV-401, a three-way equal-percentage valve that proportions the blend of main utility supply and battery-supplemented fluid entering the coil circuit.

Full PID action is warranted for Loop 1 because the process is a slow integrating thermal mass with significant dead time from coil transit. Proportional action alone produces steady-state offset. Integral action eliminates offset but winds up during batch heat-up if not clamped — hence the anti-windup logic in Appendix G. Derivative action on the process variable (not the error) reduces overshoot on batch arrival at T_{target} without introducing derivative kick on HMI setpoint changes.

The output upper limit of Loop 1 is mode-dependent as defined in Table 48. This is the primary mechanism by which the mode structure governs thermal power delivery without requiring separate PID blocks per mode.

Loop 2 — Pump Flow Rate (PI)

Loop 2 is an independent velocity controller. Its process variable is the volumetric flow velocity from FE-201. Its actuator is VFD-501. The setpoint is not fixed — it is determined by the active vessel mode and the thermal demand calculated from Loop 1's operating point. However, this setpoint is computed by the mode logic layer and presented to Loop 2 as an external input; Loop 2 does not receive its setpoint from Loop 1's output directly.

Derivative action is omitted from Loop 2. Flow velocity responds quickly and cleanly to VFD speed changes, and the electromagnetic flowmeter signal carries turbulent noise that derivative action would amplify. The EMA filter from Appendix G reduces this noise before it enters the PI calculation, but a clean PI response is sufficient for the hydraulic loop's objectives.

Two hard limits are enforced independently of the PI output:

- **Ceiling:** $v \leq v_{\text{max}}$ at all times. If the PI output would command a velocity above this, it is clamped to v_{max} and the integral increment is frozen to prevent windup against the ceiling.
- **Floor:** $v \geq v_{\text{min}}$ (0.3 m/s) at all times. If the mode logic demands a velocity below this, the floor takes precedence and Loop 1 absorbs the difference through TCV-401 modulation.

Loop 3 — Battery Recharge (PI)

Loop 3 maintains the battery thermal state against its target charge temperature $T_{\text{bat,target}}$. Its process variable is TE-102 (majority-voted from three thermocouple points). Its actuator is TCV-402, which diverts a proportion of the driver output into the battery recharge circuit.

Loop 3 carries no operating interlock against Loops 1 or 2. It runs continuously whenever the driver has surplus capacity. In Mode 1.A.2 it receives no surplus because the driver is fully committed to vessel demand, but it does not shut down — it simply has no

energy available to route and its output naturally drives toward zero. Derivative action is omitted; the battery thermal mass is very large and derivative action on a slow integrating process with thermocouple noise produces no benefit.

Loop 4 — Multi-Vessel Balancing (P only)

Loop 4 operates only in multi-vessel installations. The master PLC monitors the temperature differential $\Delta T_i = T_{\text{in},i} - T_{\text{out},i}$ across each vessel i and computes a deviation from the fleet mean. A proportional-only controller trims the VFD speed setpoint of each vessel's Loop 2 upward or downward in proportion to its deviation. Gains are intentionally set low ($K_p = 0.5\text{--}1.0$) to produce gentle redistribution rather than aggressive correction, preventing the inter-vessel hunting that integral action would introduce when all vessels compete for flow simultaneously.

13.5 F.E Control System Architecture Selection: Distributed Controllers versus Central PLC

The physical implementation of the three-loop architecture may be realised through two fundamentally different hardware strategies. This section documents the technical and economic basis for selecting between them, with cost references denominated in Euros appropriate to the Western European industrial automation market, which represents the primary expansion target for this framework.

F.E.1 Option A — Distributed Individual Controllers

In a distributed architecture, each control loop is served by a dedicated standalone controller. Loop 1 and Loop 3 are implemented on DIN-rail temperature controllers (Siemens RWD, Jumo dTRON, or equivalent). Loop 2 is implemented using the onboard PID function built into the VFD-501 drive itself, referencing FE-201 as its feedback input via a 4–20 mA signal. The four-mode vessel logic, SoC floor alarm, and HMI decision prompt have no native home in this architecture and must be implemented in a separate hardwired relay logic panel or a small standalone supervisory relay module.

What the installation looks like physically. The control panel contains three or more individual controller displays, each showing only its own PV and setpoint. There is no unified system view. Understanding the current system state requires reading three separate screens and correlating them mentally. The alarm subsystem consists of individual per-loop indicator lights and buzzers — there is no alarm priority, no timestamp, and no history. In a fault condition multiple alarms may fire simultaneously with no indication of which triggered first. The relay logic section implementing the mode switching is a row of DIN-rail relays that is opaque to any technician who was not present at commissioning. Wiring density is high because parallel cable runs service controllers that sit physically adjacent but have no electrical awareness of each other.

Repairability. Individual controllers are straightforward to replace — a DIN-rail unit swaps in under ten minutes. Hardware cost per unit is €180–450. However, configuration is stored inside the controller itself. Replacement without restoring saved parameters brings the loop up on factory defaults. Best practice requires a printed parameter sheet for each controller taped inside the panel door. The relay logic mode section carries no self-documentation; modification requires retracing wiring physically. Cross-loop fault

consequences — for example, a VFD fault that drops Loop 2 while Loop 1 continues demanding heat — are not detected by any controller because no controller has visibility beyond its own loop. The system may continue operating in a thermally unsafe low-velocity condition until a mechanical symptom appears.

F.E.2 Option B — Central PLC

In a central PLC architecture, all three loops, the mode state machine, the SoC watchdog, the alarm logic, and the HMI interface are hosted on a single programmable logic controller. The recommended platform for European deployment is the Siemens S7-1200 or S7-1500 series programmed in TIA Portal, which natively supports all signal classes in the I/O schedule defined in Section F.B. Beckhoff CX series running TwinCAT is an alternative for installations requiring sub-millisecond loop execution.

What the installation looks like physically. One HMI touchscreen on the panel face — Siemens KTP700 Basic or equivalent. One unified alarm banner showing active alarms in ISA-18.2 priority order with timestamps. The Level 1 overview screen displays the complete system state on a single display: current mode, all eight analogue inputs, SoC bar, TCV-401 and TCV-402 positions, and VFD-501 speed. When the SoC floor alarm fires in Mode 1.A.2, the HMI presents the two operator options directly on screen with the current SoC percentage and a 30-minute drain trend. The operator makes an informed decision within thirty seconds.

The panel cabinet is organised by I/O rack — AI card, AO card, DI card, DO card — with all field wiring terminating at a marshalling strip. Any PLC-trained technician can navigate the cabinet immediately. The SCADA historian logs all 25 I/O points at one-second resolution, enabling complete post-incident causal reconstruction by overlaying any combination of tags on a unified timeline.

Repairability. The most common fault is a single I/O card failure. A card failure affects only its own channel group; other cards and all three loops on unaffected cards continue normally. Card replacement takes fifteen minutes. Configuration restores automatically from CPU memory — no manual parameter re-entry. Spare card cost is €400–1,200 depending on channel count and signal class. CPU fault is the high-impact scenario: a simplex CPU failure takes all three loops to **FAIL_SAFE** simultaneously. This is mitigated by a hot-standby duplex CPU configuration, which adds €2,000–4,500 to hardware cost but eliminates the single point of failure. Switchover time is under 100 milliseconds, transparent to the process.

13.5.1 F.E.3 Feature and Operational Complexity Comparison

Table 54. Distributed Controllers versus Central PLC — Feature Comparison

Feature	Distributed Controllers	Central PLC
Mode logic (1.A.1 / 1.A.2 / B / C)	Relay rungs only; opaque to modification	Native structured text; software change only
SoC alarm + HMI decision prompt	Not available without additional hardware	Built in; presented on Level 1 screen
Cross-loop fault visibility	None; each controller is blind to others	Full; all variables visible simultaneously
Alarm management	Per-loop buzzers; no history; no priority	Unified ISA-18.2 list with timestamps
Post-incident diagnosis	Memory and paper log only	SCADA historian; full 1-second trend replay
Configuration backup	Manual printed sheets; per-controller USB	Automatic CPU SD card; full restore on swap
Repair skill required	Low; any technician	Medium; PLC-trained technician
Common fault repair time	10–30 min hardware + 30–60 min reconfiguration	15–30 min hardware + automatic reconfiguration
Remote diagnostics	Not available	Full OPC-UA remote access
Scalability to multi-vessel	Requires full hardware duplication per vessel	I/O expansion card per vessel; €400–800
Modification complexity	Rewiring required	Software edit and download; no rewiring

13.5.2 F.E.4 Cost Analysis and Break-Even (Euro Reference Market)

Table 55 provides installed cost estimates referenced to the Western European industrial automation market. Labour rates reflect German and Benelux region averages (€65–90/hour for panel build; €75–150/hour for PLC engineering and commissioning).

Table 55. Installed Cost Estimates — Single Vessel Installation (Euro, Western Europe)

Cost Element	Distributed trollers	Con-	Central PLC
<i>Hardware</i>			
Controllers / CPU	€550–1,350 (3 units)		€1,800–3,200 (S7-1200 + I/O)
HMI panel	Not included		€600–1,200 (KTP700)
Relay logic section	€300–600		Not required
Duplex CPU (optional)	Not applicable		€2,000–4,500
Cabinet and wiring materials	€800–1,400		€600–1,000
<i>Engineering and Labour</i>			
Panel build labour	€900–1,800 (high wiring density)	(high	€500–900
PLC / controller programming	€400–800 (controller setup)	(per-	€1,500–3,500
TIA Portal software licence	Not applicable		€1,200–2,500
Commissioning and loop tuning	€600–1,200		€800–1,500
Total installed (simplex)	€3,550–7,150		€7,000–13,800
Total installed (duplex CPU)	Not applicable		€9,000–18,300

Costs exclude field instrumentation (sensors, valves, cabling) which are identical for both architectures. Import duties and VAT applicable in Nepal and other non-EU markets; hardware costs increase 15–30%.

Break-even analysis. The central PLC architecture carries an upfront cost premium of approximately €3,500–6,500 over the distributed approach for a single vessel installation. This premium is recovered through lower engineering costs on subsequent modifications and through avoided downtime costs from superior fault detection. In the Western European market, PLC programming and commissioning consultancy is billed at €75–150 per hour. A non-trivial modification to the distributed relay logic — for example, adjusting the SoC floor threshold or adding a new alarm — requires physical rewiring and relay replacement, typically two to three engineering hours plus travel at an on-site rate of €800–1,500 per visit. The equivalent PLC modification is a software parameter change downloadable remotely via OPC-UA, typically one to two hours billed at €150–300.

Assuming two modifications per year and one fault diagnosis event per year requiring on-site attendance:

Annual saving (PLC vs distributed) $\approx 2 \times \text{€}900 + 1 \times \text{€}1,200 = \text{€}3,000$ per year,

$$\text{Break-even (single vessel)} \approx \frac{\text{€}5,000 \text{ cost delta}}{\text{€}3,000/\text{year}} \approx 1.5\text{--}2 \text{ years.}$$

This break-even compresses further with additional vessels. Adding a second vessel to the distributed system requires duplicating the full controller hardware at €3,500–7,000. Adding a second vessel to the PLC system requires an I/O expansion card and Modbus TCP satellite node at €600–1,200. At two vessels the PLC system is cost-equivalent to or cheaper than the distributed system at the point of installation, before any operational savings are applied.

Localisation note. The above figures represent the primary Western European expansion market. For installations in Nepal and comparable markets, hardware costs are broadly similar subject to import duties of 15–30%, but panel build and commissioning labour costs are substantially lower. This compresses the break-even point in favour of the distributed approach for single-vessel purely local deployments. However, the operational complexity, cross-loop fault blindness, and absence of remote diagnostic capability of the distributed architecture remain unchanged regardless of geography. For any installation intended for commercial operation, client handover, or future integration into a multi-vessel network, the central PLC recommendation stands.

13.5.3 F.E.5 Recommendation

The central PLC architecture is the appropriate implementation environment for the VCH control philosophy as defined in this appendix. The distributed approach is technically incapable of implementing the Mode 1.A.2 SoC decision prompt, the unified alarm philosophy of Section F.F, and the cross-loop fault detection required by the FMEA in Section F.H without significant additional hardware that erodes its cost advantage. The PLC architecture should be specified as the standard for all VCH installations. The distributed approach may be considered only for single isolated skids with no mode logic requirements and no commercial handover obligation.

13.6 F.F Updated HMI / SCADA Screen Layout and Alarm Philosophy

The HMI and SCADA layer is organised per ISA-101 across three display tiers as defined in Appendix D, Section D.E. The following amendments apply in Version 2.

Level 1 — System Overview is updated to display the Version 2 mode tag (1.A.1, 1.A.2, B, or C) rather than the Version 1 mode numbers. The SoC bar now includes a floor threshold marker at $\text{SoC}_{\text{floor}}$ so the operator can visually track proximity to the alarm condition in real time. TCV-402 position is added to the mimic diagram. When DO-104 is energised (SoC floor reached in Mode 1.A.2), a decision prompt overlay appears on the Level 1 screen presenting Option A (stop batch) and Option B (continue, accept temperature sacrifice) as tappable buttons. This overlay cannot be dismissed without an explicit selection.

Level 2 — Loop Detail Screens now include a dedicated Loop 3 screen for the battery recharge controller, which was absent in Version 1. TE-104 and TE-105 trends appear on this screen to confirm bidirectional energy exchange.

The alarm table is updated below. The SoC floor alarm is classified as a distinct **Decision Alarm (DA)** class — it requires an operator choice rather than a response action and must not be shelved or acknowledged without a selection being recorded in the SCADA historian.

Table 56. Updated Alarm Setpoint and Priority Classification Matrix — Version 2

Tag	Alarm Description	Lo / LoLo	Hi / HiHi	Priority	Required Action
TE-101	Supply temp out of band	Lo: $T_{\text{set}} - 5^{\circ}\text{C}$ LoLo: $T_{\text{set}} - 15^{\circ}\text{C}$	Hi: $T_{\text{set}} + 10^{\circ}\text{C}$ HiHi: $T_{\text{set}} + 25^{\circ}\text{C}$	P2 / P1	Investigate driver; P1 triggers balancing bypass
TE-102	Battery core temp deviation	Lo: $T_{\text{bat,min}}$ LoLo: $T_{\text{bat,min}} - 10^{\circ}\text{C}$	Hi: $T_{\text{bat,max}}$ HiHi: $T_{\text{bat,max}} + 5^{\circ}\text{C}$	P2 / P1	P1 isolates recharge; flag for inspection
TE-103	Vessel batch temp deviation	Lo: $T_{\text{proc}} - 3^{\circ}\text{C}$	Hi: $T_{\text{proc}} + 5^{\circ}\text{C}$ HiHi: $T_{\text{proc}} + 12^{\circ}\text{C}$	P3 / P2	P2 triggers setpoint review
TE-104	Battery inlet temp loss	Signal deviation from TE-101 during recharge	$> 5^{\circ}\text{C}$	P2	Check TCV-402 position and ZS-702
TE-105	Battery outlet temp loss	Signal deviation from TE-102 during discharge	$< 2^{\circ}\text{C}$	P2	Check TCV-401 blend ratio
FE-201	Velocity breach	Lo: 0.3 m/s	Hi: 2.0 m/s HiHi: 2.5 m/s	P2 / P1	P1 trips VFD; latches FAIL_SAFE
PE-301	Differential pressure	LoLo: 0.5 bar	HiHi: 6.0 bar	P1	Isolate loop; inspect coil
LE-401	SoC floor breach	$\text{SoC} \leq \text{SoC}_{\text{floor}}$ during Mode 1.A.2		DA	HMI decision prompt; Option A or B required
VFD-501	Drive fault	—	Any fault code	P1	Acknowledge; switch to bypass pump
TCV-401	Position mismatch	$ \text{Cmd} - \text{Actual} > 5\%$ for $> 10\text{ s}$		P2	Check actuator supply
TCV-402	Position mismatch	$ \text{Cmd} - \text{Actual} > 5\%$ for $> 10\text{ s}$		P2	Check actuator supply

DA = Decision Alarm. Must not be shelved without a recorded operator selection. All others per ISA-18.2. Deadband of $\pm 0.5^{\circ}\text{C}$ on all temperature alarms to suppress chattering at steady state.

13.7 F.G Updated Network Architecture and Communication Protocols

The three-layer network hierarchy defined in Appendix D, Section D.G is retained unchanged in Version 2. The following amendments apply to the device lists.

Layer 1 (Field Bus) now includes TCV-402 on the HART analogue loop in addition to the Version 1 devices. TE-104 and TE-105 are added as PT100 RTD inputs on the 4–20 mA / HART field bus.

Layer 2 (Control Bus) is unchanged. FE-201 and VFD-501 remain on Profinet IO. Satellite PLCs in multi-vessel configurations retain Modbus TCP at 500 ms poll interval.

Layer 3 (Supervisory Bus) is unchanged. OPC-UA over dedicated VLAN at 1-second historian resolution. The historian node space is extended to include the two new temperature tags and the LE-401 SoC derived tag.

Table 57. Updated Communication Protocol and Network Layer Specifications — Version 2

Layer	Protocol		Medium		Topology	Cycle Rate	Connected Devices
Layer 1	4–20 mA HART rev 7	+	STP 0.5 mm ²	2-core	Point-to-point star	Analogue continuous	TE-101/102/103/104/105, PE-301, TCV-401, TCV-402
Layer 2	Profinet RT Class 1	IO	Cat6A Ethernet	STP	Line / star hybrid	1–4 ms	FE-201, VFD-501
Layer 2	Modbus TCP		Cat6A Ethernet	STP	Star from master	500 ms poll	Satellite vessel PLCs (multi-vessel only)
Layer 3	OPC-UA IEC 62541		Cat6A	dedicated VLAN	Star from SCADA	1 s historian	HMI terminals, historian, engineering workstation

All Ethernet segments via managed switches with VLAN tagging. IEC 62443-3-3 SL2 for OPC-UA endpoints.

13.8 F.H Updated Failure Mode and Effects Analysis — Version 2

The FMEA is updated to reflect the Version 2 I/O schedule and mode structure. New rows are added for TCV-402, TE-104/TE-105 signal loss, and the Mode 1.A.2 battery drain continuation scenario. All Version 1 rows are retained and reviewed; RPN values are unchanged where the mitigation strategy is unaffected by the architecture revision. The critical RPN threshold of 100 remains the hardware interlock trigger per IEC 60812.

Table 58. Updated FMEA — Version 2 (Part 1 of 2)

Failure Mode	Root Cause	Effect	<i>S</i>	<i>O</i>	<i>D</i>	RPN	Mitigation
FE-201 signal loss	Cable fault or transmitter failure	Loop 2 loses PV; VFD runs open-loop	8	3	4	96	Watchdog on signal; fallback to $v_{\min}$ on loss
TCV-401 stuck open	Actuator or positioner failure	Uncontrolled battery discharge; overtemperature risk	9	2	3	54	TE-103 HiHi trips battery isolation; ZS-701 mismatch alarm
TCV-401 stuck closed	Positioner or pneumatic failure	Battery cannot discharge; demand unmet in Mode 1.A.2	5	2	3	30	TE-103 Lo alarm; manual bypass valve
TCV-402 stuck open	Actuator failure	Uncontrolled recharge flow; possible battery overtemperature	7	2	3	42	TE-102 HiHi trips recharge isolation; ZS-702 mismatch alarm
TCV-402 stuck closed	Positioner failure	Battery cannot recharge; SoC degrades over successive batches	5	2	4	40	TE-104 deviation alarm; manual bypass valve
VFD-501 trip	Overtemperature or overcurrent	Loss of loop circulation; coil stagnation	9	3	2	54	VFD fault to PLC DI; auto FAIL_SAFE transition
Loop over-velocity	VFD run-away or loop instability	Coil erosion; long-term hydraulic degradation	10	2	2	40	Hardware overspeed relay independent of PLC; trips VFD contactor

Continued in Part 2.

Table 58 (cont.) — Updated FMEA — Version 2 (Part 2 of 2)

Failure Mode	Root Cause	Effect	<i>S</i>	<i>O</i>	<i>D</i>	RPN	Mitigation
TE-102 multi-point failure	Thermocouple open-circuit	SoC invalid; battery over-charge risk	7	4	4	112	Critical: Redundant thermocouple pair; majority-vote logic in PLC
TE-104/105 signal loss	Cable fault or moisture ingress	Battery energy exchange unverifiable; silent recharge / discharge failure	6	3	5	90	TE-104/105 deviation alarms; verify against TE-101/102
1.A.2 drain continuation	Operator selects Option B at SoC floor	Vessel temperature sacrifice; potential product quality impact	6	3	2	36	SCADA historian logs operator selection with timestamp; post-batch review
PLC processor fault	Hardware fault or watchdog timeout	All loops to FAIL_SAFE simultaneously	10	1	2	20	Hot-standby duplex CPU; switchover <100ms
Modbus TCP link loss	Switch fault or cable break	Master loses satellite data; balancing degrades	6	3	3	54	Satellites revert to local PI on 3 missed polls
Loop pressure loss	Coil or pipe leak	Fluid escape; thermal hazard	9	2	2	36	PE-301 LoLo hardwired to isolation valve; FAIL_SAFE latch
SCADA server loss	Server crash or VLAN fault	Loss of visibility; control continues autonomously	4	3	2	24	PLC independent of SCADA; local HMI maintains Level 1 display

$RPN = S \times O \times D$. Items with $RPN \geq 100$ require hardware interlock independent of PLC software layer. New rows in Version 2: TCV-402 stuck open/closed, TE-104/105 signal loss, 1.A.2 drain continuation.

APPENDIX G

REVISED PLC IMPLEMENTATION AND CONTROL COMMISSIONING

VERSION 2 — CURRENT

14 Appendix G(Revised E)

VERSION NOTICE This appendix supersedes Appendix E (Version 1) and provides the discrete-time mathematics, IEC 61131-3 implementation code, and commissioning procedures consistent with the three-loop control architecture defined in Appendix F. Sections carried forward from Appendix E without change are identified inline. Appendix E is retained as an archived engineering decision record. This appendix should be read in conjunction with Appendix F — all instrument tags, mode references, and loop declarations in this document refer to the Version 2 architecture.

This appendix translates the three-loop control architecture of Appendix F into the discrete-time difference equations, IEC 61131-3 Structured Text (ST) code blocks, and physical commissioning procedures required for implementation on an industrial PLC platform. The content is organised in the sequence a controls engineer would encounter it: mathematics first, pre-commissioning gain estimates second, physical identification procedure third, code implementation fourth, and commissioning sequence last. Ladder logic and relay fallback guidance is retained from Appendix E for environments where ST is not available.

14.1 G.A Discrete-Time PID and PI Algorithms

Carried forward from Appendix E, Sections E.A–E.C, with the addition of Table 59 defining active controller terms per loop.

G.A.1 Velocity Form PID Algorithm

Because a PLC samples at fixed discrete scan intervals Δt , the continuous-time PID integral and derivative terms are approximated using backward Euler discretisation. The velocity (incremental) form is used throughout this framework because it provides bumpless transfer when switching between manual and automatic modes, and naturally prevents integrator windup from propagating through actuator saturation boundaries:

$$\Delta u(k) = K_p [e(k) - e(k-1)] + K_i \Delta t e(k) + \frac{K_d}{\Delta t} [e(k) - 2e(k-1) + e(k-2)].$$

The accumulated actuator output is:

$$u(k) = u(k-1) + \Delta u(k).$$

14.1.1 G.A.2 Derivative Kick Mitigation

When the operator changes the setpoint SP via the HMI, the error $e(k)$ shifts instantaneously. In the standard form above, the derivative term produces an impulsive spike that drives the actuator to saturation. To prevent this, the derivative is computed on the

process variable PV rather than the error:

$$\text{Derivative term} = -\frac{K_d}{\Delta t} [PV(k) - 2PV(k-1) + PV(k-2)].$$

This leaves the response to disturbances unchanged while eliminating the kick on commanded setpoint steps.

14.1.2 G.A.3 Integral Anti-Windup Clamping

Large thermal masses cause the integral accumulator to grow continuously during heat-up while the actuator is saturated. To prevent overshoot on arrival at setpoint, the per-scan integral increment $\Delta I(k)$ is frozen whenever the actuator is at a saturation boundary and the error would drive it further into saturation. The accumulated output $u(k)$ is not zeroed — only the increment for the current scan is suppressed:

$$\begin{aligned} \text{If } u(k) \geq u_{\max} \text{ and } e(k) > 0: \quad \Delta I(k) &= 0, \\ \text{If } u(k) \leq u_{\min} \text{ and } e(k) < 0: \quad \Delta I(k) &= 0. \end{aligned}$$

14.1.3 G.A.4 First-Order Low-Pass Sensor Filter (EMA)

Raw analogue signals from RTDs and electromagnetic flowmeters carry turbulent process noise. An exponential moving average (EMA) filter is applied to each analogue input before it enters any PID calculation block:

$$PV_{\text{filtered}}(k) = \alpha \cdot PV_{\text{raw}}(k) + (1 - \alpha) \cdot PV_{\text{filtered}}(k-1),$$

where the smoothing constant α is derived from the desired filter time constant τ_f and the loop scan interval Δt :

$$\alpha = \frac{\Delta t}{\tau_f + \Delta t}, \quad 0 < \alpha \leq 1.$$

A lower α produces heavier filtering and more phase lag. The filter time constant τ_f must be set significantly shorter than the process time constant τ to avoid artificially inflating the measured dead time θ during the step test identification procedure in Section G.D.

14.1.4 G.A.5 Active Terms per Loop

Table 59 defines which PID terms are active for each loop and the physical justification for each omission. This table is the authoritative reference for configuring native PID instruction blocks (e.g. `PID_Compact` in TIA Portal or `PIDE` in Studio 5000).

Table 59. Active Controller Terms by Loop

Loop Name			P	I	D	Justification
1	Vessel temp	batch	✓	✓	✓	Slow integrating process with dead time; all three terms required. D on PV only to prevent setpoint-step kick.
2	Pump flow rate		✓	✓	×	Fast hydraulic loop; D amplifies flowmeter turbulent noise without improving response. PI sufficient.
3	Battery recharge		✓	✓	×	Very slow large thermal mass; D amplifies thermocouple noise. PI sufficient for drift-free temperature hold.
4	Multi-vessel balance		✓	×	×	Supervisory trim loop; I causes inter-vessel windup competition. P-only with intentionally low gain.

14.2 G.B Per-Loop Scan Interval Derivation

Replaces Appendix E, Section E.D. Extends the single-loop scan interval formula to all four loops independently.

The Nyquist–Shannon criterion applied to process control requires a minimum of ten to twenty sample points per half-period of the dominant oscillation mode to maintain closed-loop stability. For a first-order-plus-dead-time process, this translates to the scan interval recommendation:

$$\Delta t_{\text{scan}} \leq \frac{\tau}{20},$$

where τ is the dominant process time constant of the loop. This criterion is applied independently to each loop because their time constants differ by orders of magnitude. Applying a single universal scan interval equal to the fastest loop (Loop 2) to all loops wastes CPU cycles on Loops 1 and 3; applying the slowest loop’s interval to Loop 2 produces an unstable hydraulic controller.

The correct implementation assigns each loop to a separate cyclic interrupt task in the PLC task configuration, each executing at its own rate. Table 60 summarises the derivation and recommended intervals.

Table 60. Per-Loop Scan Interval Derivation

Loop	Dominant τ	Source of τ	Recommended Δt	PLC Task Type	Notes
1	$S_{t,\text{vessel}}$ (hundreds to thousands of seconds)	Step 10 thermal time constant of vessel	$S_{t,\text{vessel}}/20$	Cyclic interrupt, low priority	For reference case $S_t = 6665\text{ s}$: $\Delta t \approx 330\text{ s}$
2	$\tau_{\text{pump}} = 2\text{--}8\text{ s}$ (impeller mechanical spin-up)	Pump datasheet inertia; Step 5 hydraulic characterisation	1–4 s	Cyclic interrupt, high priority	Must be faster than Profinet IO cycle; set to match FE-201 response time
3	$S_{t,\text{battery}}$ (larger than $S_{t,\text{vessel}}$)	Step 10 battery thermal mass calculation	$S_{t,\text{battery}}/20$	Cyclic interrupt, low priority	Will be slower than Loop 1 given larger battery thermal mass
4	Governed by Modbus TCP poll rate	Network configuration	500 ms	Modbus exchange task	Matched to inter-PLC communication cycle; not a dedicated control task

Loops 1 and 3 share a low-priority task tier but execute independently with their own timers.

Loop 2 must be in a higher-priority interrupt than Loops 1 and 3 to guarantee hydraulic safety response.

The EMA filter time constant τ_f for each loop must be set shorter than the loop's Δt to avoid aliasing. Recommended starting values: $\tau_f = 0.1 \times \tau_{\text{process}}$ for each loop, which gives α values that provide meaningful noise rejection without introducing sufficient phase lag to destabilise the loop.

14.3 G.C First-Principles Gain Derivation from Framework Outputs

New in Version 2. Not present in Appendix E.

Before the physical system is built, starting PID gain estimates can be derived directly from the VCH framework calculation outputs. These estimates carry uncertainty from fabrication tolerances, actual pump curve deviations, and coil fouling state, but they provide a physically grounded starting point significantly closer to the final tuned values than generic default parameters. They also serve as a validation check: if the commissioning step test (Section G.D) yields values that diverge from these estimates by more than 30%, the physical installation should be inspected for discrepancies from the design.

Every loop transfer function is approximated as a first-order process with dead time:

$$G(s) = \frac{K_{\text{proc}} e^{-\theta s}}{\tau s + 1}.$$

The three parameters K_{proc} , τ , and θ are extracted per loop as follows.

14.3.1 G.C.1 Loop 1 — Vessel Batch Temperature

Process gain $K_{\text{proc},1}$: The process gain expresses how many degrees Celsius the vessel temperature changes per percentage point of TCV-401 opening at steady state. It is estimated from the Step 6 thermal performance evaluation:

$$K_{\text{proc},1} = \frac{\dot{Q}_{\text{max}} \cdot \Delta t_{\text{scan},1}}{m_{\text{fluid}} \cdot C_p \cdot 100} \quad \left[\frac{^\circ\text{C}}{\%} \right],$$

where $\dot{Q}_{\text{max}}$ is the maximum coil heat transfer rate from Step 6, m_{fluid} is the vessel fluid mass from Step 3, C_p is the specific heat capacity from Step 2, and the factor of 100 normalises per percentage of valve opening. For a large vessel this value is typically in the range 0.05–0.20 °C/%.

Time constant τ_1 : The dominant thermal time constant is the vessel thermal time constant S_t computed in Step 10:

$$\tau_1 = S_t.$$

Dead time θ_1 : The dead time is the coil transit time plus the RTD response lag. The transit time is derived from the coil geometry and operating velocity from Steps 4 and 7:

$$\theta_{\text{transit}} = \frac{L_{\text{coil}}}{v_{\text{operating}}}.$$

Adding 5–15 seconds for PT100 RTD Class A response lag:

$$\theta_1 = \theta_{\text{transit}} + \theta_{\text{RTD}} \approx 30\text{--}120 \text{ s}.$$

IMC tuning (recommended for Loop 1): IMC tuning is preferred over Ziegler–Nichols for Loop 1 because the large dead time ratio θ_1/τ_1 makes ZN gains dangerously aggressive. The IMC formulas with desired closed-loop time constant λ are:

$$K_p = \frac{\tau_1}{K_{\text{proc},1}(\lambda + \theta_1)}, \quad T_i = \tau_1, \quad T_d = \frac{\theta_1}{2}.$$

Set $\lambda = \tau_1/3$ as the starting point for moderate aggressiveness. Increase λ if the commissioning response is oscillatory; decrease if it is excessively sluggish.

14.3.2 G.C.2 Loop 2 — Pump Flow Rate

Process gain $K_{\text{proc},2}$: The flow velocity change per percentage point of VFD speed is read directly from the pump characteristic curve obtained in Step 5:

$$K_{\text{proc},2} = \frac{\Delta v}{\Delta \text{VFD}\%} \quad \left[\frac{\text{m/s}}{\%} \right].$$

Time constant τ_2 : The pump mechanical spin-up time constant is read from the drive datasheet:

$$\tau_2 \approx 2\text{--}8 \text{ s}.$$

Dead time θ_2 : The dead time for Loop 2 is the sum of the Profinet IO cycle time and the FE-201 electromagnetic flowmeter response time:

$$\theta_2 \approx 1\text{--}3 \text{ s}.$$

Ziegler–Nichols tuning (acceptable for Loop 2): Because θ_2/τ_2 is small (typically < 0.5), the standard ZN PI formulas are adequate:

$$K_p = \frac{0.9 \tau_2}{K_{\text{proc},2} \cdot \theta_2}, \quad T_i = 3.3 \theta_2.$$

No derivative term is applied.

14.3.3 G.C.3 Loop 3 — Battery Recharge

Process gain $K_{\text{proc},3}$: The temperature change of the battery core per percentage point of TCV-402 opening, estimated from the battery thermal mass and maximum recharge power available from the driver:

$$K_{\text{proc},3} = \frac{\dot{Q}_{\text{recharge,max}} \cdot \Delta t_{\text{scan},3}}{m_{\text{battery}} \cdot C_{p,\text{fluid}} \cdot 100} \left[\frac{^\circ\text{C}}{\%} \right],$$

where m_{battery} and $\dot{Q}_{\text{recharge,max}}$ come from the Step 10 battery sizing calculation.

Time constant τ_3 : The battery thermal time constant from Step 10:

$$\tau_3 = S_{t,\text{battery}} > S_{t,\text{vessel}}.$$

Dead time θ_3 : Recharge circuit transit time plus TE-102 thermocouple response lag:

$$\theta_3 = \frac{L_{\text{recharge circuit}}}{v_{\text{recharge}}} + \theta_{\text{TC}}.$$

IMC tuning for Loop 3: Apply the same IMC formulas as Loop 1, but set $\lambda = \tau_3/2$ rather than $\tau_3/3$. The recharge loop does not need fast response; a more conservative λ produces stable drift-free control with lower risk of overshoot into battery overtemperature:

$$K_p = \frac{\tau_3}{K_{\text{proc},3}(\lambda + \theta_3)}, \quad T_i = \tau_3, \quad T_d = 0.$$

14.3.4 G.C.4 Pre-Commissioning Gain Summary Table

Table 61 records the first-principles gain estimates computed from the framework outputs. This table should be completed before commissioning and retained alongside the commissioning record as a validation baseline.

Table 61. Pre-Commissioning Gain Estimate Record (to be completed from framework outputs)

LoopName			K_{proc} source	τ source	K_p	T_i (s)	T_d (s)	Method
1	Vessel temp	batch	Step 6 $\dot{Q}_{\text{max}}$, Step 3 m_{fluid}	Step 10 S_t	<i>calc.</i>	<i>calc.</i>	<i>calc.</i>	IMC, $\lambda = \tau/3$
2	Pump rate	flow	Step 5 pump curve	Drive datasheet	<i>calc.</i>	<i>calc.</i>	0	ZN PI
3	Battery recharge		Step 10 $\dot{Q}_{\text{recharge}}$, battery mass	Step 10 $S_{t,\text{bat}}$	<i>calc.</i>	<i>calc.</i>	0	IMC, $\lambda = \tau/2$
4	Multi-vessel balance		Not applicable	Not applicable	0.5–1.0	∞ (P only)	0	Empirical

Fill in calculated values before FAT. Compare against step test results in Section G.D.

Divergence >30% between pre-commissioning estimate and step test result warrants physical inspection.

14.4 G.D Step Test Identification and Gain Refinement Procedure

New in Version 2. Not present in Appendix E.

The step test (bump test) is the physical measurement procedure by which K_{proc} , τ , and θ are measured on the real installed system and used to confirm or correct the pre-commissioning gain estimates from Section G.C. It must be performed on each loop independently in the commissioning sequence defined in Section G.F. The following procedure applies to Loops 1, 2, and 3. Loop 4 is tuned empirically and does not require a step test.

14.4.1 G.D.1 General Step Test Procedure

Step 1. Stabilise the loop. Place the controller in manual mode. Allow the process variable to reach a steady operating point — no more than $\pm 0.2^\circ\text{C}$ drift over five minutes for temperature loops, $\pm 0.02\text{ m/s}$ over thirty seconds for the flow loop. Record the steady-state PV value PV_0 and the current manual output u_0 .

Step 2. Execute the bump. Make a step change of 10–20% of the output range. The bump must be large enough to produce a PV response clearly above the noise floor, and small enough to avoid unacceptable process disturbance. Log all relevant tags from this moment at the loop’s scan interval.

Step 3. Hold until steady state. Allow the PV to travel through its full response and approach a new steady state. For Loop 1 this may take thirty minutes or more given the vessel thermal mass.

Step 4. Read the response curve. From the logged trend identify:

- θ_{measured} : time from the bump instant to when the PV first visibly departs from PV_0 . Draw a tangent at the inflection point (steepest slope) for a more precise estimate.

- τ_{measured} : time from the end of dead time to when the PV has risen to 63.2% of its total change ΔPV_{final} .
- $K_{\text{proc,measured}}$: final steady-state PV change divided by the bump size:

$$K_{\text{proc,measured}} = \frac{\Delta PV_{\text{final}}}{\Delta u_{\text{bump}}}.$$

Step 5. Recalculate gains. Substitute $K_{\text{proc,measured}}$, τ_{measured} , and θ_{measured} into the relevant tuning formula (IMC for Loops 1 and 3; ZN for Loop 2). Record the revised values in Table 61 alongside the pre-commissioning estimates.

Step 6. Restore and re-stabilise. Return the output to u_0 , allow the process to re-stabilise, then place the controller in automatic with the calculated gains before proceeding.

14.4.2 G.D.2 Loop-Specific Notes

Loop 1 — Vessel Batch Temperature. Run the step test at normal operating temperature, not cold. The cold-start heat transfer coefficient (from Step 6) differs from the steady-state value, producing a different K_{proc} at cold conditions. Gains tuned at operating temperature will be slightly aggressive at cold-start but safe — the vessel heats slightly faster initially, which is acceptable. Gains tuned cold will be sluggish at operating temperature, reducing batch throughput.

The EMA filter interaction must be respected: the step test must be run with the production value of α already loaded. A higher α than intended inflates the measured θ by adding apparent lag, causing the tuning formula to produce an under-aggressive K_p . Do not run the step test with $\alpha = 1$ (filter bypassed) and activate the filter afterwards.

Loop 2 — Pump Flow Rate. The step test for Loop 2 is fast and straightforward. The response will be visible within seconds of the bump. Ensure the vessel loop is in manual during the Loop 2 step test so that TCV-401 does not modulate in response to the resulting flow change and distort the FE-201 reading.

Loop 3 — Battery Recharge. The recharge loop step test requires the battery to be at a stable intermediate charge state — not at $T_{\text{bat,max}}$ (which produces a truncated response) and not cold (which produces an unrepresentative K_{proc}). Target 50–70% of the battery’s usable temperature range for the step test starting point.

14.4.3 G.D.3 Acceptance Criteria

A step test result is acceptable when:

- The measured K_{proc} , τ , and θ agree with the pre-commissioning estimates to within $\pm 30\%$.
- The PV response curve is smooth and monotonic with no oscillation during the open-loop bump.
- The signal-to-noise ratio of ΔPV_{final} to the steady-state noise band is at least 5:1, confirming the bump size was sufficient.

If these criteria are not met, Step 4 of Section G.D.1 should be repeated with the bump size adjusted, or the physical installation inspected for discrepancies from the design.

14.5 G.E IEC 61131-3 Structured Text Implementation — Three Loop Blocks

Replaces Appendix E, Section E.E. Three independent ST blocks are provided, one per controlled loop. Each block is self-contained with its own VAR declarations and scan interval parameter. Blocks are intended for implementation in separate cyclic interrupt tasks as defined in Section G.B.

14.5.1 G.E.1 Loop 1 — Vessel Batch Temperature (PID)

Listing G.1 — Loop 1: Vessel Batch Temperature PID — TE-103 → TCV-401

```

VAR
    // --- Inputs ---
    PV_Raw_TE103      : REAL;    // Raw analogue input from TE-103 (deg C)
    SP_Vessel         : REAL;    // Setpoint: T_target or T_hib per active mode
    Output_UpperLim   : REAL;    // Mode-dependent upper limit (% TCV-401
                                // Set by mode logic layer: Appendix F, Table
                                // F.2
    // --- Tuning Parameters (from Sections G.C.1 and G.D, Loop 1) ---
    Kp1               : REAL;    // Proportional gain
    Ti1               : REAL;    // Integral time constant (s) = tau_1
    Td1               : REAL;    // Derivative time constant (s) = theta_1 / 2
    dt1               : REAL;    // Scan interval (s) = S_t_vessel / 20
    Alpha1            : REAL;    // EMA filter constant = dt1 / (tau_f1 + dt1)

    // --- Memory (RETAIN: preserved across PLC power cycle) ---
    PV1_Filtered      : REAL;
    PV1_Prev1         : REAL;
    PV1_Prev2         : REAL;
    Error1_Prev       : REAL;
    TCV401_Out        : REAL;    // Accumulated actuator output (% opening)

    // --- Intermediates ---
    Error1            : REAL;
    Delta_U1          : REAL;
    P1_Term           : REAL;
    I1_Term           : REAL;
    D1_Term           : REAL;
END_VAR

// --- 1. EMA Sensor Filter ---
PV1_Filtered := (Alpha1 * PV_Raw_TE103)
               + ((1.0 - Alpha1) * PV1_Filtered);

// --- 2. Error ---
Error1 := SP_Vessel - PV1_Filtered;

// --- 3. Proportional Term ---

```

```

P1_Term := Kp1 * (Error1 - Error1_Prev);

// --- 4. Derivative Term (PV-based: no setpoint-step kick) ---
D1_Term := -(Kp1 * Td1 / dt1)
           * (PV1_Filtered - (2.0 * PV1_Prev1) + PV1_Prev2);

// --- 5. Integral with Anti-Windup ---
IF (TCV401_Out >= Output_UpperLim AND Error1 > 0.0) OR
   (TCV401_Out <= 0.0 AND Error1 < 0.0) THEN
    I1_Term := 0.0;
ELSE
    I1_Term := Kp1 * (dt1 / Ti1) * Error1;
END_IF;

// --- 6. Velocity Form Accumulation ---
Delta_U1 := P1_Term + I1_Term + D1_Term;
TCV401_Out := TCV401_Out + Delta_U1;

// --- 7. Output Saturation (mode-dependent upper limit) ---
IF TCV401_Out > Output_UpperLim THEN TCV401_Out := Output_UpperLim; END_IF;
IF TCV401_Out < 0.0 THEN TCV401_Out := 0.0; END_IF;

// --- 8. Memory Update ---
Error1_Prev := Error1;
PV1_Prev2 := PV1_Prev1;
PV1_Prev1 := PV1_Filtered;

// --- 9. Heartbeat for Software Watchdog (Section G.I) ---
WD_Heartbeat := NOT WD_Heartbeat;

```

14.5.2 G.E.2 Loop 2 — Pump Flow Rate (PI)

Listing G.2 — Loop 2: Pump Flow Rate PI — FE-201 → VFD-501

```

VAR
  // --- Inputs ---
  PV_Raw_FE201      : REAL;          // Raw velocity signal from FE-201 (m/s)
  SP_Velocity       : REAL;          // Velocity setpoint from mode logic
  (m/s)
  V_Max             : REAL := 2.0;    // Erosion ceiling (m/s) --- hard limit
  V_Min             : REAL := 0.3;    // Stagnation floor (m/s) --- hard limit

  // --- Tuning Parameters (from Sections G.C.2 and G.D, Loop 2) ---
  Kp2               : REAL;          // Proportional gain
  Ti2               : REAL;          // Integral time constant (s) = 3.3 * theta_2
  dt2               : REAL;          // Scan interval (s): 1--4 s per Section G.B
  Alpha2            : REAL;          // EMA filter constant

  // --- Memory ---
  PV2_Filtered      : REAL;
  Error2_Prev       : REAL;
  VFD501_Out        : REAL;          // Accumulated actuator output (% VFD speed)

  // --- Intermediates ---
  Error2            : REAL;
  Delta_U2          : REAL;
  P2_Term           : REAL;
  I2_Term           : REAL;
END_VAR

// --- 1. EMA Sensor Filter ---
PV2_Filtered := (Alpha2 * PV_Raw_FE201)
               + ((1.0 - Alpha2) * PV2_Filtered);

// --- 2. Error ---
Error2 := SP_Velocity - PV2_Filtered;

// --- 3. Proportional Term ---
P2_Term := Kp2 * (Error2 - Error2_Prev);

// --- 4. Integral with Anti-Windup ---
IF (VFD501_Out >= 100.0 AND Error2 > 0.0) OR
   (VFD501_Out <= 20.0 AND Error2 < 0.0) THEN
  I2_Term := 0.0;
ELSE
  I2_Term := Kp2 * (dt2 / Ti2) * Error2;
END_IF;

// --- 5. Velocity Form Accumulation (PI: no D term) ---
Delta_U2 := P2_Term + I2_Term;
VFD501_Out := VFD501_Out + Delta_U2;

```

```
// --- 6. Output Saturation ---  
IF VFD501_Out > 100.0 THEN VFD501_Out := 100.0; END_IF;  
IF VFD501_Out < 20.0 THEN VFD501_Out := 20.0; END_IF;  
// 20% minimum prevents coil stagnation regardless of setpoint  
  
// --- 7. Hard Velocity Ceiling Check (independent of PID output) ---  
IF PV2_Filtered > V_Max THEN  
    VFD501_Out := VFD501_Out - (Kp2 * (PV2_Filtered - V_Max));  
    IF VFD501_Out < 20.0 THEN VFD501_Out := 20.0; END_IF;  
END_IF;  
  
// --- 8. Memory Update ---  
Error2_Prev := Error2;
```

14.5.3 G.E.3 Loop 3 — Battery Recharge (PI)

Listing G.3 — Loop 3: Battery Recharge PI — TE-102 → TCV-402

```

VAR
  // --- Inputs ---
  PV_Raw_TE102      : REAL;    // Majority-voted TE-102 signal (deg C)
  SP_Battery         : REAL;    // Battery target charge temp T_bat_target
  T_Bat_Max          : REAL;    // Battery overtemperature ceiling (hard
                                limit)

  // --- Tuning Parameters (from Sections G.C.3 and G.D, Loop 3) ---
  Kp3                : REAL;    // Proportional gain
  Ti3                : REAL;    // Integral time constant (s) = tau_3
  dt3                : REAL;    // Scan interval (s) = S_t_battery / 20
  Alpha3             : REAL;    // EMA filter constant

  // --- Memory ---
  PV3_Filtered       : REAL;
  Error3_Prev        : REAL;
  TCV402_Out         : REAL;    // Accumulated actuator output (% TCV-402
                                opening)

  // --- Intermediates ---
  Error3             : REAL;
  Delta_U3           : REAL;
  P3_Term            : REAL;
  I3_Term            : REAL;
END_VAR

// --- 1. EMA Sensor Filter (majority-vote assumed pre-processed) ---
PV3_Filtered := (Alpha3 * PV_Raw_TE102)
               + ((1.0 - Alpha3) * PV3_Filtered);

// --- 2. Error ---
Error3 := SP_Battery - PV3_Filtered;

// --- 3. Proportional Term ---
P3_Term := Kp3 * (Error3 - Error3_Prev);

// --- 4. Integral with Anti-Windup ---
IF (TCV402_Out >= 100.0 AND Error3 > 0.0) OR
   (TCV402_Out <= 0.0 AND Error3 < 0.0) THEN
  I3_Term := 0.0;
ELSE
  I3_Term := Kp3 * (dt3 / Ti3) * Error3;
END_IF;

// --- 5. Velocity Form Accumulation (PI: no D term) ---
Delta_U3 := P3_Term + I3_Term;
TCV402_Out := TCV402_Out + Delta_U3;

```

```

// --- 6. Output Saturation ---
IF TCV402_Out > 100.0 THEN TCV402_Out := 100.0; END_IF;
IF TCV402_Out < 0.0 THEN TCV402_Out := 0.0; END_IF;

// --- 7. Hard Overtemperature Ceiling ---
// If battery core exceeds T_Bat_Max, close recharge valve immediately.
IF PV3_Filtered >= T_Bat_Max THEN
    TCV402_Out := 0.0;
END_IF;

// --- 8. Memory Update ---
Error3_Prev := Error3;

```

14.6 G.F Loop Commissioning Sequence and Mode Transition Testing

New in Version 2. Not present in Appendix E.

Tuning all three loops simultaneously causes interaction effects that prevent stable convergence. The correct commissioning sequence is mandatory: each loop must be stable before the next is activated.

14.6.1 G.F.1 Mandatory Commissioning Order

Phase 1 — Loop 2 (Pump Flow Rate, PI). Loop 2 is commissioned first because it is the actuator environment in which Loop 1 operates. With Loops 1 and 3 in manual and TCV-401 held at a fixed mid-range opening, activate Loop 2 in automatic with the calculated gains. Execute the Loop 2 step test per Section G.D and refine gains. Verify that the velocity setpoint is tracked within ± 0.05 m/s under steady conditions and that the hard ceiling and floor limits engage correctly. Loop 2 is considered commissioned when it holds a commanded velocity step without overshoot beyond 10% and settles within three scan cycles.

Phase 2 — Loop 3 (Battery Recharge, PI). With Loop 2 commissioned and running, activate Loop 3 in automatic. Loop 3 is slow and benign; it does not interact with Loop 2 during normal operation. Execute the Loop 3 step test with the battery at 50–70% charge state per Section G.D.2. Verify that TCV-402 responds smoothly to setpoint changes and that the hard overtemperature ceiling at $T_{\text{bat,max}}$ closes the valve immediately when triggered. Loop 3 is considered commissioned when it holds $T_{\text{bat,target}}$ with steady-state error below 0.5°C over a 30-minute hold period.

Phase 3 — Loop 1 (Vessel Batch Temperature, PID). With Loops 2 and 3 both commissioned and running in automatic, activate Loop 1. The step test for Loop 1 must now be performed in this coupled condition — Loop 2 will respond to the flow changes induced by TCV-401 modulation, and Loop 3 will continue running in the background. This is the correct commissioning environment because it represents real operating conditions. Refine gains and verify the following acceptance criteria:

- Vessel temperature reaches T_{target} with overshoot $\leq 2^{\circ}\text{C}$.

- Steady-state error in Mode B hold $\leq 0.3^{\circ}\text{C}$ over 60 minutes.
- No sustained oscillation after setpoint step under any mode.

Phase 4 — Mode Transition Testing. With all three loops commissioned, the mode transitions defined in Appendix F, Section F.C must be tested in the sequence defined in Section G.F.2.

14.6.2 G.F.2 Mode Transition Test Sequence

- T1. STANDBY $\rightarrow$ MODE_1A1.** Issue batch start command via HS-602. Confirm that Loop 2 ramps to demand velocity, Loop 1 begins heating toward T_{target} , and Loop 3 continues recharging. Verify no velocity ceiling breach during ramp-up.
- T2. MODE_1A1 $\rightarrow$ MODE_B.** Allow the vessel to reach T_{target} . Confirm that Loop 1 setpoint remains at T_{target} and TCV-401 throttles smoothly to a low steady-state opening. Confirm that Loop 2 setpoint drops to the $\dot{Q}_{\text{loss}}$ velocity. **Watch for overshoot on this transition** — the sudden reduction in Loop 1's output upper limit at arrival can cause the integrator to kick if the arrival deadband δ is set too small. If overshoot exceeds 2°C , increase δ or reduce K_p by 15% and repeat.
- T3. MODE_B $\rightarrow$ MODE_C.** Issue hibernate command with $\phi = 0.5$ as the test case. Confirm that T_{hib} is computed correctly from $\phi \cdot \dot{Q}_{\text{hold}}$, that Loop 1 setpoint updates to T_{hib} , and that the vessel cools to T_{hib} without oscillation. Confirm Loop 2 setpoint scales proportionally downward.
- T4. MODE_C $\rightarrow$ MODE_B (Wake-up).** Issue wake command. Confirm that T_{target} is restored as Loop 1 setpoint and the system re-enters Mode 1.A.1 or 1.A.2 correctly based on the $\dot{Q}_{\text{pump,max}}$ evaluation.
- T5. MODE_1A2 SoC Floor Alarm Test.** Simulate the SoC floor condition by manually setting $\text{SoC}_{\text{floor}}$ to the current SoC value. Confirm that AL-001 and AL-002 activate, D0-104 energises, and the HMI decision prompt appears on Level 1 with both options available. Confirm that no automatic state transition occurs. Test both Option A (graceful shutdown) and Option B (continue with reduced output limit) and verify the correct PLC response to each.
- T6. FAIL_SAFE Latch Test.** Simulate a velocity ceiling breach by commanding VFD-501 to a speed that produces $v > v_{\text{HiHi}}$ in a controlled test condition. Confirm that FAIL_SAFE latches within one watchdog scan cycle, all outputs move to de-energised positions, D0-103 pulse halts, D0-101 de-energises, and the field alarm beacons activate. Confirm that the latch cannot be cleared by any automated logic path and requires manual HMI acknowledgement. Log the transition timestamp from the SCADA historian and verify it matches the physical alarm event within one scan cycle.

All six transition tests must pass before the system is considered commissioned. Results must be recorded in the Factory Acceptance Test (FAT) documentation and signed off before site delivery.

14.7 G.G Ladder Logic (LD) Implementation Guidelines

Carried forward from Appendix E, Section E.F, without change. The following guidelines apply to environments where IEC 61131-3 Structured Text is not available. The three-loop structure of Version 2 maps to three separate LD program blocks, each residing in its own cyclic interrupt task at the scan intervals defined in Section G.B.

When implementing the VCH control loops in LD, the following structural guidelines apply:

- **Cyclic Execution via Interrupts.** The LD network for each loop should reside inside a dedicated cyclic hardware interrupt block (e.g. OB30 in Siemens TIA Portal or a Periodic Task in Rockwell RSLogix). The block execution rate must be statically configured to the per-loop Δt_{scan} values from Section G.B. A single global interrupt block for all loops is not acceptable in Version 2 — the loop scan rates differ by orders of magnitude.
- **Mathematical Compute Blocks.** The EMA sensor filter cannot be executed with basic add/multiply coils. A dedicated Compute (CPT) instruction block should calculate the $(1 - \alpha)$ filtering formula in a single rung per loop.
- **Native PID Instruction Utilisation.** Use native PID instruction blocks (PID_Compact, PIDE, or equivalent). Configure each block to read the *filtered* PV rather than the raw physical analogue channel. For Loop 1 configure derivative-on-PV mode explicitly; for Loops 2 and 3 disable the derivative term entirely in the instruction block parameters.
- **Boolean Anti-Windup Triggers.** Use XIC/XIO contact structures to evaluate actuator saturation. Route a parallel rung branch to a MOV instruction that zeros the integral increment register when saturation and error direction conditions are simultaneously true. Do not zero the accumulated output register — only the increment.
- **Output Limit Rungs.** The mode-dependent upper limit for Loop 1 (Output_UpperLim) must be written by the mode logic rung before the Loop 1 PID rung executes in each scan. Rung order within the LD program is therefore significant for Loop 1.

14.8 G.H Hardwired Relay-Based Control (Non-PLC Architecture)

Carried forward from Appendix E, Section E.G, without change. The operational limitations noted below remain disqualifying for any installation requiring Mode 1.A.2, the SoC decision alarm, or multi-vessel balancing (Loop 4).

For smaller, isolated hydronic skids, or as a robust fail-safe override to the primary PLC, a traditional hardwired electromechanical control architecture can be employed without microprocessor intervention.

- **Bang-Bang (On/Off) Control Profiles.** Proportional-Integral modulation is impossible without a dedicated controller. Continuous analogue temperature sensors (RTDs) are replaced with mechanical thermostats or liquid-filled capillary thermal switches that snap closed when bulk fluid temperature drops below the physical setpoint.

- **Relay Logic and Contactors.** Thermostat closure energises a 24 VDC or 120 VAC Master Control Relay (MCR) coil. The MCR pulls in the heavy-duty main motor contactor, providing three-phase power directly to the primary utility heater or fixed-speed circulation pump.
- **Hardware Safety Interlocks.** Physical flow switches (paddle or thermal dispersion type) and high-limit pressure switches (capsule type) are wired in series with the MCR coil. Loss of flow or pressure surge breaks the series circuit instantly, dropping the contactor and halting the loop.
- **Operational Limitations.** Hardwired relay logic is immune to software faults, network crashes, and EMI, but suffers from thermal hysteresis and cannot implement dynamic load balancing, thermal battery peak shaving, the Mode 1.A.2 SoC decision prompt, or the hibernation mode. A hardwired network restricts the framework to baseline direct charging and standby states only. It must not be specified for any commercial installation requiring the full four-mode vessel architecture defined in Appendix F.

14.9 G.I Watchdog Timer and FAIL_SAFE State Implementation

Carried forward from Appendix E, Section E.H, with amendments to tag references and the addition of the Loop 1 heartbeat toggle in Section G.E.1.

Two complementary watchdog mechanisms are required. A software watchdog detects PLC scan-cycle failures and task overruns. A hardware watchdog provides a fail-safe physically independent of the PLC processor and output cards.

Software Watchdog: Loop 1's ST block (Section G.E.1) toggles `WD_Heartbeat` on every execution. A separate high-priority watchdog task monitors this flag. If the heartbeat is not refreshed within two consecutive watchdog periods, all control outputs are forced to de-energised positions and `FAIL_SAFE_Active` is latched.

Hardware Watchdog: D0-103 outputs a 1 Hz pulse train to an external hardware watchdog relay module. This relay holds D0-101 (VFD-501 run enable) closed only while the pulse train is active. If the pulse train stops for any reason — PLC halt, output card failure, or loss of 24 VDC control supply — the relay drops out and the VFD loses run permission independently of the PLC software state.

Listing G.4 — Software Watchdog and FAIL_SAFE Latch — High-Priority 1-second Task

```
VAR
    WD_Heartbeat      : BOOL;           // Toggled by Loop 1 each execution
    WD_Prev           : BOOL;
    WD_MissCount       : UINT := 0;
    WD_TripThreshold   : UINT := 2;      // FAIL_SAFE after 2 missed scans
    Velocity_Live      : REAL;           // From FE-201 signal chain
    Velocity_HiHi      : REAL := 2.5;    // Erosion HiHi per Appendix F, F.F
    HW_WD_Toggle       : BOOL;
    HW_WD_Pulse        : BOOL;           // Physical DO to hardware WD relay
    FAIL_SAFE_Active    : BOOL := FALSE;
    PLC_State          : UINT;
    TCV401_Demand      : REAL;
```

```

    TCV402_Demand      : REAL;           // New in Version 2
    VFD501_Demand      : REAL;
    Alarm_Beacon       : BOOL;
END_VAR

// --- Block 1: Hardware Watchdog Pulse (1 Hz) ---
HW_WD_Toggle := NOT HW_WD_Toggle;
HW_WD_Pulse  := HW_WD_Toggle;

// --- Block 2: Software Heartbeat Monitor ---
IF WD_Heartbeat = WD_Prev THEN
    WD_MissCount := WD_MissCount + 1;
ELSE
    WD_MissCount := 0;
END_IF;
WD_Prev := WD_Heartbeat;

// --- Block 3: FAIL_SAFE Trigger Evaluation ---
IF (WD_MissCount >= WD_TripThreshold) OR
   (Velocity_Live > Velocity_HiHi) THEN
    FAIL_SAFE_Active := TRUE;
END_IF;

// --- Block 4: Safe-State Output Enforcement ---
IF FAIL_SAFE_Active THEN
    PLC_State      := 99;           // FAIL_SAFE state code
    TCV401_Demand := 0.0;          // Battery discharge isolated
    TCV402_Demand := 0.0;          // Recharge isolated (Version 2 addition)
    VFD501_Demand := 0.0;          // Loop circulation stopped
    Alarm_Beacon  := TRUE;
    HW_WD_Pulse   := FALSE;        // Pulse halted; WD relay drops; VFD loses
                                   enable
END_IF;

```

Two FAT verification requirements apply before commissioning sign-off. First, the hardware watchdog relay drop-out time must be shorter than the VFD-501 deceleration ramp time, ensuring the drive reaches zero speed before any mechanical isolation valve actuates. Second, `FAIL_SAFE_Active` must be confirmed as non-resettable by any automated logic path; reset requires explicit operator acknowledgement on the Level 3 HMI screen defined in Appendix F, Section F.F.

APPENDIX H
WORKED DESIGN EXAMPLE
PALM OIL STORAGE AND BATCH PROCESSING SYSTEM
KATHMANDU, NEPAL
VERSION 1 — ARCHIVED

15 Appendix H (Archived)

Methodology Supersession Notice. This worked example was developed under the First Edition VCH Sizing Framework methodology and is retained in full as an archived record of the original calculation approach, for engineering history and portfolio documentation purposes only.

The specific technical issue: the governing process load established in Step 1 is correctly resolved into three thermal phases — sensible heating of solid (Q_{s1}), latent heat of fusion at constant T_{melt} (Q_{lat}), and sensible heating of liquid (Q_{s2}) — together accounting for the full transient batch energy. However, Step 6, Section 6.4 (the NTU-Effectiveness Check) treats the *entire* batch process as a single isothermal phase-change condition at constant T_{melt} , applying a steady-state, single-phase NTU-effectiveness formulation across the whole batch. This is only strictly valid for the latent-heat portion of the load (Phase 2, $\approx 82\%$ of total energy). It does not correctly represent Phases 1 and 3, during which the bulk process temperature is rising continuously and the coil driving force ΔT varies with time rather than holding constant. Treating a genuinely multi-phase (sensible–latent–sensible) transient problem as a single isothermal snapshot misrepresents the achievable heat transfer rate during the two sensible phases, and every downstream figure computed from that coil capability — Step 6.4's $\dot{Q}_{coil,max}$, Step 7's optimisation conclusion, Step 9's batch time, Step 10's Mode 1.A.2 depletion and benefit figures, and Gate V5 in Step 11 — inherits this simplification.

This appendix has been superseded in its entirety by **Appendix I — Revised Worked Design Example (Phase-Resolved Transient Coil Sizing Method)**, which replaces the single isothermal NTU-effectiveness snapshot with a phase-resolved transient energy balance integrating coil heat transfer rate separately across each of the three thermal phases, using the correct time-varying ΔT for each. The following sections of this document contain content that conflicts with the Appendix I methodology and must not be used as a calculation template for new designs:

- **Step 6, Section 6.4** — The NTU-effectiveness check assumes an isothermal process side for the full batch. Superseded by the phase-resolved coil performance verification in Appendix I, Section I.6.
- **Step 7** — The "no optimisation required" conclusion is drawn directly from the single-phase $\dot{Q}_{coil,max} = 72.9$ kW figure. Re-evaluated in Appendix I,

Section I.7.

- **Step 9, Sections 9.1 and 9.4** — The batch time calculation ($t_{batch} = Q_{design,25}/\dot{Q}_{net,batch}$) divides total lumped energy by a constant net power, implicitly assuming a constant driving force across the entire batch. Superseded by the phase-by-phase batch time summation in Appendix I, Section I.9.
- **Step 10, Section 10.1** — The Mode 1.A.2 emergency draw, battery depletion time, and batch-time benefit figures are all derived from the single-phase $\dot{Q}_{coil,max}$ value and inherit its limitation. Recalculated in Appendix I, Section I.10.
- **Step 11, Gate V5** — The batch-time-vs-brief verification is computed from the superseded Step 9 result. Re-verified in Appendix I, Section I.11.

Steps **1 through 5, 8**, and the underlying **coil/vessel/hydraulic geometry** established in Steps 3, 4, and 8 do not depend on the single-phase coil-capability assumption and remain dimensionally valid; they are carried forward into Appendix I without re-derivation. Where any discrepancy exists between this appendix and Appendix I, the Appendix I version takes precedence.

Superseded by: Appendix I — Revised Worked Design Example (Phase-Resolved Transient Coil Sizing Method) *Retention rationale: Engineering decision record / portfolio documentation*

Note — Steps 1 through 5 and Step 8. The process load definition (Step 1), fluid selection and properties (Step 2), vessel geometry (Step 3), NPS selection and coil configuration (Step 4), hydraulic head and pump selection (Step 5), and coil displacement compensation (Step 8) are unaffected by the single-phase coil-performance simplification described above. These sections are not flagged as conflicting and may be read as background reference, including by Appendix I.

ABOUT THIS APPENDIX

This appendix presents a complete eleven-step application of the VCH Sizing Framework to a real industrial design problem. All calculations are shown in full. Engineering choices made at each decision point are documented alongside their technical justification. This example may be used as a commissioning reference for the specific installation described and as a numerical template for future applications of the framework, **subject to the methodology caveat above for Step 6.4 onward.**

15.1 Problem Statement and Design Brief

A food-grade palm oil storage and processing facility in Kathmandu, Nepal requires two heated storage vessels for RBD (Refined, Bleached, Deodorised) palm oil used in biscuit and confectionery manufacturing. Palm oil solidifies below 35°C and must be maintained in liquid state for processing and dispensing. The facility has limited floor space and is located outdoors in a sheltered enclosure. The following design brief was established prior to framework application:

Table 62. Design Brief Summary

Parameter	Specification
Process fluid	RBD palm oil (food grade, biscuit manufacturing)
Tank 1 capacity	10,000 litres (10 KL) — holding duty only
Tank 2 capacity	25,000 litres (25 KL) — batch processing and hibernation
Operating temperature	45°C (above solidification, below degradation threshold)
Cold feed condition	Fully solid at 25°C (cold storage delivery)
Batch requirement	2,000 kg processed to 45°C within 4 hours
HTF specification	Hot water, 60°C $\pm$ 5°C (product quality constraint)
Site ambient (worst case)	5°C (Kathmandu winter)
Installation	Outdoors, sheltered; limited footprint
Vessel geometry	Vertical cylindrical with cone bottom (drainage requirement)
Coil and vessel material	SS 316L seamless (food grade)

KEY DESIGN CHARACTERISTIC

Palm oil has a latent heat of fusion of 187 kJ/kg and melts over the range 33–39°C. With an HTF supply temperature of 55°C (worst case) and a process melting temperature of 36°C, the maximum available driving force is only 19°C. This narrow temperature differential is the defining constraint of this design and governs coil area selection throughout the framework. The latent heat phase accounts for 82% of total batch energy.

15.2 Step 1 Establishment of the Design Thermal Target

15.2.1 1.1 Palm Oil Thermophysical Constants

Table 63. RBD Palm Oil Thermophysical Properties

Property	Symbol	Value	Phase
Density (liquid at 45°C)	ρ_{oil}	891 kg/m ³	Liquid
Specific heat capacity	$C_{p,solid}$	1.98 kJ/kg·K	Solid
Specific heat capacity	$C_{p,liquid}$	2.15 kJ/kg·K	Liquid
Latent heat of fusion	h_{sf}	187 kJ/kg	Phase change
Melting point midpoint	T_{melt}	36°C	—
Starting temperature	T_{cold}	25°C	Cold storage
Target temperature	T_{target}	45°C	Process
Worst-case ambient	T_{amb}	5°C	Kathmandu winter

15.2.2 1.2 10 KL Tank — Holding Duty Only

The 10 KL tank maintains palm oil in liquid state continuously. Its governing thermal load is environmental heat loss only — no batch heating requirement. The balance load calculation is deferred to Step 5 where vessel surface area is established. The 10 KL tank does not contribute to the batch thermal target.

15.2.3 1.3 25 KL Tank — Batch Thermal Target

The 25 KL tank processes 2,000 kg sub-batches of solid palm oil. The tank starts cold (full 25 KL at 25°C solid) and only the 2,000 kg sub-batch must reach 45°C within 4 hours. The surrounding cold mass acts as a thermal sink that partially absorbs energy from the coil during the batch window.

Sub-batch energy — three phases:

Mass of sub-batch: $m_{batch} = 2,000$ kg

Phase 1 — Sensible heating of solid from T_{cold} to T_{melt} :

$$Q_{s1} = m_{batch} \cdot C_{p,solid} \cdot (T_{melt} - T_{cold}) = 2000 \times 1.98 \times (36 - 25) = \mathbf{43,560 \text{ kJ}}$$

Phase 2 — Latent heat of fusion at $T_{melt} = 36^\circ\text{C}$:

$$Q_{lat} = m_{batch} \cdot h_{sf} = 2000 \times 187 = \mathbf{374,000 \text{ kJ}}$$

Phase 3 — Sensible heating of liquid from T_{melt} to T_{target} :

$$Q_{s2} = m_{batch} \cdot C_{p,liquid} \cdot (T_{target} - T_{melt}) = 2000 \times 2.15 \times (45 - 36) = \mathbf{38,700 \text{ kJ}}$$

$$Q_{batch,2t} = 43,560 + 374,000 + 38,700 = \mathbf{456,260 \text{ kJ}}$$

Surrounding cold mass thermal sink:

Remaining mass: $m_{surrounding} = 22,275 - 2,000 = 20,275$ kg

The surrounding solid oil heats from 25°C toward 36°C during the batch window, absorbing energy from the coil. A 40% participation factor is applied, reflecting the limited thermal conductivity of solid palm oil ($k_{solid} \approx 0.20 \text{ W/m}\cdot\text{K}$) which slows heat penetration outward from the coil surface:

$$\begin{aligned} Q_{sink, effective} &= 0.40 \times m_{surrounding} \times C_{p, solid} \times (T_{melt} - T_{cold}) \\ &= 0.40 \times 20,275 \times 1.98 \times 11 = \mathbf{176,636 \text{ kJ}} \end{aligned}$$

Total governing thermal energy:

$$Q_{design, 25} = Q_{batch, 2t} + Q_{sink, effective} = 456,260 + 176,636 = \mathbf{632,896 \text{ kJ}}$$

Required thermal power over 4-hour batch window (reference only):

$$\dot{Q}_{batch, 25} = \frac{Q_{design, 25}}{t_{batch}} = \frac{632,896}{4 \times 3600} = \mathbf{44.0 \text{ kW}}$$

STEP 1 RESULT

$Q_{design, 25} = 632,896 \text{ kJ}$ $\dot{Q}_{batch, 25} = 44.0 \text{ kW}$ (reference power for 4-hour target)
 Latent heat fraction: 82% of total batch energy
 10 KL tank: holding duty only, thermal load governed by Step 5 loss calculation.

15.3 Step 2 Thermofluid Selection and Property Characterisation

2.1 HTF Selection

Hot water at $60^\circ\text{C} \pm 5^\circ\text{C}$ is specified to prevent thermal degradation and discolouration of food-grade palm oil. The governing worst-case supply temperature is $T_{HTF, in} = 55^\circ\text{C}$. All sizing calculations use this conservative value; actual operation at 60°C provides additional thermal margin.

15.3.1 2.2 HTF Properties — Water at 55°C

Table 64. Hot Water HTF Thermophysical Properties at 55°C

Property	Symbol	Value	Units
Density	ρ_{HTF}	985.7	kg/m ³
Specific heat capacity	$C_{p, HTF}$	4,183	J/kg·K
Dynamic viscosity	μ_{HTF}	0.504×10^{-3}	Pa·s
Thermal conductivity	k_{HTF}	0.648	W/m·K
Prandtl number	Pr_{HTF}	3.26	—

15.3.2 2.3 Process Fluid Properties — RBD Palm Oil at 36°C

The design condition for coil sizing is the melting phase, where the process side is isothermal at $T_{melt} = 36^\circ\text{C}$. The high Prandtl number ($Pr = 489$) confirms that natural convection governs heat transfer on the process side — no mechanical agitation is assumed.

Table 65. RBD Palm Oil Properties at Melting Condition (36°C)

Property	Symbol	Value	Units
Density (liquid at wall)	ρ_{oil}	895	kg/m ³
Specific heat capacity	$C_{p,oil}$	2,150	J/kg·K
Dynamic viscosity	μ_{oil}	38.2×10^{-3}	Pa·s
Thermal conductivity	k_{oil}	0.168	W/m·K
Prandtl number	Pr_{oil}	489	—
Thermal expansion coeff.	β	7.2×10^{-4}	K ⁻¹

15.3.3 2.4 Material Compatibility

From Appendix C, Table C.3: SS 316L is rated EXCELLENT for both RBD palm oil (food grade) and hot water HTF. Carbon steel is rejected (product tainting), copper is rejected (catalyses oil oxidation). SS 316L is confirmed for all wetted surfaces including coil, vessel shell, fittings, and gaskets.

Governing temperature differences at worst-case condition:

$$\Delta T_{max} = T_{HTF,in} - T_{melt} = 55 - 36 = \mathbf{19^\circ C}$$

STEP 2 RESULT

HTF: Hot water, $T_{HTF,in} = 55^\circ\text{C}$ (worst case). Maximum $\Delta T = 19^\circ\text{C}$. Process side $Pr = 489$ confirms natural convection dominance on oil side. Coil and vessel material: SS 316L seamless throughout. Thermal conductivity of coil wall: $k_{wall} = 15.0 \text{ W/m}\cdot\text{K}$.

15.4 Step 3 Conceptual Vessel Geometry and Baseline Dimensional Framework

Vertical cylindrical vessels with cone bottom are selected for both tanks. The cone half-angle is standardised at 60° included (30° from vertical) for complete gravity drainage of viscous palm oil. Aspect ratio H/D is targeted between 1.5 and 2.5 to minimise footprint while maintaining structural stability.

A 15% freeboard allowance is applied to working volume for thermal expansion and safe operating headspace.

15.4.1 3.1 Governing Volume Equations

For a vertical cylinder with cone bottom:

$$V_{cone} = \frac{\pi r^3 \sqrt{3}}{3}, \quad V_{cylinder} = \pi r^2 H_{cyl}, \quad H_{cyl} = \frac{V_{design} - V_{cone}}{\pi r^2}$$

15.4.2 3.2 10 KL Tank Dimensions

Design volume: $V_{design,10} = 10.0 \times 1.15 = 11.5 \text{ m}^3$. Diameter selected: $D = 2.2 \text{ m}$ ($r = 1.1 \text{ m}$).

$$\begin{aligned}
V_{cone,10} &= \frac{\pi \times 1.1^3 \times \sqrt{3}}{3} = \frac{\pi \times 2.327}{3} = 2.436 \text{ m}^3 \\
H_{cyl,10} &= \frac{11.5 - 2.436}{\pi \times 1.1^2} = \frac{9.064}{3.801} = 2.384 \text{ m} \\
h_{cone,10} &= r\sqrt{3} = 1.1 \times 1.732 = 1.905 \text{ m} \\
H_{total,10} &= 2.384 + 1.905 = 4.289 \text{ m}, \quad H/D = 4.289/2.2 = 1.95 \quad \checkmark
\end{aligned}$$

15.4.3 3.3 25 KL Tank Dimensions

Design volume: $V_{design,25} = 25.0 \times 1.15 = 28.75 \text{ m}^3$. Diameter selected: $D = 3.0 \text{ m}$ ($r = 1.5 \text{ m}$).

$$\begin{aligned}
V_{cone,25} &= \frac{\pi \times 1.5^3 \times \sqrt{3}}{3} = \frac{\pi \times 5.845}{3} = 6.119 \text{ m}^3 \\
H_{cyl,25} &= \frac{28.75 - 6.119}{\pi \times 1.5^2} = \frac{22.631}{7.069} = 3.202 \text{ m} \\
h_{cone,25} &= 1.5 \times 1.732 = 2.598 \text{ m} \\
H_{total,25} &= 3.202 + 2.598 = 5.800 \text{ m}, \quad H/D = 5.800/3.0 = 1.93 \quad \checkmark
\end{aligned}$$

15.4.4 3.4 External Surface Areas for Heat Loss

Table 66. Vessel External Surface Areas

Surface	10 KL Tank	25 KL Tank
Cylinder lateral: $A_{cyl} = \pi D H_{cyl}$	16.48 m ²	30.18 m ²
Cone lateral: $A_{cone} = \pi r \sqrt{r^2 + h_{cone}^2}$	7.60 m ²	14.14 m ²
Top head: $A_{top} = \pi r^2$	3.80 m ²	7.07 m ²
Total external area A_{ext}	27.88 m²	51.39 m²

STEP 3 RESULT

10 KL: $D = 2.2 \text{ m}$, $H_{total} = 4.289 \text{ m}$, $A_{ext} = 27.88 \text{ m}^2$.

25 KL: $D = 3.0 \text{ m}$, $H_{total} = 5.800 \text{ m}$, $A_{ext} = 51.39 \text{ m}^2$.

Both vessels within $H/D = 1.93$ – 1.95 . Cone bottom at 30° from vertical confirmed for complete palm oil drainage.

Note: Cone zone (18–21% of working volume) relies on natural convection from heated liquid above — no direct coil coverage in cone section.

15.5 Step 4 Nominal Pipe Size Selection and Maximum Coil Configuration

Design limits for SS 316L in hydronic service: erosion ceiling ($v_{\max} = 2.0 \text{ m/s}$) and stagnation floor ($v_{\min} = 0.3 \text{ m/s}$).

15.5.1 4.1 25 KL Batch Tank — NPS Selection

Required HTF flow rate at $\dot{Q} = 44.0 \text{ kW}$ and $\Delta T_{\text{HTF}} = 5 \text{ }^\circ\text{C}$:

$$\dot{m}_{\text{HTF}} = \frac{\dot{Q}}{C_{p,\text{HTF}} \cdot \Delta T_{\text{HTF}}} = \frac{44,000}{4183 \times 5} = 2.104 \text{ kg/s} \quad (237)$$

$$\dot{V}_{\text{HTF}} = \frac{\dot{m}_{\text{HTF}}}{\rho_{\text{HTF}}} = \frac{2.104}{985.7} = 2.133 \times 10^{-3} \text{ m}^3/\text{s} \approx 7.68 \text{ m}^3/\text{hr} \quad (238)$$

Calculated minimum cross-sectional area:

$$A_{\min} = \frac{\dot{V}_{\text{HTF}}}{v_{\max}} = \frac{2.133 \times 10^{-3}}{2.0} = 1.067 \times 10^{-3} \text{ m}^2 \quad (239)$$

Selection: NPS 1.5" Schedule 10S SS 316L ($A_{\text{in}} = 1.433 \times 10^{-3} \text{ m}^2 > A_{\min}$). Actual velocity and Reynolds number:

$$v_{\text{actual},25} = \frac{\dot{V}_{\text{HTF}}}{A_{\text{in}}} = 1.489 \text{ m/s} \quad (25.6\% \text{ below erosion ceiling}) \quad (240)$$

$$Re_{25} = \frac{\rho \cdot v \cdot d_{\text{in}}}{\mu} = \frac{985.7 \times 1.489 \times 0.04272}{0.504 \times 10^{-3}} = 124,271 \quad (\text{Turbulent}) \quad (241)$$

15.5.2 4.2 10 KL Holding Tank — NPS Selection

The holding tank is governed by the stagnation floor ($v_{\min} = 0.3 \text{ m/s}$). Selecting NPS 1" Schedule 10S ($A_{\text{in}} = 0.610 \times 10^{-3} \text{ m}^2$):

$$\dot{V}_{\text{hold}} = v_{\min} \cdot A_{\text{in}} = 1.830 \times 10^{-4} \text{ m}^3/\text{s} \approx 0.659 \text{ m}^3/\text{hr} \quad (242)$$

15.5.3 4.3 Coil Configuration

Structural wall clearance is set to 100 mm per side. Pitch is defined as $1.25 \times d_{\text{out}}$ with a packing factor $\eta_{\text{pack}} = 0.85$.

Table 67. Coil Configuration Summary

Parameter	10 KL Holding	25 KL Batch
Pipe Specification	NPS 1" Sch 10S SS 316L	NPS 1.5" Sch 10S SS 316L
Inner diameter d_{in}	27.86 mm	42.72 mm
Outer diameter d_{out}	33.40 mm	48.26 mm
Helix diameter D_{helix}	1.967 m	2.752 m
Coil pitch p	41.75 mm	60.33 mm
Active turns N_{active}	46	42
Developed length L_{coil}	284.2 m	363.2 m
External area A_s	29.81 m ²	55.05 m ²
Operating velocity	0.30 m/s	1.489 m/s
Reynolds number	16,362	124,271

Minimum bend radius ($R_{min} = 2.5 \times d_{out}$) verification:

$$R_{min,1.5''} = 0.121 \text{ m} \ll (D_{helix}/2 = 1.376 \text{ m}) \quad \checkmark \quad (243)$$

$$R_{min,1''} = 0.084 \text{ m} \ll (D_{helix}/2 = 0.984 \text{ m}) \quad \checkmark \quad (244)$$

STEP 4 RESULT

25 KL: NPS 1.5" Sch 10S, 42 active turns, $L_{coil} = 363.2 \text{ m}$, $A_s = 55.05 \text{ m}^2$.

10 KL: NPS 1" Sch 10S, 46 active turns, $L_{coil} = 284.2 \text{ m}$, $A_s = 29.81 \text{ m}^2$.

Both coils satisfy bend radius constraints and packing efficiency requirements.

15.6 Step 5 Hydraulic Resistance Characterisation and Pump Selection

5.1 25 KL Tank — System Head

Darcy friction factor (Petukhov, $Re = 124,271$):

$$f_{straight,25} = (0.790 \ln(124,271) - 1.64)^{-2} = (0.790 \times 11.730 - 1.64)^{-2} = (7.627)^{-2} = 0.01718$$

Helical correction factor:

$$f_{helix,25} = f_{straight} \times \left[1 + 3.54 \sqrt{\frac{d_{in}}{D_{helix}}} \right] = 0.01718 \times \left[1 + 3.54 \sqrt{\frac{0.04272}{2.752}} \right] = 0.01718 \times 1.441 = 0.02475$$

Coil friction head:

$$H_{coil,25} = f_{helix} \cdot \frac{L_{coil}}{d_{in}} \cdot \frac{v^2}{2g} = 0.02475 \times \frac{363.2}{0.04272} \times \frac{1.489^2}{19.62} = \mathbf{23.81 \text{ m}}$$

Fittings equivalent length (4× 90° LR elbow, 2× gate valve, 2× ball valve, 2× tee run, 1× swing check, per Appendix C Table C.4):

$$L_{eq,25} = (4 \times 16 + 2 \times 8 + 2 \times 3 + 2 \times 20 + 1 \times 100) \times 0.04272 = 226 \times 0.04272 = 9.655 \text{ m}$$

$$H_{fittings,25} = 0.02475 \times \frac{9.655}{0.04272} \times 0.1131 = \mathbf{0.633 \text{ m}}$$

Static head: $H_{static,25} = H_{total,25} = 5.800 \text{ m}$

Safety margin (20% on friction losses): $H_{safety,25} = 0.20 \times (23.81 + 0.633) = 4.889 \text{ m}$

$$H_{total,25} = 23.81 + 0.633 + 5.800 + 4.889 = \mathbf{35.13 \text{ m}}$$

15.6.1 5.2 10 KL Tank — System Head

At $v = 0.3 \text{ m/s}$, $Re = 16,362$, $f_{straight,10} \approx 0.028$:

$$f_{helix,10} = 0.028 \times \left[1 + 3.54 \sqrt{\frac{0.02786}{1.967}} \right] = 0.028 \times 1.421 = 0.03979$$

$$H_{coil,10} = 0.03979 \times \frac{284.2}{0.02786} \times \frac{0.3^2}{19.62} = \mathbf{1.863 \text{ m}}$$

$$L_{eq,10} = 226 \times 0.02786 = 6.296 \text{ m}, \quad H_{fittings,10} = \mathbf{0.413 \text{ m}}$$

$$H_{total,10} = 1.863 + 0.413 + 4.289 + 0.455 = \mathbf{7.02 \text{ m}}$$

15.6.2 5.3 Environmental Heat Loss

Insulation specification: 75 mm PIR foam jacket ($k_{ins} = 0.024 \text{ W/m}\cdot\text{K}$), aluminium sheet cladding. External convection (sheltered outdoor): $h_{ext} = 15 \text{ W/m}^2\text{K}$.

$$R_{total} = \frac{t_{ins}}{k_{ins}} + \frac{1}{h_{ext}} = \frac{0.075}{0.024} + \frac{1}{15} = 3.125 + 0.067 = 3.192 \text{ m}^2\text{K/W}$$

10 KL holding tank (continuous hold at 45°C):

$$\dot{Q}_{loss,10} = \frac{T_{target} - T_{amb}}{R_{total}} \times A_{ext,10} = \frac{45 - 5}{3.192} \times 27.88 = \mathbf{349 \text{ W} = 0.35 \text{ kW}}$$

25 KL tank hibernation (hold at 38°C, just above melt point):

$$\dot{Q}_{loss,25,hib} = \frac{38 - 5}{3.192} \times 51.39 = \mathbf{531 \text{ W} = 0.53 \text{ kW}}$$

Combined steady-state heat loss (both tanks):

$$\dot{Q}_{loss,total} = \dot{Q}_{loss,10} + \dot{Q}_{loss,25,hib} = 0.35 + 0.53 = \mathbf{0.88 \text{ kW}}$$

This combined figure is the absolute minimum continuous thermal output any driver must supply to prevent both tanks from cooling below their respective set-points. It forms the floor for the Step 9 driver power envelope under the heat-loss-only sizing basis.

15.6.3 5.4 Pump Selection

Table 68. Pump Duty Points and Selection

Parameter	10 KL Holding	25 KL Batch
Flow rate $\dot{V}$	0.659 m ³ /hr	7.68 m ³ /hr
Total head H	7.02 m	35.13 m
Selected pump	KSB Rio-Eco 25-40	KSB Etanorm SYT 040-025-200
Motor power	0.04 kW	2.2 kW
Nepal supplier	General Market	General Market
Est. price ex-works Europe	€180 – €260	€750 – €980
Nepal landed (incl. 20% import)	€216 – €312	€900 – €1,176
VFD (Siemens G120C)	€280 – €380	€420 – €560

Prices are reference estimates as of 2025. Confirm current pricing with present date.

STEP 5 RESULT

25 KL: $H_{total} = 35.13$ m at 7.68 m³/hr. KSB Etanorm SYT 040-025-200.

10 KL: $H_{total} = 7.02$ m at 0.659 m³/hr. KSB Rio-Eco 25-40.

Steady-state heat losses: 0.35 kW (10 KL) + 0.53 kW (25 KL hibernation) = **0.88 kW** combined. This is the driver sizing floor.

15.7 Step 6 Thermal Performance Evaluation

The governing condition is the melting phase: process side isothermal at $T_{melt} = 36^\circ\text{C}$, HTF at $T_{HTF,in} = 55^\circ\text{C}$, $\Delta T_{max} = 19^\circ\text{C}$.

15.7.1 6.1 Inside Convection Coefficient h_i — 25 KL Coil

Gnielinski correlation with helical Dean flow enhancement:

Straight pipe Nusselt number ($Re = 124,271$, $Pr = 3.26$, $f/8 = 2.148 \times 10^{-3}$):

$$Nu_{straight} = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)} = \frac{2.148 \times 10^{-3} \times 123,271 \times 3.26}{1.7254} = 500.9$$

Helical correction:

$$Nu_{helix,25} = 500.9 \times \left[1 + 3.54 \left(\frac{d_{in}}{D_{helix}} \right)^{0.868} \right] = 500.9 \times [1 + 3.54 \times (0.01553)^{0.868}] = 500.9 \times 1.0958 = 548.9$$

$$h_{i,25} = \frac{Nu_{helix,25} \times k_{HTF}}{d_{in}} = \frac{548.9 \times 0.648}{0.04272} = \mathbf{8,326 \text{ W/m}^2\text{K}}$$

15.7.2 6.2 Outside Convection Coefficient h_o — Natural Convection in Palm Oil

Churchill-Chu correlation for natural convection from horizontal cylinder. Mean film temperature 40°C , $\Delta T_{wall} = 14^\circ\text{C}$:

$$Ra_{25} = \frac{g\beta\Delta T_{wall}d_{out}^3}{\nu\alpha} = \frac{9.81 \times 7.2 \times 10^{-4} \times 14 \times (0.04826)^3}{4.268 \times 10^{-5} \times 8.735 \times 10^{-8}} = 2.975 \times 10^6$$

$$Nu_o = \left[0.60 + \frac{0.387 \times Ra^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right]^2 = \left[0.60 + \frac{0.387 \times 11.99}{1.00648} \right]^2 = [5.208]^2 = 27.12$$

$$h_{o,25} = \frac{27.12 \times 0.168}{0.04826} = \mathbf{94.4 \text{ W/m}^2\text{K}}$$

15.7.3 6.3 Overall Heat Transfer Coefficient U

Wall conduction term for NPS 1.5" SS 316L ($k_{wall} = 15.0 \text{ W/m}\cdot\text{K}$):

$$\frac{1}{U_{wall}} = \frac{d_{out} \ln(d_{out}/d_{in})}{2k_{wall}} = \frac{0.04826 \times \ln(1.1297)}{30.0} = \frac{0.04826 \times 0.12201}{30.0} = 1.962 \times 10^{-4}$$

$$\frac{1}{U_{25}} = \frac{1}{h_o} + \frac{1}{U_{wall}} + \frac{d_{out}}{d_{in}} \cdot \frac{1}{h_i} = \frac{1}{94.4} + 1.962 \times 10^{-4} + \frac{0.04826}{0.04272} \times \frac{1}{8326} = 0.010912 \text{ m}^2\text{K/W}$$

$$U_{25} = \mathbf{91.6 \text{ W/m}^2\text{K}}$$

Note: $h_o = 94.4 \text{ W/m}^2\text{K}$ accounts for 97% of total thermal resistance. $h_i = 8,326 \text{ W/m}^2\text{K}$ contributes less than 2%. Outside natural convection in viscous oil is the dominant resistance — confirmed as expected from $Pr_{oil} = 489$.

Section Conflict — 6.4. The NTU-effectiveness check below treats the entire batch as a single isothermal phase-change condition at constant T_{melt} . This is a valid representation only of the latent-heat phase (Q_{lat} , $\approx 82\%$ of total load) established in Step 1.3; it does not correctly represent the two sensible-heating phases (Q_{s1} , Q_{s2}), during which the process-side temperature is not constant. The resulting $\dot{Q}_{coil,max}$ figure, and every value derived from it in Steps 7, 9, 10, and Gate V5 of Step 11, is superseded by the phase-resolved transient method in Appendix I, Section I.6. Do not use this section as a calculation template.

15.7.4 6.4 NTU-Effectiveness Check — 25 KL

For an isothermal phase-change process (melting at constant T_{melt}):

$$NTU_{25} = \frac{U_{25} \times A_{s,25}}{\dot{m}_{HTF} \times C_{p,HTF}} = \frac{91.6 \times 55.05}{2.104 \times 4183} = \frac{5,043}{8,801} = 0.5730$$

$$\varepsilon_{25} = 1 - e^{-NTU} = 1 - e^{-0.5730} = 0.4364$$

$$\dot{Q}_{max} = \dot{m}_{HTF} \cdot C_{p,HTF} \cdot (T_{HTF,in} - T_{melt}) = 2.104 \times 4183 \times 19 = 167,115 \text{ W}$$

$$\dot{Q}_{actual,25} = \varepsilon_{25} \times \dot{Q}_{max} = 0.4364 \times 167,115 = \mathbf{72,918\text{ W} = 72.9\text{ kW}}$$

The coil is capable of transferring 72.9 kW to the process. The actual heat delivered is limited by whichever is smaller: the coil capability or the driver rated output. With the 36 kW driver selected in Step 9, the driver is the binding constraint and the coil operates well within its thermal capacity.

15.7.5 6.5 10 KL Holding Tank Thermal Check

At $v = 0.3\text{ m/s}$ (transitional-turbulent), conservative $h_i \approx 800\text{ W/m}^2\text{K}$, $h_o \approx 85\text{ W/m}^2\text{K}$, giving $U_{10} \approx 74.3\text{ W/m}^2\text{K}$.

Required coil area for 0.35 kW holding load:

$$A_{required,10} = \frac{\dot{Q}_{loss,10}}{U_{10} \times \Delta T} = \frac{350}{74.3 \times 40} = \mathbf{0.118\text{ m}^2}$$

Available: 29.81 m^2 . Ratio: $253\times$ oversized by thermal area requirement. Coil geometry is governed by structural and hydraulic integrity requirements, not heat transfer area.

STEP 6 RESULT

25 KL coil capability: $U = 91.6\text{ W/m}^2\text{K}$, $\dot{Q}_{coil,max} = 72.9\text{ kW}$ (Section 6.4 figure — see conflict note above; superseded by Appendix I, Section I.6). With the 36 kW driver selected at Step 9, the driver is the binding constraint. The coil has substantial reserve capacity for fouling degradation.

10 KL: $U = 74.3\text{ W/m}^2\text{K}$, holding load trivially satisfied.

15.8 Step 7 Coil Geometry Optimisation

Section Conflict — Step 7. The conclusion below rests on the single-phase $\dot{Q}_{coil,max} = 72.9\text{ kW}$ figure from Step 6.4. Re-evaluated against the phase-resolved coil performance in Appendix I, Section I.7.

No geometric optimisation is required for either vessel.

The 25 KL single helix is capable of transferring 72.9 kW, well above the 36 kW driver output. The coil is not the limiting element. Reducing coil turns would erode the fouling degradation buffer and is not recommended for a food-grade oil application where coil cleaning frequency is limited. The 42-turn configuration is confirmed as final.

The 10 KL coil is structurally and hydraulically driven. The 46-turn configuration provides uniform fluid distribution throughout the vessel and supports the minimum circulation integrity required by the VFD stagnation floor constraint.

Final confirmed coil specifications:

Table 69. Final Coil Specifications

Parameter	10 KL Holding	25 KL Batch
Configuration	Single helix	Single helix
Pipe specification	NPS 1" Sch 10S SS 316L	NPS 1.5" Sch 10S SS 316L
Active turns N_{active}	46	42
Helix diameter D_{helix}	1.967 m	2.752 m
Pitch	41.75 mm	60.33 mm
Developed length L_{coil}	284.2 m	363.2 m
External surface area A_s	29.81 m ²	55.05 m ²
Overall U (melting phase)	74.3 W/m ² K	91.6 W/m ² K
Coil capability $\dot{Q}_{coil,max}$	1.76 kW	72.9 kW
Driver-limited $\dot{Q}_{delivered}$	0.35 kW	35.12 kW (net)
Thermal margin (coil vs driver)	$\gg$ driver	+102%

15.9 Step 8 Volumetric Displacement Compensation and Final Vessel Dimensioning

The coil occupies internal vessel volume that displaces working oil. This must be verified against the working volume requirement.

15.9.1 8.1 25 KL Tank — Coil Displacement

Pipe wall cross-section (NPS 1.5"):

$$A_{wall,25} = \frac{\pi}{4}(d_{out}^2 - d_{in}^2) = \frac{\pi}{4}(0.04826^2 - 0.04272^2) = 3.958 \times 10^{-4} \text{ m}^2$$

Total coil displacement (pipe wall + HTF inside coil):

$$V_{coil,25} = (A_{wall} + A_{in}) \times L_{coil} = (3.958 \times 10^{-4} + 1.433 \times 10^{-3}) \times 363.2 = 1.829 \times 10^{-3} \times 363.2 = \mathbf{0.664 \text{ m}^3}$$

$$\frac{V_{coil,25}}{V_{working,25}} = \frac{0.664}{25.0} = 2.66\% \leq 5\% \quad \checkmark$$

15.9.2 8.2 10 KL Tank — Coil Displacement

$$V_{coil,10} = (A_{wall,1"} + A_{in,1"}) \times L_{coil,10} = (4.54 \times 10^{-4} + 0.610 \times 10^{-3}) \times 284.2 = 1.064 \times 10^{-3} \times 284.2 = \mathbf{0.302 \text{ m}^3}$$

$$\frac{V_{coil,10}}{V_{working,10}} = \frac{0.302}{10.0} = 3.02\% \leq 5\% \quad \checkmark$$

No vessel resize is required for either tank.

Table 70. Final Vessel Dimensions After Displacement Compensation

Parameter	10 KL Tank	25 KL Tank
Working volume	10.0 m ³	25.0 m ³
Coil displacement	0.302 m ³ (3.0%)	0.664 m ³ (2.7%)
Net oil capacity	9.698 m ³	24.336 m ³
Design volume (with freeboard)	11.5 m ³	28.75 m ³
Vessel diameter D	2.2 m	3.0 m
Cylindrical height H_{cyl}	2.384 m	3.202 m
Cone height h_{cone}	1.905 m	2.598 m
Total vessel height H_{total}	4.289 m	5.800 m
Footprint area	3.80 m ²	7.07 m ²

15.10 Step 9 Thermal Driver Sizing

Section Conflict — Step 9. Sections 9.1 and 9.4 compute batch time as total lumped energy divided by a constant net power, implicitly assuming a constant driving force across the entire batch. This is superseded by the phase-by-phase batch time summation in Appendix I, Section I.9. The driver power envelope and selection (Sections 9.2–9.3) are unaffected and remain valid.

SIZING BASIS — THIS STEP

The driver minimum floor is set by the combined steady-state heat loss of both tanks only. It does not incorporate the batch thermal power requirement. This reflects the operational decision that batch cycle time is a scheduling variable rather than a hard constraint, and that a reduced-capital driver with a longer batch window is acceptable for this installation. The batch time consequence is calculated explicitly below and documented in the commissioning notes.

15.10.1 9.1 Minimum Driver Power — Heat Loss Basis

The minimum continuous output required to prevent either vessel cooling below its set-point under worst-case ambient ($T_{amb} = 5^\circ\text{C}$) is the sum of both tank steady-state heat losses from Step 5:

$$\dot{Q}_{driver,min} = \dot{Q}_{loss,10} + \dot{Q}_{loss,25,hib} = 0.35 + 0.53 = \mathbf{0.88\text{ kW}}$$

Any driver delivering less than 0.88 kW continuously cannot maintain both tanks above their set-points under worst-case conditions. This is the hard floor.

Note on batch capability: A driver at the 0.88 kW floor has zero net power available for batch heating after covering losses. Useful batch capability requires the driver to exceed the floor. Net batch power and resultant batch time are:

$$\dot{Q}_{net,batch} = \dot{Q}_{driver} - \dot{Q}_{driver,min}$$

$$t_{batch} = \frac{Q_{design,25}}{\dot{Q}_{net,batch}} = \frac{632,896 \text{ kJ}}{\dot{Q}_{net,batch}}$$

15.10.2 9.2 Maximum Driver Power

The maximum is constrained by two independent physical limits. The more restrictive governs.

Limit A — HTF temperature ceiling: The product quality constraint fixes $T_{HTF,max} = 60^\circ\text{C}$ to prevent thermal degradation of RBD palm oil. At erosion velocity ceiling $v_{max} = 2.0 \text{ m/s}$ through NPS 1.5" Sch 10S:

$$\dot{m}_{HTF,max} = \rho_{HTF} \cdot v_{max} \cdot A_{in} = 985.7 \times 2.0 \times 1.433 \times 10^{-3} = 2.825 \text{ kg/s}$$

$$\dot{Q}_{Limit A} = \dot{m}_{HTF,max} \cdot C_{p,HTF} \cdot (T_{HTF,max} - T_{HTF,return}) = 2.825 \times 4183 \times (60 - 55) = \mathbf{59.1 \text{ kW}}$$

Limit B — Coil thermal saturation: The 25 KL coil maximum capability is 72.9 kW. Limit B is therefore $\dot{Q}_{Limit B} = 72.9 \text{ kW}$.

Governing limit: Limit A at 59.1 kW governs:

$$\dot{Q}_{driver,max} = \min(59.1, 72.9) = \mathbf{59.1 \text{ kW}}$$

15.10.3 9.3 Design Envelope

$$0.88 \text{ kW} \leq \dot{Q}_{driver} \leq 59.1 \text{ kW}$$

15.10.4 9.4 Driver Selection — 36 kW

Selected: 36 kW electric immersion heater bank (three $\times$ 12 kW SS 316L elements, 3-phase 415 V, 50 Hz)

$$\dot{Q}_{driver} = 36 \text{ kW} \quad \in \quad [0.88, 59.1 \text{ kW}] \quad \checkmark$$

Batch time at 36 kW:

$$\dot{Q}_{net,batch} = 36.0 - 0.88 = \mathbf{35.12 \text{ kW}}$$

$$t_{batch} = \frac{632,896 \times 10^3 \text{ J}}{35,120 \text{ W}} = 18,021 \text{ s} = \mathbf{5.01 \text{ hr}}$$

The 36 kW driver delivers a batch time of 5.01 hours, which exceeds the original 4-hour brief by 1.01 hours (25%). This is a deliberate design trade-off: the 36 kW selection reduces installed electrical capacity and capital cost relative to the original 60 kW specification, at the cost of a longer batch window. The commissioning schedule must reflect the 5-hour batch cycle.

*Batch time calculated at worst-case 55°C HTF supply, including 0.88 kW steady-state loss deduction. Batch time figures inherit the Section 6.4 single-phase limitation noted above. **Framework output***

Table 71. Driver Power Envelope and Standard Size Assessment (Revised)

Standard Size	Within Envelope	Net batch power	Batch time	Engineering note
0.88 kW	Floor	0 kW	∞	Survival only; no batch capability
12 kW	✓	11.12 kW	15.8 hr	Very slow; one 12 kW element
24 kW	✓	23.12 kW	7.6 hr	Moderate; two 12 kW elements
36 kW	✓	35.12 kW	5.01 hr	Selected. Three 12 kW elements.
48 kW	✓	47.12 kW	3.73 hr	Meets 4-hr SLA; four 12 kW
55 kW	✓	54.12 kW	3.25 hr	Fast batch; near original spec
60 kW	Warning	59.12 kW	2.98 hr	At HTF ceiling; original design

to designer:

- Selected rated output: **36 kW** (3×12 kW elements) at 60°C flow temperature
- Minimum floor (heat loss only): **0.88 kW** at 60°C flow temperature
- Maximum ceiling (HTF temperature limit): **59.1 kW**
- Supply / return temperatures: 60°C / 55°C ($\Delta T_{HTF} = 5^\circ\text{C}$)
- Supply voltage: 3-phase 415 V, 50 Hz (Nepal standard)
- Batch time consequence: **5.01 hr** per 2,000 kg sub-batch (worst-case 55°C)
- Control interface: 0–10 V or 4–20 mA modulating input for SCR integration with PLC Loop 1 per Appendix G

STEP 9 RESULT

Driver floor (heat-loss basis): **0.88 kW**.

HTF ceiling: 59.1 kW. Envelope: [0.88, 59.1] kW.

Selected: 36 kW (3×12 kW SS 316L immersion elements, 415 V 3-phase).

Net batch power: 35.12 kW. Worst-case batch time: **5.01 hr** (exceeds 4-hr brief by 25%; figure superseded by Appendix I, Section I.9 — see conflict note above).

Commissioning schedules must be updated to reflect 5-hour batch cycle pending the Appendix I recalculation.

15.11 Step 10 Thermal Battery Sizing

Section Conflict — Step 10.1. The Mode 1.A.2 emergency draw, battery depletion time, and batch-time benefit figures below are all derived from the single-phase $\dot{Q}_{coil,max} = 72.9$ kW value from Step 6.4. Recalculated in Appendix I, Section I.10. Section 10.2 and 10.3 (battery energy requirement and thermal time constant) depend only on steady-state heat loss and vessel geometry, and remain valid.

15.11.1 10.1 Mode 1.A.2 Assessment

Under normal PLC-controlled operation the 25 KL pump runs at its design setpoint (1.489 m/s, 7.68 m³/hr), and the coil transfers 35.12 kW net to the batch — exactly what the 36 kW driver can sustain after covering losses. Mode 1.A.2 is not triggered under this condition.

However, the coil is physically capable of transferring up to 72.9 kW when the pump operates at full rated speed (2.0 m/s, erosion ceiling). If the VFD is commanded to maximum flow — deliberately or in an emergency rapid-heat scenario — the coil demand rises to 72.9 kW while the driver can only supply 36 kW. The deficit must be drawn from the thermal battery:

$$\dot{Q}_{bat,draw} = \dot{Q}_{coil,max} - \dot{Q}_{driver} = 72.9 - 36.0 = \mathbf{36.9\text{ kW}}$$

Battery depletion under Mode 1.A.2 emergency draw:

The 200 L buffer vessel holds useful energy between $T_{supply} = 60^\circ\text{C}$ and $T_{min} = 45^\circ\text{C}$ (the process set-point; below this temperature the buffer can no longer drive heat into the process):

$$Q_{bat,useful} = m_{bat} \cdot C_{p,HTF} \cdot (T_{supply} - T_{min}) = 197.1 \times 4183 \times (60 - 45) = \mathbf{12,370\text{ kJ}}$$

Time for battery to reach minimum useful temperature ($T_{min} = 45^\circ\text{C}$):

$$t_{dep} = \frac{Q_{bat,useful}}{\dot{Q}_{bat,draw}} = \frac{12,370 \times 10^3\text{ J}}{36,900\text{ W}} = \mathbf{335\text{ s} \approx 5.6\text{ min}}$$

This is a short burst. After 5.6 minutes the HTF temperature in the buffer falls to 45°C , the driving force across the coil collapses to zero, and heat transfer to the process stops entirely. The PLC **must** enforce a hard time limit on Mode 1.A.2 operation. The recommended interlock is: if $T_{bat} \leq 47^\circ\text{C}$ (2°C above minimum, allowing sensor lag), the PLC commands the VFD back to normal setpoint immediately and locks out further emergency-flow requests for a configurable recovery period (suggested minimum: 30 minutes, to allow the driver to reheat the buffer vessel).

Batch time benefit of Mode 1.A.2: During the 5.6-minute emergency window the process receives 72.9 kW net (minus 0.88 kW losses = 72.02 kW effective). After the window the system reverts to 35.12 kW net. The combined effect:

$$\begin{aligned} E_{emergency} &= 72,020\text{ W} \times 335\text{ s} = 24,127\text{ kJ} \\ E_{remaining} &= 632,896 - 24,127 = 608,769\text{ kJ} \\ t_{remaining} &= \frac{608,769 \times 10^3}{35,120} = 17,333\text{ s} = 4.81\text{ hr} \\ t_{total,Mode\ 1.A.2} &= \frac{335}{3600} + 4.81 = \mathbf{4.90\text{ hr}} \end{aligned}$$

Mode 1.A.2 reduces the worst-case batch time by approximately 6 minutes (from 5.01 hr to 4.90 hr). The battery is not sized for sustained Mode 1.A.2 duty — it is sized for resilience only — so this benefit is a one-shot burst per recovery cycle, not a repeatable acceleration tool.

15.11.2 10.2 Battery Energy Requirement

Battery sized for a 2-hour backup window sustaining both vessels in holding mode during a power interruption or driver maintenance outage:

$$Q_{battery} = (\dot{Q}_{loss,10} + \dot{Q}_{loss,25,hold}) \times t_{backup} = (0.35 + 0.53) \times 2 \times 3600 = \mathbf{6,336 \text{ kJ}}$$

Minimum buffer vessel volume at $\Delta T = T_{supply} - T_{target} = 60 - 45 = 15^\circ\text{C}$:

$$V_{battery,min} = \frac{Q_{battery}}{\rho_{HTF} \cdot C_{p,HTF} \cdot \Delta T} = \frac{6,336,000}{985.7 \times 4183 \times 15} = \mathbf{0.102 \text{ m}^3} = 102 \text{ L}$$

Specified: 200 L insulated buffer vessel (2× margin on minimum volume). Standard off-the-shelf unit with 75 mm PIR insulation jacket. Battery sizing is unchanged from the original design because the heat-loss loads that govern resilience are independent of the driver selection.

15.11.3 10.3 Battery Thermal Time Constant

For a 200 L cylindrical vessel ($D \approx 0.5 \text{ m}$, $H \approx 1.0 \text{ m}$), $A_{bat} \approx 2.36 \text{ m}^2$, $R_{ins} = 3.0 \text{ m}^2\text{K/W}$:

$$S_{t,battery} = \frac{m_{bat} \cdot C_{p,HTF}}{U_{bat} \cdot A_{bat}} = \frac{200 \times 4183}{(1/3.0) \times 2.36} = \frac{836,600}{0.786} = \mathbf{1,064,000 \text{ s} \approx 295 \text{ hours}}$$

Loop 3 scan interval per Appendix G, Section G.B: $\Delta t_{scan,3} = S_{t,battery}/20 = 53,200 \text{ s}$. In practice Loop 3 is configured at a 30-minute practical minimum interval given the extreme time constant.

STEP 10 RESULT

Mode 1.A.2 is **not triggered** under normal PLC setpoint operation.

Mode 1.A.2 **can be triggered** deliberately by commanding the VFD to maximum flow (2.0 m/s). At full pump speed the coil draws 72.9 kW; the driver covers 36 kW; the battery must supply the 36.9 kW deficit (figures superseded by Appendix I, Section I.10 — see conflict note above).

Battery depletion time ($60^\circ\text{C} \rightarrow 45^\circ\text{C}$ minimum useful): **5.6 min**.

Batch time benefit: -6 min ($5.01 \text{ hr} \rightarrow 4.90 \text{ hr}$).

PLC interlock required: revert VFD to normal setpoint when $T_{bat} \leq 47^\circ\text{C}$.

Battery is sized for resilience only (2-hr outage at 0.88 kW loss rate), not for sustained Mode 1.A.2 duty. Minimum volume: 102 L. Specified: 200 L. $S_{t,battery} = 295 \text{ hours}$.

15.12 Step 11 Design Verification and Release

Section Conflict — Gate V5. The batch-time verification below is computed from the superseded Step 9 result. Re-verified in Appendix I, Section I.11. Gates V1–V4 and V6–V8 do not depend on the single-phase coil-performance assumption and remain valid.

All eight verification gates are evaluated below. Gates V1–V4 and V6–V8 are unaffected by the driver selection. Gate V5 (batch time) and Gate V6 (driver capacity) are revised.

Table 72. Design Verification Gate Summary — 36 kW Driver

Gat	Criterion	Verification	Result	Margin Note	/
V1	Velocity ceiling	$v_{25} = 1.489 \leq v_{max} = 2.0$ m/s	PASS	–25.6%	
V2	Velocity floor	$v_{10} = 0.30 \geq v_{min} = 0.30$ m/s	PASS	At limit	
V3	Thermal performance 25 KL	Coil capability 72.9 kW $\gg$ driver 36 kW	PASS	Coil not limiting	
V4	Thermal performance 10 KL	$\dot{Q}_{avail} \gg \dot{Q}_{req} = 0.35$ kW	PASS	+400%	
V5	Batch time	$t_{actual} = 5.01$ hr $t_{brief} = 4$ hr	PASS*	+25% over brief	
V6	Driver vs heat-loss floor	$\dot{Q}_{driver} = 36$ kW $\dot{Q}_{floor} = 0.88$ kW	$\gg$ PASS	+3,991%	
V7	Coil displacement	$V_{coil}/V_{working} \leq 3.1\%$ both tanks $\leq 5\%$ limit	PASS	Within limit	
V8	Material compatibility	SS 316L with RBD palm oil + hot water: EXCELLENT	PASS	—	

* V5 passes against the heat-loss-only sizing basis using the superseded Step 9 figure. See design note below and Appendix I, Section I.11.

Battery sized for resilience; not for sustained Mode 1.A.2 duty. See Step 10.1.

STEP 11 DESIGN NOTE — BATCH TIME AND BRIEF COMPLIANCE

The original design brief specified a 2,000 kg batch to 45°C within **4 hours**. The 36 kW driver delivers a worst-case batch time of **5.01 hours** at 55°C HTF supply, exceeding the brief by 1.01 hours (25%).

This outcome is accepted under the revised sizing basis where the driver floor is governed by heat loss only, not by the batch time requirement. Gate V5 is marked PASS* to indicate a deliberate informed deviation from the original brief, not an oversight. **This batch-time figure is itself superseded by the phase-resolved recalculation in Appendix I — see the Methodology Supersession Notice at the head of this appendix.**

At the nominal HTF supply of 60°C the batch time reduces to approximately 4.5 hours due to the increased driving force. The 4-hour window can be recovered without hardware change by one or more of the following operational measures:

1. Accept palm oil deliveries in a pre-softened or partially melted state (reduced latent heat load).
2. Maintain the 25 KL tank at hibernation temperature (38°C) so the next batch starts above the melting point — eliminating Phase 1 and Phase 2 entirely, reducing batch energy to 38,700 kJ and batch time to 0.31 hours.
3. Install a third 12 kW element (upgrade to 48 kW) which achieves a 3.73 hr batch time, meeting the original 4-hour SLA without any change to the vessel or coil design.

The commissioning schedule must reflect the 5-hour worst-case cycle until one of the above measures is adopted, or until Appendix I's recalculated figure is available.

ALL VERIFICATION GATES PASSED — DESIGN RELEASED (ARCHIVED, VERSION 1)

Gate V5 carries a deliberate brief deviation notation. See design note above. **This release record is archived; current design verification must reference Appendix I.**

15.13 Complete Design Summary

Archived summary. The table below reflects the Version 1 methodology described in the Methodology Supersession Notice at the head of this appendix. Rows describing batch window, coil capability, and thermal margin are computed from the single-phase Step 6.4 figure and are superseded by Appendix I. All other rows (vessel, coil geometry, hydraulics, and shared infrastructure sizing) remain valid.

Table 73. Complete VCH Design Output — Palm Oil Storage System, Kathmandu

Parameter	10 KL Holding Tank	25 KL Batch Tank
<i>Duty</i>		
Function	Constant liquid hold	Batch processing + hibernation
Operating modes	Mode B / Mode C	Mode 1.A.1 / Mode B / Mode C
Governing thermal load	0.35 kW (heat loss)	35.12 kW (net batch)
Batch window (worst case)	—	5.01 hr
<i>Vessel</i>		
Geometry	Vertical cyl. + cone bottom	Vertical cyl. + cone bottom
Diameter	2.2 m	3.0 m
Total height	4.289 m	5.800 m
Net oil capacity	9.698 m ³	24.336 m ³
Footprint	3.80 m ²	7.07 m ²
Material	SS 316L	SS 316L
Insulation	75 mm PIR + Al cladding	75 mm PIR + Al cladding
<i>Coil</i>		
Configuration	Single helix	Single helix
Pipe	NPS 1" Sch 10S	NPS 1.5" Sch 10S
Active turns	46	42
Developed length	284.2 m	363.2 m
External surface area	29.81 m ²	55.05 m ²
Overall U	74.3 W/m ² K	91.6 W/m ² K
Dominant resistance	Outside nat. convection	Outside nat. convection
<i>Hydraulics</i>		
HTF flow rate	0.659 m ³ /hr	7.68 m ³ /hr
Operating velocity	0.30 m/s	1.489 m/s
System head	7.02 m	35.13 m
Pump model	KSB Rio-Eco 25-40	KSB Etanorm SYT 040-025-200
VFD	Siemens G120C	Siemens G120C

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Table 73 — continued from previous page

Parameter	10 KL Holding Tank	25 KL Batch Tank
<i>Shared Infrastructure</i>		
Thermal driver	Electric immersion heater bank, 3 × 12 kW (36 kW total)	
Driver floor	0.88 kW (steady-state losses)	
Driver ceiling	59.1 kW	
Thermal battery	200 L buffer; 2-hr outage; Mode 1.A.2: 5.6 min	
HTF specification	Hot water, 55–60°C, closed loop	
Control system	Central PLC, three-loop architecture	
<i>Indicative Hardware Cost (Euro, Nepal landed)</i>		
Pumps (both)	€1,116 – €1,488	
VFDs (both)	€840 – €1,128	
Thermal driver	€520 – €750	

Vessel fabrication, coil bending, insulation, instrumentation, PLC, and civil works costs are site-specific and excluded from this summary.

DESIGN NOTES FOR COMMISSIONING (ARCHIVED — VERSION 1)

1. The melting phase ($T_{melt} = 36^{\circ}\text{C}$, isothermal process side) is the critical thermal condition. PID tuning per Appendix G must be performed at this condition, not at the final target temperature of 45°C .
2. The worst-case batch time is **5.01 hours** at 55°C HTF supply for a 2,000 kg sub-batch starting fully solid at 25°C , per the Version 1 single-phase method. **Refer to Appendix I for the phase-resolved recalculation before finalising production schedules.** At nominal 60°C HTF the Version 1 batch time estimate reduces to approximately 4.5 hours.
3. The fastest upgrade path to recover the original 4-hour batch window is to add a **fourth 12 kW element** (total 48 kW, *no vessel or coil changes required*), achieving a Version 1-estimated batch time of 3.73 hours. The heater bank enclosure and electrical supply should be sized for 48 kW at installation even if only 36 kW of elements are initially fitted.
4. Outside natural convection ($h_o = 94.4 \text{ W/m}^2\text{K}$) accounts for 97% of total thermal resistance. Even gentle agitation of the vessel contents would dramatically increase h_o and reduce batch time — this is a potential throughput upgrade path requiring no driver change.
5. The 25 KL coil is thermally capable of 72.9 kW under the Version 1 isothermal assumption (see Appendix I for the corrected figure). The driver at 36 kW is the binding constraint; the coil has substantial reserve for fouling degradation. Monitor TCV-401 opening trend in the SCADA historian. A progressive increase in required valve opening to maintain batch rate is the primary fouling indicator. Schedule coil inspection when required opening exceeds 80% of maximum.
6. The cone bottom zone (21% of 25 KL working volume) has no direct coil coverage. Ensure the outlet drain valve is opened first when starting a batch to allow warm liquid from the top of the vessel to flow into the cone zone by gravity before pumping begins.